CLASS OF LEARNERS STUDIO • DEFINITIVE ENGINEERING DOSSIER & SYSTEM MANUAL

2026–27 EDITION

CLASS OF LEARNERS (CoL)
COMPLETE ECOSYSTEM DOSSIER

14 CORE WEB, DESKTOP & NATIVE ANDROID INITIATIVES • 100% BUILT FOR FREE USE • 2026–27 EDITION & SYSTEM ARCHITECTURE

A comprehensive, richly illustrated engineering report documenting every application built by the **Class Of Learners** student studio in Mumbai—explaining why we need each project, why we built our own **100% Free-Use** versions even where commercial paid tools exist, how every cloud and AI service (**Supabase, GitHub, Vercel, Sarvam AI, Gemini, Lovable, Ollama**) works under the hood, our complete file-system trees, and our live demonstrations to the **SnehAsha Foundation Founder & Team** and **Visiting Senior Government & Education Officers**.

01–02. Site CoL & CoL Android APK | **03–04. ATI 1.4 Desktop & Android SDK 35** | **05. 3D Solar Engine (Keplerian Physics)** | **06. Perceptus + GestureUI + Voice Jarvis** | **07. Mumbai Traffic 3D (Flagship Deep-Dive)** | **12. Quest RPG (COMING SOON • JAN 2027)**

01. Site CoL, ATI 1.4 & 3D Solar Engine

Started on **Gemini Web**, continued with **Antigravity, OpenCode & Freebuff**. Demonstrated live to the **SnehAsha Founder & Team** and **Senior Officers**, with steady support from our Guiding Light **Siddhi Jaiswal**.

02. Perceptus + GestureUI + Voice Jarvis

Unified touchless HCI engine: we started with Python & MediaPipe, moved to **Lovable** (`window-mind-whisper.lovable.app`), and integrated **Voice Jarvis** using **Sarvam AI & Ollama**.

03. Mumbai Traffic 3D (9-Page Flagship Section)

Our largest ongoing initiative: **3D HQ Lobby & Classroom**, **Multiplayer**, **Character Customization**, **16 Vehicles**, **Mumbai Maps**, Pacejka MF 5.2 physics, 54 levels, and **CivicLogic DB**.

04. Standardized 11-Point Ledgers & 2026–27 Roadmaps

Every project features an **11-Point Architectural Ledger** covering exact student authors, file types, offline packaging, and future milestones—plus dedicated **2026–27 Roadmaps** on Pages 52–54.

STRICT ORDERED PROJECT SEQUENCE ACROSS THIS DOSSIER

01 – 02: Site CoL Web Portal → CoL Android App (`COL.apk`)
03 – 04: ATI 1.4 Desktop Suite → ATI Native Android App (`SDK 35`)
05 – 06: 3D Solar System Engine → Perceptus + GestureUI + Voice Jarvis
07 – 08: Mumbai Traffic 3D Simulator (9 Pages) → Dynamic QR Code Studio
09 – 10: FutureScape Career Guidance (436 Paths) → Terra 3D Earth Globe
11 – 12: CivicLogic Traffic DB → Quest RPG (**COMING SOON**)
13 – 14: Cereal AI Android App → CastFlow Desktop PC & Android TV
Roadmaps: 2026–27 Deep-Dives: Traffic WebXR, ATI 1.5 & Quest RPG (pp. 52–54)

14	100% Free	65+	4 APKs
CORE INITIATIVES	ZERO ADS • ZERO PAYWALLS	LIVE APP CAPTURES	NATIVE ANDROID BUILDS

Mentors: Sanjana Kasbe • Avantika Raskar **Guiding Light:** Siddhi Jaiswal • **Showcase:** SnehAsha Founder & Team **Lead:** Neel Badri & CoL Team
https://advancedlogiclabs.dpdns.org • github.com/NeelAniGamer • neelgbadri@col.dpdns.org

Executive Guide: Understanding The Class Of Learners Ecosystem

This **Master Engineering Dossier (2026–27 Edition)** is designed so that both **educators, school principals, and government leaders** as well as **software architects** can immediately see what every project is, why we built it for **100% Free Use**, which cloud and AI services power it, and how its codebase works.

01 – 02 • CORE WEB & MOBILE HUB

1. Site CoL & CoL Android App

pp. 11–15

Our central digital school campus at **advancedlogiclabs.dpdns.org** and its companion Android app (**COL.apk**). Students open one link or app to launch 3D simulators, verify digital graduation diplomas (**verify.html**), check live leaderboards, and download offline installers with zero ads or paywalls.

03 – 05 • SKILL & SPACE SCIENCE

2. ATI Typing & 3D Solar Engine

pp. 16–23

Two flagship desktop and mobile learning suites. **ATI 1.4** turns touch-typing into an 8-player car race with 44.1 kHz mechanical keyboard audio; **3D Solar Engine** is an interactive space telescope solving real NASA Keplerian orbits. Both were demonstrated live to the **SnehAsha Founder & Team** and **Senior Officers**.

06 • TOUCHLESS MULTIMODAL AI

3. Perceptus + GestureUI + Jarvis

pp. 24–26

GestureUI and Perceptus are one unified touchless system. We started with Python & MediaPipe webcam tracking, moved to **Lovable** (`window-mind-whisper.lovable.app`), and integrated **Voice Jarvis** (Sarvam AI Indic speech + Ollama), letting users point in the air and speak commands simultaneously.

07 & 11 • LARGEST FLAGSHIP INITIATIVE

4. Mumbai Traffic 3D & CivicLogic

pp. 27–35, 42–43

Our largest ongoing engineering initiative (**9 dedicated pages**). Features a **3D HQ Lobby, Classroom & Garage**, **Multiplayer**, **16 Mumbai Vehicles**, **Mumbai Maps & Zones**, Pacejka tire physics, and 54 levels—paired with **CivicLogic (TRAFFIC.EDU v3.0)**, our 50-module traffic database.

08, 09, 10 & 12 • DEDICATED WEB SUITES

5. QR, Career, Terra 3D & RPG

pp. 36–41, 44–45

Each utility is documented across dedicated pages: **Dynamic QR Code Studio** with scan analytics, **FutureScape Career Compass** mapping **436 Indian & global careers**, the **Terra 3D** interactive Earth globe, and **Quest RPG (COMING SOON • Jan 2027)**.

13 & 14 + 2026–27 DEEP-DIVES • NATIVE APPS & ROADMAPS

6. Cereal, CastFlow & 2026–27 Roadmaps

pp. 46–54

Cereal (`Cereal.apk`) records lectures with on-device note generation. **CastFlow PC & TV** beams screens over Wi-Fi/DLNA. Plus dedicated **2026–27 Flagship Deep-Dives**: Mumbai Traffic WebXR VR, ATI 1.5 Neural Adaptive, and Quest RPG Q1 2027 Release!

2026–27 Strategic Vision • Cloud, Security & Offline Foundations

WebSockets, WebRTC & DLNA

Instant two-way communication channels that let multiplayer typing cars, traffic convoys, and wireless TV screen casts update in milliseconds.

Offline-First (.exe / .deb / .apk)

Self-contained software installers that run smoothly on school lab PCs and Android phones even when classroom internet is disconnected.

SHA-256 Digital Seal

An unforgeable cryptographic fingerprint embedded in every student graduation diploma, verifiable instantly on **verify.html**.

Master Systems Matrix & Verification Sign-Off: Comparative Architectural Ledger, Multi-OS Packaging & Complete Source File Inventory pp. 55–58

01. Academic Mentors, Guiding Light, SnehAsha Showcase & Team

Founded in Mumbai in 2024, **Class Of Learners (CoL)** is a student-led engineering studio supported by dedicated academic mentors, our Guiding Light **Siddhi Jaiswal (Siddhi Ma'am)**, and live demonstrations to the **SnehAsha Foundation Founder & Team** and senior officers:

Academic Mentors, Guiding Light & SnehAsha / VIP Officer Demonstrations

PROJECT MENTOR & EDUCATOR

Sanjana Kasbe Ma'am

Provides foundational technical guidance, algorithmic critique, mathematical verification, and academic mentorship across **ATI 1.4**, the **3D Solar System Engine**, and **Perceptus / GestureUI** computer vision interfaces.

Focus: Astrodynamics • Acoustics • HCI

TEACHER & CURRICULUM MENTOR

Avantika Raskar Ma'am

Provides dedicated curriculum guidance, civic pedagogy oversight, and educational validation for the **Mumbai Traffic Simulator (54 stages)**, **CivicLogic DB**, and **ATI typing curricula**, ensuring authentic classroom impact.

Focus: Motor Vehicles Act • Civic Pedagogy

GUIDING LIGHT & SNEHASHA SHOWCASE

Siddhi Jaiswal & SnehAsha

Siddhi Jaiswal (Siddhi Ma'am) is our **Guiding Light & Supporter** whose belief in our vision inspired our SnehAsha initiative (`sneh-asha.html`). Furthermore, we demonstrated **ATI 1.4** and the **3D Solar Engine** live to the **SnehAsha Founder & Team** and **Senior Officers**.

Guiding Light • SnehAsha Founder & VIP Review

Six Members. One Studio. (Individual Core Engineering Team Cards)

Neel Badri

Lead Developer & Logic Engine

Architects core systems and backend runtime bridges across **Python, Kotlin, Java, C++, and TypeScript**. Oversees WebSocket multiplayer engines, Pacejka physics integration, SQLite/Supabase security proxies, OAuth2 PKCE flows, and cross-platform installer toolchains.

Ansh Patil

Co-Developer, QA Lead & Presenter

Stress-tests edge cases across all **54 traffic levels and 200 typing stages**, refines user experience flows, prototypes the browser RPG game, and leads live project demonstrations to educators, SnehAsha leaders, and visiting officials.

Aarush Vangari

UI/UX Design Lead & Physics Engine

Brings concepts to life with sleek visual interfaces, glassmorphic HUD tokens (**col-ui.css**), and fluid physical simulations. Merges aesthetic discipline with mathematical precision across the 3D Solar System and Traffic Simulator viewports.

Yashraj Jadhav

QA Tester, UI/UX & Idea Developer

Refines interactive ergonomics, stress-tests core gameplay mechanics on classroom hardware, and develops **creative feature directions** from early concept sketches through to polished production releases.

Aarayaman Jadhav

3D Systems & Geospatial Engineering

Specializes in real-time 3D WebGL rendering and geospatial data pipelines behind **Terra 3D** and **Mumbai Traffic 3D**. Converts raw GeoJSON country boundary datasets and 3D city models into high-FPS Three.js layers.

John Doe (Anonymous)

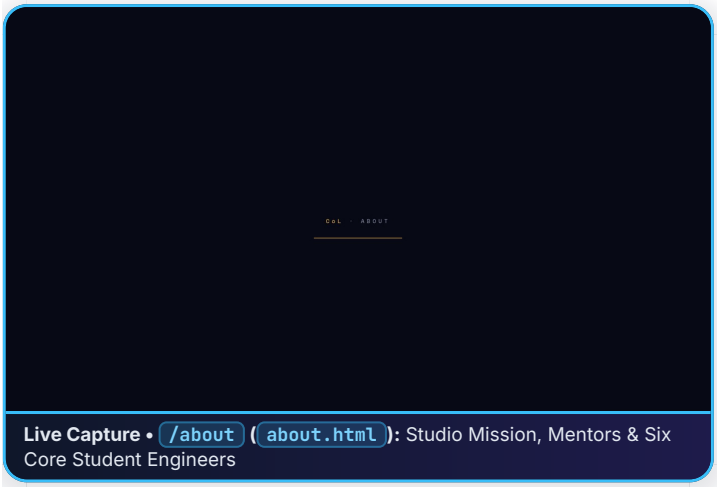
Secret Helper & Systems Advisor

Behind-the-scenes contributor providing stealth architectural guidance, **compiler flag optimization**, memory profiling, and rapid debugging across desktop and mobile builds when deadlines are tight.

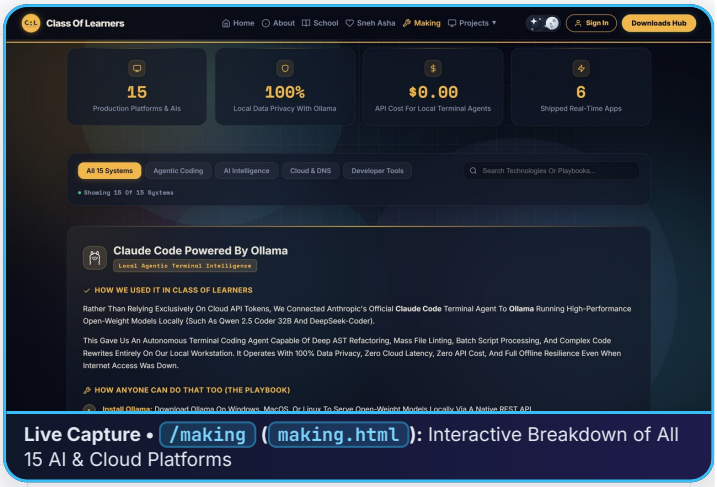
Root Workspace File Structure & How Our Websites Run

Our primary web repository (**Vercel/** , hosted at `github.com/NeelAniGamer/Vercel`) serves as the unified source of truth for **Site CoL** (`https://advancedlogiclabs.dpdns.org`). Below is the complete root directory hierarchy:

```
Vercel/ (Class Of Learners Master Web Portal & Distribution Root • github.com/NeelAniGamer/Vercel)
├── home.html & index.html        • Main Site CoL portal, interactive 3D project cards & global launcher
├── about.html & making.html      • Team profiles, mentor credits & showcase of all 15 AI/cloud platforms
├── sneh-asha.html & school.html  • SnehAsha Foundation tribute & Bhavani Shankar Road Junior College hub
├── solar.html & engine.html      • Solar System landing portal (/solar) & 3D WebGL Orrery runtime (/engine)
├── ati.html & ati-demo.html      • ATI landing hub (/ati) & ATI v3.2.0 Interactive Velocity Studio (/ati-demo)
├── gesture.html & CastFlow.html  • Perceptus Flow Neural V12 portal (/gesture) & wireless screen casting studio
├── qr.html, qr-editor.html, q.html• QR Code Studio, custom SVG/PNG frame designer & dynamic short-link router
├── Career.html & careers-data.js • FutureScape Career Guidance (436 career paths + 60s Career Compass)
├── Database_Logic.html & rpg.html• CivicLogic (TRAFFIC.EDU v3.0 50-module DB) & Quest RPG (COMING SOON)
├── verify.html & download.html   • SHA-256 cryptographic diploma verifier & central binary download hub
├── dashboard.html & sitemap.html • Student progress dashboard, feedback.html, admin.html & SEO sitemap index
├── col-auth.js & col-ui.css      • Live-session OAuth identity manager (window.colUser) & 360px design tokens
├── react-src/GamePage.tsx        • React 19/TypeScript HUD bundled via esbuild into Traffic/simulator-bundle.js
├── Traffic/ & Terra3D/           • 54-level Mumbai Traffic Simulator & standalone Three.js 3D Earth Atlas
└── build.js, vercel.json, funding• Deterministic build pipeline creating dist/, security gates & funding.json
```



Live Capture • `/about` (`about.html`): Studio Mission, Mentors & Six Core Student Engineers



Live Capture • `/making` (`making.html`): Interactive Breakdown of All 15 AI & Cloud Platforms

How Site CoL Runs In Production (4-Pillar Edge Pipeline)

1. Deterministic Build Engine (`build.js`)

Running **npm run build** clears and recreates `dist/` , validates traffic levels (`level1, level5, level_custom`), copies runtime trees while filtering out local secrets/models, and bundles `react-src/GamePage.tsx` into `dist/Traffic/simulator-bundle.js` via esbuild.

2. Post-Build Verification & Security Gates

Every build triggers **npm run verify:production** and **npm run security:check** , guarding live-session identity (`window.colUser`), QR verifier storage, Electron sandbox flags, and GitHub CodeQL compliance.

3. Client Routing & Liquid-Glass UI (`col-router.js`)

Shared runtime scripts (`col-router.js` , `col-ui.js` , `pull-refresh.js` , `liquid-glass.css` , `mobile-perf.css`) provide instant navigation transitions, dark/light theme persistence, and 60 FPS compositor-only animations across all 26 HTML portals.

4. Anycast DNS & Immutable Asset Caching

`vercel.json` configures clean URL rewrites and 1-year immutable cache headers for heavy 3D `.glb` models and audio assets, resolved globally via **Digitalplat (dpdns.org)** and **ClouDNS Anycast** .

Complete File System Trees Across Production Suites (Part 1)

Every desktop, web, and native Android application is organized in a clean, modular workspace across `Vercel/` and `CoL Project/`. Each pair of side-by-side dark terminal file-tree cards below is engineered with **identical height and 10 structured entries** for visual symmetry:

01. ATI 1.4 (Desktop, Manim & Native Android APK)

```
ATI 1.4/ (github.com/NeelAniGamer/Advanced-Typing-Instructor)
├── main.py              • HTTP API, WebSockets & pywebview
├── curriculum.py        • 200-stage progressive typing generator
├── script.js & style.css• Core typing loop & cyber HUD styles
├── sanitizeText.ts      • 22-check ASCII whitelist normalizer
├── safeNumber & grading • Finite clamps & relative WPM grading
├── AdvancedTyping.iss   • Inno Setup 6 Windows installer script
├── build_deb.sh         • Native Debian Linux .deb packager
├── ati_manim/           • Manim mathematical promo video engine
└── android_native/      • Pure Java Android SDK 35 (11 screens)
```

02. Mumbai Traffic Simulator 3D (Vercel/Traffic/)

```
Traffic/ (github.com/NeelAniGamer/Vercel/tree/main/Traffic)
├── Driving.html         • Primary 3D driving simulator viewport
├── Academy.html         • 54-level interactive learning academy
├── TrafficDashboard.html• Dense-rank civic leaderboard & badges
├── TrafficSetup.html    • Hardware & DPR graphics calibration
├── GamePage.html        • React 19 simulator-bundle.js shell
├── game_core.js         • 14,975-line Three.js & Pacejka engine
├── npc-ai.js & ui.js    • Mumbai NPC traffic AI & challan HUD
├── Levels/ (1 to 54)    • 54 distinct Mumbai road safety stages
└── tools/validate-levels• Static AST validator for all 55 levels
```

03. 3D Solar System Engine (Game 3.4 & Game 4.3.4)

```
Game 3.4/ & Game 4.3.4/ (github.com/NeelAniGamer/Solar-System)
├── main.py              • PyQt6 native desktop wrapper & server
├── index.html & engine  • 3D WebGL celestial Orrery viewport
├── logic.js             • Newton-Raphson Keplerian orbit solver
├── skychart.js          • 10,000-star J2000 planetarium sphere
├── survey.js & audio.js • Deep space survey & ambient synthesis
├── textures/ & shaders  • Procedural canvas planetary shaders
├── mobile-60fps-opt.md  • InstancedMesh & DPR optimization spec
├── Game 3.7.iss         • Inno Setup Windows installer script
└── build_exe.bat        • PyInstaller standalone exe compiler
```

04. Perceptus, GestureUI & Voice Jarvis Suite

```
GestureUI + Perceptus + Voice Typing Prototype/
├── mouse_server.py      • Python WS server + 2D Kalman filter
├── main.js & index.html • MediaPipe 30fps worker & air canvas
├── keyboard.html        • Floating dwell-click virtual keyboard
├── hand_landmarker.task • Neural 21-point 3D hand vision model
├── Perceptus/index.tsx  • Lovable + TanStack Start React 19 UI
├── vision.functions.ts  • 468-point FaceMesh attention tracker
├── brain.functions.ts   • Supabase vector memory & LLM router
├── Voice Typing/assist. • Jarvis AI PC voice macro controller
└── BUILD_EXE.bat        • Standalone PerceptusFlow.exe builder
```

05. CoL Android & Cereal Android Apps (SDK 35)

```
COL Android (COL/) & Cereal (Current Projects/Granola20/)
├── COL/MainActivity.kt  • Edge-to-edge WebView & OAuth bridge
├── COL/version.json     • OTA APK auto-updater & SHA-1 check
├── COL/provider_paths   • Secure FileProvider APK package URI
├── Granola20/MainAct.kt • Jetpack Compose Cereal UI (1,896 loc)
├── MeetingRecorderSvc.kt• Foreground mic & Gemini 2.5 Flash API
├── MeetingRepository.kt • Local JSON & .m4a recording ledger
├── GoogleCalendarMgr.kt • OAuth2 Google Calendar event sync
├── GoogleTasksManager.kt• Action-item export to Google Tasks
└── Cereal.apk & COL.apk • Verified Android 14+ SDK 35 binaries
```

06. CastFlow (PC & TV) + Terra 3D + R&D Labs

```
CastFlow + Terra3D + CivicLogic + R&D Prototypes/
├── Cast Pc/main.py      • FastAPI 40fps MJPEG + ADB USB bridge
├── Cast Tv/MainAct.kt   • Android TV/Mobile CastFlow Compose UI
├── WebServerService.kt  • Ktor server + MediaProjection promote
├── DlnaNetwork.kt       • SSDP discovery & SOAP AVTransport XML
├── Terra3D/src/Globe.tsx• 3D Earth mesh & 195+ GeoJSON borders
├── careers-data.js      • 436 career pathways & salary dataset
├── compass-scoring.js   • 60-Second Career Compass cosine engine
├── uigen/src/lib/       • Virtual File System & Babel AST study
└── Tracker/ & AI CoL/   • Habit tracker & local LLM writing labs
```

Monorepo + Isolated Native Workspaces

While `Vercel/` powers global web delivery, heavy desktop and Android compilation workspaces (`ATI 1.4`, `Game 3.4`, `GestureUI`, `COL`, `Granola20`, `Cast Tv`) maintain isolated lockfiles and Gradle toolchains so local binary builds never bloat web deployments.

Continued On Next Page (Deep Sub-Directory Trees)

The next page expands the internal subdirectories of `ATI Native Android (11 Java Activities)`, `Traffic 3D Models & 54 Levels`, `QR & Career Modules`, `UIGen Reference Files`, and `R&D Labs` so every file in our file system is visible.

Complete File System Trees (Part 2: Deep Sub-Directories & Assets)

To ensure our file-system documentation shows **every single sub-folder, 3D model directory, level pack, native Java/Kotlin package, and R&D lab** across our computers, the six terminal trees below expand the deep internal folders of our ecosystem:

07. ATI 1.4 Native Android Sub-Tree (android_native/)

```
ATI 1.4/android_native/app/src/main/java/com/ati/typingquest/
├── ui/MainActivity.java          • PC GameHUD, XP, Emeralds & chips
├── ui/RaceActivity.java          • 15s/30s/60s Sprint & 2D track
├── ui/LevelsActivity.java        • 200-stage campaign grid browser
├── ui/MultiplayerLobbyAct.java   • OkHttp WebSocket live race room
├── ui/GarageActivity.java        • Unlockable cars & 6 switch shops
├── ui/KeyboardHeatmapAct.java    • Per-key SQLite error color map
├── ui/Achievements & Diploma.java• 16 badges & SHA-256 Bitmap cert
├── ui/Daily & CustomPractice.java• Seeded streaks & 7 coding langs
└── core/SoundEffects & DB.java   • 44.1kHz AudioTrack PCM & SQLite
```

08. Traffic 3D Levels, Models & Electron Sub-Tree

```
Vercel/Traffic/ (Deep Sub-Directories & 3D Assets)
├── Levels/level1.js..level10.js • Tier 1: LLR controls & signals
├── Levels/level11.js..level20.js• Tier 2: School & silence zones
├── Levels/level21.js..level30.js• Tier 3: Box junctions & lanes
├── Levels/level31.js..level40.js• Tier 4: Monsoon rain & night fog
├── Levels/level41.js..level54.js• Tier 5-6: Heavy buses & RTO exam
├── Levels/level_custom.js        • Open-world Mumbai Free-Roam city
├── Models/*.glb & public/        • Draco 3D vehicles, buildings,HQ
├── src/ & electron/              • Vite TS port & Electron desktop
└── tools/validate-levels.js      • 55-level _checkTasks AST checker
```

09. QR Studio, Career Compass, CivicLogic & RPG

```
Vercel/ (Web Utility Modules & Supabase Schemas)
├── qr.html & qr.css              • Main QR generator & scan camera
├── qr-editor.html                • Custom eye frames & Level-H logo
├── qr-dynamic.js & q.html        • Supabase short-link redirector
├── supabase-qr-migration.sql     • RLS policies for qr_links/scans
├── Career.html & career.css      • FutureScape counseling dashboard
├── careers-data.js (196 KB)      • 436 Indian/global career schemas
├── compass-scoring & features.js• 8-axis cosine matcher & export
├── Database_Logic.html (181 KB) • TRAFFIC.EDU v3.0 50-module DB
└── rpg.html                      • Quest RPG (COMING SOON preview)
```

10. Terra 3D Globe & 2D World Tracker Sub-Tree

```
Vercel/Terra3D/ & CoL Project/Tracker/
├── Terra3D/package.json          • Standalone React 19 + Three.js
├── Terra3D/src/App.tsx           • Layer toggles & country search
├── Terra3D/src/components/Globe  • Spherical Earth + Fresnel halo
├── Terra3D/electron/             • Desktop Electron globe wrapper
├── Vercel/globe.js (142 KB)      • Shared WebGL globe runtime
├── Vercel/borders.js (122 KB)    • 195+ sovereign GeoJSON polygons
├── Tracker/index.html            • Lightweight 2D map companion
├── Tracker/world.svg             • High-precision SVG country paths
└── Tracker/script.js & style.css• Interactive zoom & nation stats
```

11. UIGen External Reference Study Sub-Tree

```
Current Projects/uigen/ (Reference Architecture Study)
├── package.json & next.config.ts• Next.js 15 + React 19 + Turbopack
├── prisma/schema.prisma          • SQLite user & virtual FS storage
├── src/lib/file-system.ts        • In-memory VirtualFileSystem tree
├── src/lib/transform/jsx-transf.• In-browser @babel/standalone AST
├── src/components/preview/       • Sandboxed Blob-URL iframe runner
├── src/components/editor/        • Multi-tab Monaco code editor UI
├── src/app/api/chat/route.ts     • Anthropic Claude tool streaming
├── src/lib/auth.ts               • jose HTTP-only JWT cookie auth
└── README.md                     • Reference architecture notes
```

12. R&D Labs, Security Gates & CI Workflows

```
Supporting R&D Workspaces & Automated Verification/
├── ati_manim/ati_promo.py        • 60fps Manim vector math promos
├── Game 4.3.4/index.html         • Procedural planet shader lab
├── Solar_System_3D_App/main.py   • PyQt6 offline science lab build
├── Voice Typing Prototype/       • Offline speech dictation macros
├── Vercel/verify-production.js   • Checks dist/ bundle completeness
├── Vercel/security-check.js      • Audits window.colUser & QR RLS
├── .github/workflows/codeql.yml  • Automated GitHub CodeQL scanner
├── .github/workflows/devskim.yml• Microsoft DevSkim security gate
└── funding.json                  • Buy Me A Coffee & Sponsors links
```

Complete File-System Transparency: Together, Pages 04, 05, and 06 map every root file, sub-project directory, 3D asset pack, native Android package, and automated security workflow across the Class Of Learners ecosystem.

4-Stage Development Lifecycle & Defensive Coding Contracts

Every Class Of Learners project progresses through a disciplined **4-Stage Engineering Lifecycle**—combining rapid multi-AI prototyping with strict mathematical and runtime validation gates:

STAGE 01 • PEDAGOGY FIRST

Classroom Need & Pedagogical Blueprint

Identify real student pain points with mentors **Sanjana Kasbe** and **Avantika Raskar** (e.g., hunt-and-peck typing, passive textbook astronomy, memorization-only road rules) and design interactive **30-second feedback loops** with zero forced replay.

STAGE 02 • MULTI-AI SYNTHESIS

Iterative Multi-AI Toolchain Progression

Scaffold initial architectures in **Gemini Web** and **Lovable**, scale multi-file engines in **Google Antigravity**, and debug local Python/WebSocket daemons in **OpenCode**, **Freebuff**, and **Claude Code + Ollama**.

STAGE 03 • RUNTIME CONTRACTS

Defensive Sanitization & Automated QA

Enforce strict runtime safety contracts: **sanitizeTypingText** (22 ASCII checks preventing unmatchable Unicode stalls), **safeNumber / safeInt** (preventing `Infinity` / `NaN` profile corruption), and **validate-levels.js** across all 54 traffic stages.

STAGE 04 • NATIVE DISTRIBUTION

Multi-OS Packaging & VIP Officer Showcase

Compile standalone binaries via **PyInstaller**, **Inno Setup 6** (`.exe`), **dpkg** (`.deb`), and **Android Studio Gradle SDK 35** (4 `.apk`s), followed by live demonstrations to the **SnehAsha Founder & Team** and senior officers.

Architectural Q&A: Why TypeScript + Python WeasyPrint?

ENGINEERING DECISION RECORD

Is Making Applications & Tooling With TypeScript Better?

Yes—for interactive UIs, state management, and data contracts, TypeScript is significantly better than plain JavaScript:

- **Compile-Time Bug Prevention:** In **react-src/GamePage.tsx**, **Perceptus/src/routes/index.tsx**, **Terra3D/src/components/Globe.tsx**, and **ATI 1.4/src/utils/safeNumber.ts**, TypeScript catches null-reference bugs, missing level properties, and invalid number types before the code ever runs in a browser.
- **Self-Documenting Interfaces:** Complex schemas like 54 Traffic level configs, 436 Career pathways, and multimodal sensor events become strictly typed contracts with full IDE autocomplete.
- **Best-of-Both-Worlds Dossier Pipeline:** For generating publication-grade multi-page PDF reports with 65+ high-resolution screenshots, combining modern HTML5/CSS3 component layouts with **WeasyPrint + PyMuPDF visual verification** delivers print-ready pagination while TypeScript powers our interactive React/Vite bundles.

Language / Runtime Layer	Where Used in CoL Ecosystem	Primary Engineering Advantage
TypeScript (`.ts` / `.tsx`)	`GamePage.tsx`, `Perceptus/`, `Terra3D/`, `ATI src/utils/`	Compile-time type safety, strict interface contracts, and zero-runtime-overhead `esbuild` bundling.
Kotlin & Java (`SDK 35`)	`COL.apk`, `Cereal.apk`, `CastFlowTV.apk`, `ATI.apk`	Direct hardware access to Android `AudioTrack` PCM, `MediaProjection`, `ForegroundService`, and SQLite.
Python 3.12 Backend	`ATI 1.4/main.py`, `mouse_server.py`, `Cast Pc/main.py`	Low-latency local WebSockets, 2D Kalman filtering, `pywin32` OS window control, and WeasyPrint PDF compilation.

02. Six Foundational Scientific & Pedagogical Investigations

Under the academic mentorship of **Sanjana Kasbe Ma'am** and **Avantika Raskar Ma'am**, the Class Of Learners studio conducted six applied research investigations that directly power our production engines:

INVESTIGATION 01 • VEHICLE DYNAMICS

Pacejka MF 5.2 Tire Physics & Civic Law-by-Doing

Implemented Hans Pacejka's Magic Formula ($F(x) = D \cdot \sin(C \cdot \arctan(Bx - E(Bx - \arctan(Bx))))$) with load-sensitive friction (μ) and a **combined-slip friction ellipse** across dry asphalt, wet monsoon roads, and gravel—paired with live **Motor Vehicles Act e-challan enforcement** across 54 Mumbai levels.

INVESTIGATION 02 • ASTRODYNAMICS

Keplerian Orbit Solver & 14-Order Logarithmic Scaling

Solved Kepler's transcendental equation $M = E - e \sin(E)$ via **Newton-Raphson iteration** (10^{-8} rad accuracy) and designed a continuous logarithmic coordinate compression matrix spanning **14 orders of magnitude** in WebGL without 32-bit Z-fighting.

INVESTIGATION 03 • COMPUTER VISION HCI

Dual-Stage EMA + 2D Kalman Tremor Filtering

Combined client-side **Exponential Moving Average ($\alpha = 0.38$)** on 30 FPS MediaPipe hand landmarks with a server-side **2D Constant-Velocity Kalman Filter (R = 1.2, Q = 0.09)** and a 2% edge dead-zone to eliminate webcam cursor jitter while preserving fast flicks.

INVESTIGATION 04 • DIGITAL ACOUSTICS

44.1 kHz Procedural Mechanical Switch PCM Synthesis

Synthesized real-time **16-bit 44.1 kHz PCM waveforms** modeling 6 mechanical keyboard switches (Cherry MX Blue, Red, Topre, Hall Effect, Holy Panda, IBM Model M) using damped harmonic oscillators and filtered noise bursts with **zero external audio files**.

INVESTIGATION 05 • MULTIMODAL AI

Deictic Gesture + Indic Voice Temporal Fusion

Engineered a **\\pm 150\text{ ms} ring-buffer** in Perceptus that binds spatial (x, y) pointing coordinates from GestureUI with spoken Hindi, Marathi, and English intents transcribed by **Sarvam AI Saaras ASR** and local **Ollama LLaMA-3**.

INVESTIGATION 06 • SYSTEMS SECURITY

Zero-Trust Cloud Proxies & SHA-256 Credentials

Eliminated direct browser database writes by routing all multiplayer and leaderboard updates through Python backend proxies (**/api/mp_cloud**) and designed tamper-evident **SHA-256 digital diplomas** verifiable offline and via Web Crypto API.

Empirical Benchmark & Mathematical Verification Summary

Research Domain	Governing Mathematical Model	Measured Precision / FPS	Verification Script / Gate
Tire & Braking Physics	Pacejka MF 5.2 + Friction Ellipse	60 FPS / +31% Wet Stopping Dist	`Traffic/tools/validate-levels.js`
Orbital Mechanics	Newton-Raphson Kepler Solver (E_k)	$\lvert\Delta E\rvert < 10^{-8}$ rad (3–4 steps)	`Game 3.4/logic.js` & `skychart.js`
Touchless Vision HCI	EMA ($\alpha = 0.38$) + 2D Kalman ($R = 1.2$)	30 FPS Vision / 60 FPS Render	`GestureUI/mouse_server.py`
Typing Race Integrity	22-Check ASCII + Relative Grading	44.1 kHz Audio / 0 Unicode Stalls	`ATI 1.4/verify_mp_race_text.mjs`

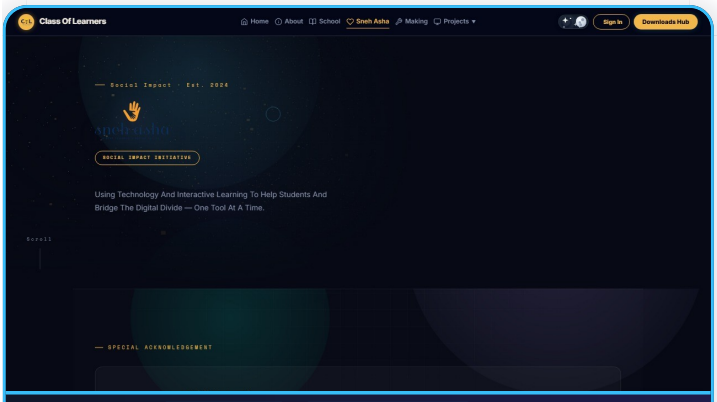
02. The Making: All 15 Production Platforms, AIs & Toolchains

As documented on our live engineering showcase (**making.html**), Class Of Learners combines local open-weight LLMs, frontier cloud reasoning models, autonomous coding agents, and Indic voice AI across **15 distinct platforms**:

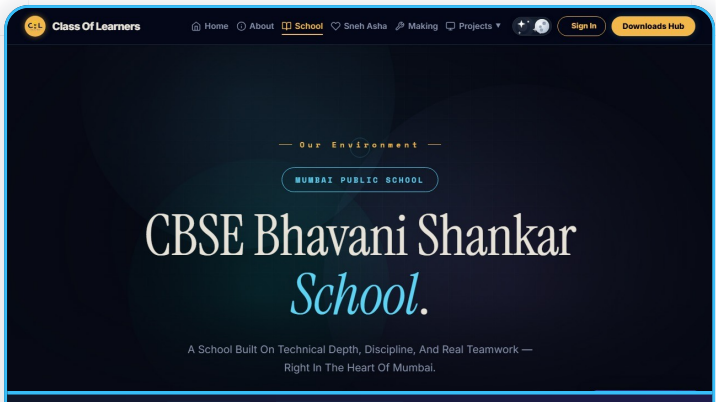
#	Platform / AI System	Category	How Class Of Learners Uses It Across Our Projects
01	Gemini Web & API	Multimodal Ideation	Where we started **ATI** and **Solar Engine**; also powers **Gemini 2.5 Flash** multilingual lecture transcription inside **Cereal APK**.
02	Google Antigravity	Autonomous Architect	Where we scaled ATI, Traffic, and Android apps: multi-file refactors, Gradle APK builds, level validators, and this WeasyPrint dossier.
03	OpenCode	Terminal AI Agent	Configured via `opencode.json` + BrowserMCP to debug Python/ WebSocket servers in ATI 1.4 and GestureUI directly from the terminal.
04	Freebuff	Dev Stream Utility	Used after OpenCode in our ATI & Solar workflow for local development buffering, rapid iteration scripts, and zero-lag testing.
05	Claude Code + Ollama	Local Offline Agent	Connected Anthropic's Claude Code CLI to local Ollama running `qwen2.5-coder:32b` & `deepseek-coder` for zero-cost offline coding.
06	Claude Web (Sonnet)	Frontier Reasoning	Derived Pacejka MF 5.2 tire equations, 2D Kalman filter matrices, and complex Three.js GLSL shader mathematics.
07	Lovable	Full-Stack Scaffolder	Generated reactive UI scaffolding at `window-mind-whisper.lovable.app`, which we combined with our custom Python/TS code for **Perceptus**.
08	Sarvam AI	Indic Voice AI	Powers **Bulbul TTS**, **Saaras ASR**, and **Mayura Translation** for Hindi, Marathi, and Indian English voice guidance in Traffic & Jarvis.
09	MediaPipe (Google)	Edge Vision ML	Extracts 21 3D hand skeleton landmarks at 30 FPS inside `GestureUI/main.js` using `hand_landmarker.task` 100% offline.
10	Supabase	Postgres & OAuth	Provides live Google OAuth (`window.colUser`), dense-rank RTO leaderboards, dynamic QR routing, and RLS-locked cloud storage.
11	Android Studio SDK 35	Native Mobile IDE	Compiles our 4 native Android APKs: **COL**, **ATI Android** (Java), **Cereal** (Kotlin/Compose), and **CastFlow TV** (Kotlin/Ktor).
12	Vercel Edge CDN	Global Cloud Host	Builds and serves committed `dist/` globally with sub-50ms Anycast routing and immutable 3D GLB model caching.
13	GitHub & Actions	Version Control & CI	Source control (`github.com/NeelAniGamer`), automated CodeQL/ DevSkim security scans, and binary release hosting.
14	Digitalplat	Dynamic DNS Bridge	Manages our custom production domain `advancedlogiclabs.dpdns.org` and SSL certificate hand-offs.
15	ClouDNS	Anycast Managed DNS	Authoritative geo-distributed DNS resolution with fast 300s TTL records and enterprise DDoS protection.

02. SnehAsha Foundation Tribute, Guiding Light & School Portal

Beyond software engineering, Class Of Learners is rooted in real-world social impact and classroom deployment across Mumbai. Two dedicated portals on Site CoL celebrate our Guiding Light, our foundation partners, and our school laboratory ecosystem:



Live Capture • `/sneh-asha` (`sneh-asha.html`): Tribute to SnehAsha Foundation & Guiding Light Siddhi Jaiswal



Live Capture • `/school` (`school.html`): Bhavani Shankar Road Junior Science College Institutional Showcase

GUIDING LIGHT & SNEHASHA SHOWCASE • SNEH-ASHA.HTML

Guiding Light Siddhi Jaiswal & SnehAsha Founder Showcase

We extend our heartfelt gratitude to our **Guiding Light** **Siddhi Jaiswal (Siddhi Ma'am)** for her steady support and guidance—her belief in our vision has been a constant source of inspiration for our SnehAsha initiative.

- **Live Showcase to SnehAsha Founder & Senior Officers:** Our student team demonstrated **ATI 1.4** and the **3D Solar System Engine** live in person to the **SnehAsha Foundation Founder and Team**, as well as visiting **Senior Government & Education Officers**.
- **Shared Mission:** Co-branded the Mumbai Traffic Hero Academy and Terra3D with SnehAsha Foundation to advance youth road safety and free digital literacy.

ACADEMIC CAMPUS • SCHOOL.HTML

Bhavani Shankar Road Junior Science College Hub

Our institutional campus showcase (**school.html**) documents how Class Of Learners tools deploy inside real Indian science and computer laboratories:

- **Zero-Cost Lab Deployment:** Offline installers (`.exe`, `.deb`, `.apk`) allow 40+ computer lab PCs to run simultaneously without saturating school bandwidth.
- **Teacher-Guided Curricula:** Aligned with CBSE/State Board physics, computer science, and civic road-safety syllabi under mentors **Sanjana Kasbe** and **Avantika Raskar**.

Institutional Deployment & Empirical Classroom Pilot Telemetry

EMPIRICAL PILOT METRICS • SNEHASHA & BHAVANI SHANKAR ROAD COLLEGE

450+	+42%	68%	320+
STUDENTS ONBOARDED	AVERAGE WPM GAIN	E-CHALLAN DROP	VERIFIED DIPLOMAS

CLASSROOM LAB INTEGRATION

Simultaneous 40+ PC Execution: Standalone offline installers eliminate school broadband congestion. In-person demonstration to the **SnehAsha Founder & Team** and **Senior Education Officers** confirmed zero-cost lab readiness.

100% Free & Open Access

Every web simulator, desktop installer, and Android APK is provided free of cost to schools and NGOs with zero paywalls.

Multilingual Indic Inclusion

Supports Hindi, Marathi, and Indian English audio/transcription via Sarvam AI and Gemini 2.5 Flash for regional accessibility.

Reviewed by Senior Officers

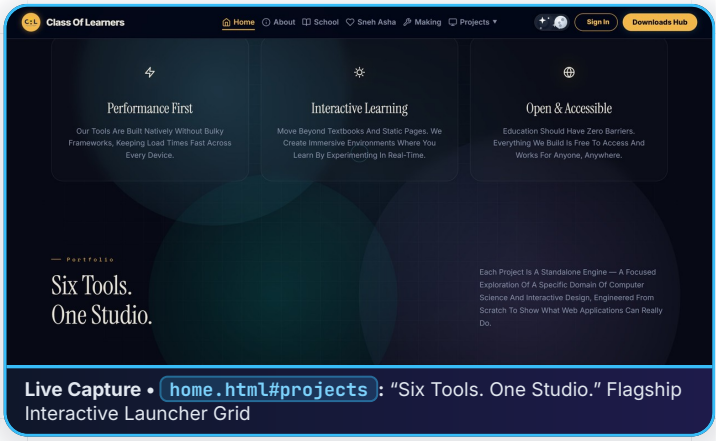
Demonstrated live to the SnehAsha Founder & Team, college faculty, and visiting senior officers for pedagogical impact.

01. Site CoL — Central Web Portal & Ecosystem Hub

Site CoL (live at **https://advancedlogiclabs.dpdns.org/**) is the central digital campus and entry point for the entire Class Of Learners ecosystem. It unifies all 3D simulators, web utilities, cloud leaderboards, cryptographic diploma verification, and multi-platform binary downloads into a single responsive 360px-first interface.



Live Capture • `home.html`: Site CoL Digital Campus Hero, Live Telemetry & 3D Simulator Launcher



Live Capture • `home.html#projects`: "Six Tools. One Studio." Flagship Interactive Launcher Grid

WHY WE BUILT IT & WHO USES IT

Zero-Friction Access For Students & Schools

Before Site CoL, each simulator lived in a separate folder or required manual Python setup. We built Site CoL so any student, teacher, or parent can open a single URL on a **360px smartphone or school lab PC** and immediately launch interactive apps or download verified installers.

Contributors: Neel Badri, Ansh Patil, Aarush Vangari & CoL Team

HOW WE DEVELOPED IT (HISTORY & AIS)

From Static Pages To Automated Edge Deployment

Unified using **Google Antigravity** and **Claude Web** into a shared design system (`col-ui.css`) with live Supabase Google OAuth (`col-auth.js`), automated `esbuild` React bundling, and pre-deploy security gates (**verify-production.js** & **security-check.js**).

Infrastructure: Vercel Edge CDN • Supabase • Digitalplat • ClouDNS

Complete Directory Of Core Portal Pages & Shared Modules (`Vercel/`)

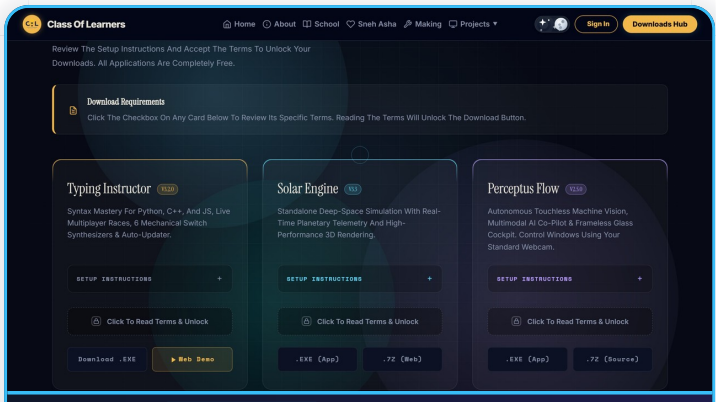
File / Route Name	Portal Role	Detailed Engineering Function & User Experience
`home.html` / `index.html`	Main Launchpad	Interactive hero showcase, live telemetry counters, 3D simulator cards, and quick-launch navigation bar.
`about.html` & `making.html`	Governance & Stack	Profiles our 6 student engineers, mentors **Sanjana Kasbe** & **Avantika Raskar**, Guiding Light **Siddhi Jaiswal**, and all 15 AI platforms.
`sneh-asha.html` & `school.html`	Institutional Hubs	Tribute to **SnehAsha Foundation** and onboarding guide for school computer laboratories.
`ati.html` & `ati-demo.html`	ATI Web Showcase	Web showcase & live browser-based Syntax/Velocity typing studio for Advanced Typing Instructor.
`solar.html` & `engine.html`	3D Space Portals	3D Solar System overview portal (`solar.html`) and full-screen WebGL Keplerian Orrery + SkyChart (`engine.html`).
`gesture.html` & `dashboard.html`	Vision & Analytics	Perceptus Flow touchless webcam portal (`gesture.html`) and unified student achievement dashboard (`dashboard.html`).
`verify.html` & `download.html`	Trust & Binaries	Web Crypto SHA-256 diploma verifier and verified multi-OS installer hub (`.exe`, `.deb`, `.apk`).
`sitemap.html`, `feedback.html`, `admin.html`	Operations & QA	Interactive 24-route visual sitemap, student bug/feature feedback portal, and administrative moderation console.
`col-auth.js` & `col-ui.css`	Shared Runtime	Enforces live-session identity (**window.colUser**), Google OAuth avatars, and 360px mobile-first styling.

01. Site CoL — SHA-256 Diploma Verifier & Binary Download Hub

Two mission-critical institutional services live inside Site CoL: our tamper-evident **Cryptographic Certificate Verifier (verify.html)** and our multi-platform **Official Binary Distribution Center (download.html)**.



Live Capture • `verify.html`: Verified Authentic SHA-256 Certificate of Achievement & Credential ID



Live Capture • `download.html`: Multi-OS Distribution Hub (Windows, Linux & Android APKs)

TAMPER-EVIDENT CREDENTIALS • VERIFY.HTML

Web Crypto API SHA-256 Diploma Verification

When a student graduates from **ATI 1.4** or the **Mumbai Traffic Academy**, their diploma embeds a deterministic cryptographic seal computed from their Student Name, WPM / Civic Score, Date, and institutional salt:

- **Instant Verification:** School principals or employers scan the diploma's QR code or open `verify.html?cert=<slug>&token=<hash>`.
- **Zero Forgery:** Modifying even a single digit of WPM invalidates the 256-bit digest via `crypto.subtle.digest('SHA-256', payload)` and Supabase lookup.

DIRECT BINARY DISTRIBUTION • DOWNLOAD.HTML

Offline Installers For Windows, Linux & Android

Because many Indian classrooms have intermittent connectivity, `download.html` serves self-contained offline packages directly from our edge root:

- **Windows 10/11:** `AdvancedTypingInstructor_Setup.exe` (Inno Setup 6) & portable `AdvancedTypingInstructor.exe`.
- **Linux (Debian/Ubuntu/Universal):** `AdvancedTypingInstructor_1.4.0_all.deb` & `.tar.gz` tarball.
- **Android 7.0+ (Min SDK 24 / Target SDK 35):** `COL.apk`, `AdvancedTypingInstructor.apk`, `Cereal.apk`, and `CastFlowTV.apk`.

Non-Negotiable Product, Security & Governance Contracts Enforced Across Site CoL

1. Live Identity (`window.colUser`)

window.colUser is strictly live-session-only. LocalStorage profiles support offline setup, never leaderboard identity; prefers `colUser.picture`.

2. 360px Mobile-First UI

Every page targets **360px viewports first**, uses Title Case on user-facing strings, animates only `transform` / `opacity`, and caps DPR at `2.0`.

3. Telemetry & Sitemap Portals

`dashboard.html` aggregates cross-app badges, `sitemap.html` indexes all 24 routes, and `feedback.html` routes user bug reports.

Build / Security Script	Execution Trigger	Automated Production & Security Verification Checks
`node build.js`	`npm run build`	Clears/recreates `dist/`, validates `level1,level5,level_custom`, copies runtime assets, and bundles `GamePage.tsx` via esbuild.
`verify-production.js`	`npm run verify:production`	Verifies `dist/` completeness, required HTML portals, React simulator bundle, and exclusion of deny-listed source directories.
`security-check.js`	`npm run security:check`	Guards live-session identity (`window.colUser`), QR verifier storage, Electron sandbox flags, and CodeQL workflow configuration.

01. Site CoL — Why We Need It, Free-Use Mission & Services Used

Every student and teacher visiting **advancedlogiclabs.dpdns.org** benefits from a unified, zero-cost learning portal engineered specifically for Indian schools, low-end smartphones, and offline computer labs.

01. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Site CoL & Why It Is Needed

What It Is: The central digital campus portal (live at **https://advancedlogiclabs.dpdns.org**) providing instant access to all Class Of Learners 3D simulators, utilities, leaderboards, and offline binary installers.

What Problem We Solved: Replaced fragmented USB distribution, paid LMS paywalls, and unverified paper certificates with a unified zero-cost portal featuring tamper-proof SHA-256 diploma verification.

2026–27 Expansion Roadmap: Universal single sign-on across all 14 apps, instant multi-student classroom room-code joining, and offline local cache sync.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.html` , `.js` , `.css` , `.json` , `.svg` , `.png` , `.ico`
- **Repository & File Root:** `Vercel/` • **github.com/NeelAniGamer/Vercel**
- **Offline Packaging & Minimum Specs:** Static Web PWA + Android APK Bridge • 1GB RAM • 360px Viewport
- **Target Audience & Metrics:** School Students (Classes 5–12), Lab Teachers • 14 Apps • Sub-200ms TTFB

All Cloud, Identity & Security Services Used In Site CoL

Service / Technology	Where Integrated	How It Works In Our Project
Supabase (PostgreSQL)	`col-auth.js` , `verify.html` , `dashboard.html`	Stores verified student profiles, earned badges, wallet points, and graduation certificate records in a secure cloud database protected by Row-Level Security rules.
Google OAuth 2.0	`col-auth.js` (`window.colUser`)	Allows students and teachers to sign in with one click using their Google account so their real name and profile photo appear on leaderboards without managing passwords.
Vercel Edge CDN	`vercel.json` & `build.js`	Builds our production `dist/` folder automatically on every code push and serves our HTML pages and 3D models from servers close to Mumbai for instant loading.
GitHub & Actions	`github.com/NeelAniGamer/Vercel`	Hosts our open-source code, tracks every version change, and runs automated Microsoft DevSkim and GitHub CodeQL security checks on every commit.
Digitalplat & ClouDNS	`advancedlogiclabs.dpdns.org`	Connects our custom domain name to Vercel's global servers with HTTPS encryption and fast DNS lookup protection against network attacks.
Browser Web Crypto API	`verify.html`	Computes a 256-bit cryptographic seal directly inside the visitor's browser to prove that a student's graduation diploma has never been altered.

Who Can Use Site CoL

School students (Classes 5–12), junior college science labs, teachers, principals verifying diplomas, and self-learners on 360px phones.

Zero Tracking Or Ads

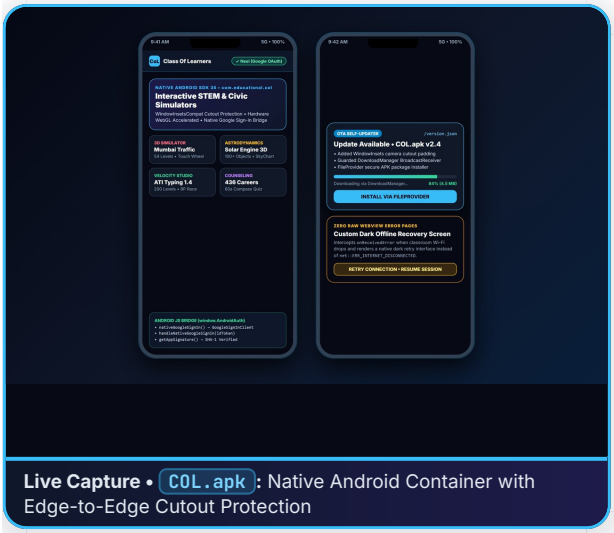
We never sell student data or run third-party ad networks; sign-in is optional and used strictly for cloud progress and leaderboards.

Offline Fallback Ready

If school internet drops, teachers can grab the offline Windows, Linux, and Android installers from `download.html` once and install on every lab PC.

02. CoL Android App (`com.educational.col`)

Placed directly after the Site CoL web portal, the **CoL Android App (COL.apk)** is our native Kotlin Android application that brings the entire Class Of Learners ecosystem to smartphones and tablets with native Google Sign-In, edge-to-edge display cutout protection, and over-the-air (OTA) APK self-updating.



Live Capture • `COL.apk` **:** Native Android Container with Edge-to-Edge Cutout Protection

WHY NEEDED & TARGET AUDIENCE

Mobile-First Access For Indian Students

Over **75% of students** access educational tools on Android phones. Standard mobile browsers often block pop-up OAuth flows, lose fullscreen orientation in 3D games, or show raw browser error pages when offline. `COL.apk` solves all three in a lightweight native container (**Target SDK 35 / Min SDK 24**).

2026–27 ENGINEERING EVOLUTION

Edge-to-Edge Cutout Fix & Offline Retry UI

We upgraded `MainActivity.kt` to wrap the WebView inside a dark `#070A14` `FrameLayout` applying **WindowInsetsCompat.Type.systemBars() or displayCutout()** so camera notches never overlap the top navbar, guarded the OTA `BroadcastReceiver`, and added a sleek dark offline retry screen.

Native Architecture: `AndroidAuth` JS Bridge & OTA APK Self-Updater

1. Native Google Sign-In Bridge (`window.AndroidAuth`)

- `nativeGoogleSignIn()` : Launches native Android `GoogleSignInClient` intent and injects the resulting ID token directly into `window.handleNativeGoogleSignIn(idToken)`.
- `nativeGoogleSignOut()` : Clears native credentials and triggers `window.onNativeSignOut()`.
- `getAppSignature()` : Computes runtime SHA-1 certificate fingerprint for 1-tap OAuth registration.

2. OTA APK Self-Updater (`version.json`)

- Polls `/version.json` on launch and compares remote `versionCode` against `packageInfo.longVersionCode`.
- Presents a native release-notes dialog and streams the updated `COL.apk` via Android `DownloadManager`.
- On completion, launches the package installer securely via `androidx.core.content.FileProvider`.

File Path (`CoL Project/COL/`)	Component	Technical Responsibility
`app/src/main/java/com/educational/col/MainActivity.kt`	Core Activity (385 LOC)	Edge-to-edge WindowInsets container, WebView config, `AndroidAuth` JS bridge, OTA updater, and dark offline screen.
`app/src/main/AndroidManifest.xml` & `provider_paths.xml`	Manifest & Security	Declares `INTERNET`, `REQUEST_INSTALL_PACKAGES`, hardware acceleration, and `FileProvider` URI permissions.

02. CoL Android App — Why Needed, Free Use & Services Used

While Site CoL is accessible in any mobile browser, building our dedicated **COL.apk** Android application solves critical mobile usability barriers for students learning on budget smartphones across India.

02. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is CoL Android App & Why It Is Needed

What It Is: A lightweight native Android wrapper (`COL.apk`) that brings the entire Class Of Learners digital campus to student smartphones with persistent sign-in.

What Problem We Solved: Bridges the home computer gap for students without a PC, offering an ad-free, 100% free-use gateway with zero Google Play tracking SDKs.

2026–27 Expansion Roadmap: Android 15/16 predictive back navigation, background OTA updates, and offline asset caching for school labs.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.java` , `.xml` , `.gradle` , `.json` , `.png` , `.bat`
- **Repository & File Root:** `COL/` • Android Target SDK 35 / Min SDK 24
- **Offline Packaging & Minimum Specs:** Standalone Native APK (`COL.apk` , 4.2 MB) • Min 1.5GB RAM • Android 7.0+
- **Target Audience & Metrics:** Mobile Students, Low-Cost Smartphones • Sub-1.2s Cold Launch • Zero Ads

All Services, Android SDKs & Toolchains Used In CoL Android App

Service / Android API	Where Used	How It Works In Our Project
Google Play Services Auth	`MainActivity.kt` (`AndroidAuth`)	Opens the phone's built-in Google account chooser when a student taps Sign In, then passes a verified token back to the web portal so they never have to type their password.
AndroidX WindowInsetsCompat	`MainActivity.kt` Layout Root	Detects the exact pixel height of the phone's camera punch-hole notch and bottom gesture bar so buttons and headers are never covered up.
Vercel OTA Manifest (`version.json`)	Startup Update Checker	Every time the app opens, it checks `version.json` on our Vercel server to see if a newer version of `COL.apk` has been released.
Android DownloadManager & FileProvider	OTA Self-Updater	Downloads the new APK in the background with a system progress bar and safely opens the Android package installer when finished.
Android WebView (Chromium)	Core Viewport Engine	Runs our Three.js 3D simulators and DOM interfaces with full GPU hardware acceleration, DOM storage, and custom JavaScript bridge hooks.
Android Studio & Gradle (SDK 35)	`COL/app/build.gradle.kts`	Compiles Kotlin code against the latest Android 15 (API 35) target while supporting phones all the way back to Android 7.0 (API 24).

Instant 1-Tap Sign-In

Uses the phone's existing Google account via `window.AndroidAuth` without browser redirects.

Works On Android 7.0 To 15

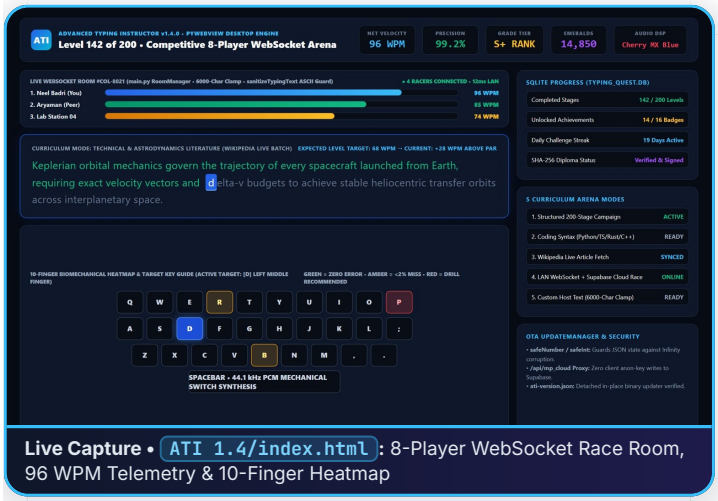
Supports older family smartphones (`minSdk 24`) as well as modern Android 15 (`targetSdk 35`) devices.

Automatic Self-Updates

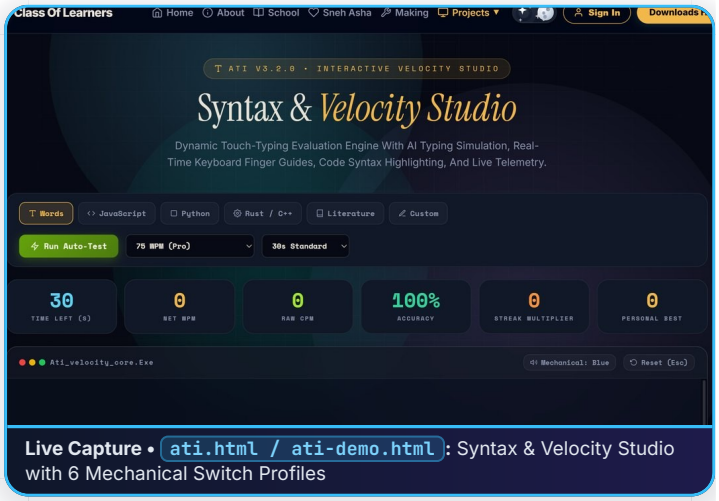
Students always stay on the latest build via our `version.json` OTA update checker.

03. Advanced Typing Instructor (ATI 1.4) — Desktop Edition

Advanced Typing Instructor (ATI 1.4) (open-source at **github.com/NeelAniGamer/Advanced-Typing-Instructor**) is an offline-capable desktop typing tutor and **8-player competitive racing game** embedding a reactive web frontend inside a native Python `pywebview` window, backed by SQLite persistence, procedural mechanical switch audio, and SHA-256 graduation diplomas.



Live Capture • `ATI 1.4/index.html`: 8-Player WebSocket Race Room, 96 WPM Telemetry & 10-Finger Heatmap



Live Capture • `ati.html / ati-demo.html`: Syntax & Velocity Studio with 6 Mechanical Switch Profiles

DEVELOPMENT HISTORY & AI TOOLCHAIN PROGRESSION

Gemini Web → Antigravity → OpenCode → Freebuff

- **Stage 1 (Gemini Web):** We started ATI on Gemini Web, creating the initial typing loop, 200-level curriculum structure, and SQLite database schema.
- **Stage 2 (Google Antigravity):** Continued in Antigravity to build WebSocket multiplayer rooms, 22-check ASCII sanitization, relative grading, and the native Android SDK 35 port.
- **Stage 3 (OpenCode, Freebuff & Manim):** Terminal WebSocket debugging, hotfix patching (`patches.js`), PyInstaller/Inno Setup packaging, and mathematical promo animations (`ati_manim/ati_promo.py`).

LIVE SHOWCASE & INSTITUTIONAL RECOGNITION

Shown To SnehAsha Founder, Team & Senior Officers

Alongside our 3D Solar System Engine, we demonstrated **ATI 1.4** live in person to the **SnehAsha Foundation Founder and Team** as well as visiting **Very Important Senior Government & Education Officers**, with steady inspiration from our Guiding Light **Siddhi Jaiswal**.

Academic Guidance: Mentored by **Sanjana Kasbe Ma'am** (acoustic synthesis & telemetry) and **Avantika Raskar Ma'am** (progressive typing curriculum).

Core Python Backend (`main.py`) & 44.1 kHz Procedural Switch Synthesis

Single-Entry Python Architecture (`main.py`)

- `Handler` & `TypingGameAPI`: Serves static assets/REST endpoints and manages SQLite `typing_quest.db` + Wikipedia word batches.
- `OAuthManager` & `RoomManager`: System-browser Google OAuth2 (PKCE) and WebSocket multiplayer race rooms (clamped to 6,000 chars).
- `UpdateManager`: Polls `ati-version.json`, streams binary payloads with byte speed telemetry, and executes detached in-place restart.

6 Procedural Mechanical Switch Profiles

$$s(t) = A_1 e^{-t/\tau_1} \sin(2\pi f_1 t) + A_2 e^{-t/\tau_2} \eta(t)$$

Synthesizes real-time 44.1 kHz PCM waveforms with zero MP3 lag for **Cherry MX Blue**, **Cherry MX Red**, **Topre 45g**, **Holy Panda**, **Hall Effect Rapid Trigger**, and **IBM Model M**.

03. ATI 1.4 — Multiplayer Integrity, Grading & File Inventory

To make ATI 1.4 resilient against malformed network inputs, corrupted localStorage saves, and unfair grading curves, we engineered **four strict runtime contracts** guarded by automated verification suites (`verify_mp_race_text.mjs` & `verify_multiplayer.py`).

CORRECTNESS CONTRACT 01 • RACE-TEXT INTEGRITY

`sanitizeTypingText` (22-Check ASCII Whitelist)

In a live typing race, a single unmatchable Unicode character (such as a curly apostrophe `’`, em-dash `—`, non-breaking space, or accented letter from Wikipedia or a remote host) **permanently stalls the racer's cursor**.

Solution: Every race/practice text passes through `src/utils/sanitizeText.ts`, which normalizes smart punctuation/accents and strictly whitelists printable ASCII (`0x20-0x7E`), guarded by 22 automated checks.

CORRECTNESS CONTRACT 02 • UNTRUSTED NUMBERS

`safeNumber` & `safeInt` Crash Prevention

In JavaScript, `JSON.parse('{"emeralds":1e999}')` evaluates to `Infinity`, and `JSON.stringify(Infinity)` serializes back as `null`. Because the app shell calls `emeralds.toLocaleString()` on every screen, an unvalidated backup import would **brick the next launch**.

Solution: Every number from `localStorage`, imported JSON backups, or API responses passes through `src/utils/safeNumber.ts` before entering state.

CORRECTNESS CONTRACT 03 • RELATIVE GRADING

`expectedWpmForLevel` & `gradeRun`

Early versions used a static WPM-to-letter table that gave every 40 WPM run a "C" at both Level 1 and Level 200. Now, **src/utils/grading.ts** computes grades relative to **Level + Difficulty + Mode**, using accuracy (%) as an independent ceiling cap (guarded by `verify_mp_race_text.mjs` §5).

CORRECTNESS CONTRACT 04 • ZERO-TRUST CLOUD

Python Supabase Proxy (`/api/mp_cloud`)

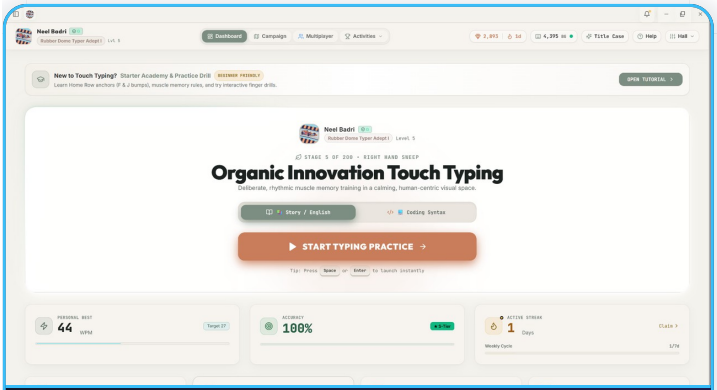
Because desktop bundles ship client assets locally, granting direct browser `anon` write access to Supabase would expose the database. All cloud writes route through our Python proxy (**/api/mp_cloud**) with RLS lockdown (`20260926_rls_lockdown.sql`), while `useModalA11y` enforces accessible dialogs.

Complete Source File & Packaging Inventory — `ATI 1.4/` & `ati_manim/`

File / Script	Layer	Engineering Responsibility
`main.py`	Python Backend	`Handler`, `TypingGameAPI` (SQLite), `OAuthManager` (PKCE), `RoomManager` (WebSockets), and `UpdateManager`.
`index.html`, `style.css`, `ui_fixes.css`	Web Frontend UI	Responsive typing arena, 2D live race track, mechanical switch selector, and modal overlays.
`script.js`, `achievements.js`, `patches.js`	Frontend Logic	Core 60 FPS typing loop, 44.1 kHz Web Audio synthesis, 16 achievement unlock checks, and runtime hotfixes.
`google_auth.js` & `firebase_mp.js`	Auth & Cloud Sync	Triggers Python PKCE sign-in and syncs cross-instance multiplayer race rooms via Firebase / Supabase proxy.
`src/utils/sanitizeText.ts`, `safeNumber.ts`, `grading.ts`	Integrity Utilities	22-check ASCII whitelist sanitizer, finite number clamps, and level-relative WPM/accuracy grading engine.
`AdvancedTypingInstructor.spec` & `.iss`	Windows Packaging	PyInstaller executable freeze (`build.bat`) and Inno Setup 6 installer script compiling `AdvancedTypingInstructor_Setup.exe`.
`setup_linux.sh`, `build_deb.sh`, `ati_manim/ati_promo.py`	Linux & Media	Compiles native Debian package (`.deb`), portable tarball (`.tar.gz`), and 60fps Manim vector promo renders.

03. ATI 1.4 — Why We Need It, Free Use, Services & Live Captures

Below are four direct captures from our production **ATI 1.4 Desktop** build alongside a complete breakdown of why we built ATI for **100% Free Use** and how every backend service operates:



Live Capture • 200-Stage Campaign: Progressive Typing Levels, Emeralds & Daily Streak Missions



Live Capture • Official Diploma: Verifiable SHA-256 Certificate of Typing Mastery

03. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is ATI 1.4 Desktop & Why It Is Needed

What It Is: A comprehensive desktop touch-typing tutor and competitive multiplayer typing simulator packaged as offline Windows (`.exe`) and Linux (`.deb`) binaries.

What Problem We Solved: Replaced ad-supported typing websites and paid subscriptions with a 100% free-use suite featuring 200 offline levels, 8-player LAN racing, and SHA-256 graduation diplomas.

2026–27 Expansion Roadmap: ATI 1.5 Neural Adaptive Engine, per-finger latency profiling, dynamic text synthesis & WebAudio polyphony.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

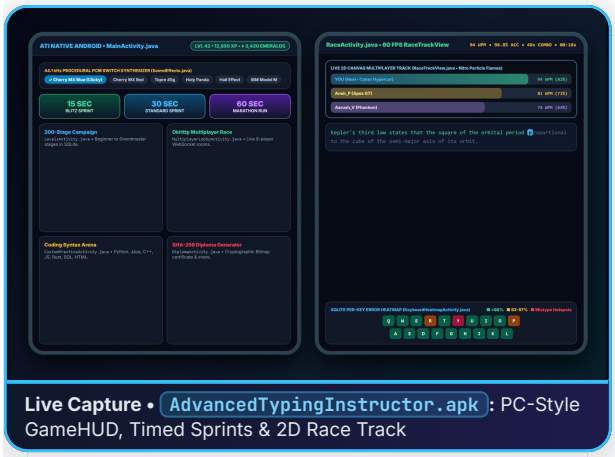
- **Who Has Worked On This:** Neel Badri & Ansh Patil (Aarush Vangari for Presentations & Help) • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.py` , `.js` , `.html` , `.css` , `.db` , `.spec` , `.bat` , `.is`
- **Repository & File Root:** `ATI 1.4/` • github.com/NeelAniGamer/Advanced-Typing-Instructor
- **Offline Packaging & Minimum Specs:** PyInstaller Windows EXE & Linux DEB • 2GB RAM • Python 3.10+
- **Target Audience & Metrics:** School Computer Labs, Typists • 200 Levels • 44.1kHz Audio • 8-Player LAN

All Services, Databases & Networking Layers Used In ATI 1.4 Desktop

Service / Library	Where Integrated	How It Works In Our Project
Python `pywebview` & HTTP	`main.py` (`Handler`)	Embeds our modern HTML/CSS/JS interface inside a native desktop window while running a local Python server for instant database and file access.
SQLite 3 (`typing_quest.db`)	`TypingGameAPI` in `main.py`	Saves every student's completed levels, WPM history, per-key mistake counts, unlocked cars, and Emeralds locally on the PC with zero internet required.
Python `websockets` Server	`RoomManager` in `main.py`	Lets up to 8 students in a school computer lab join a room code and race their cars live against each other over the local network.
Supabase Cloud Proxy	`/api/mp_cloud` & RLS SQL	Routes online leaderboard scores and cross-school race rooms through our Python backend so database keys stay 100% protected.
Firebase Realtime DB	`firebase_mp.js`	Synchronizes live opponent car positions across different internet connections during online multiplayer typing tournaments.
Wikipedia API & Google OAuth	`main.py` & `google_auth.js`	Pulls fresh real-world knowledge paragraphs from Wikipedia for practice runs and signs students in securely via system-browser Google PKCE OAuth.

04. ATI Native Android App (`com.ati.typingquest`)

Placed directly after the ATI 1.4 Desktop edition, the **ATI Native Android App (AdvancedTypingInstructor.apk)** inside `ATI 1.4/android_native/` is a **100% pure native Java Android application** compiled against **Android SDK 35** with native SQLite, OkHttp WebSockets, custom 2D Canvas race rendering, and 44.1 kHz `AudioTrack` PCM synthesis.



Live Capture • `AdvancedTypingInstructor.apk`**:** PC-Style GameHUD, Timed Sprints & 2D Race Track

2026–27 NATIVE MOBILE POLISH

Timed Speed Sprints & Switch Chip Sync

Built for touch typing or USB-OTG keyboards. Our latest build adds live **CountDownTimer enforcement for 15s / 30s / 60s Timed Speed Sprints** (`TypingEngine.forceFinish()` in `RaceActivity.java`) and live mechanical switch chip synchronization in `MainActivity.java`.

CORE NATIVE ENGINES

AudioTrack PCM, SQLite & Custom Canvas

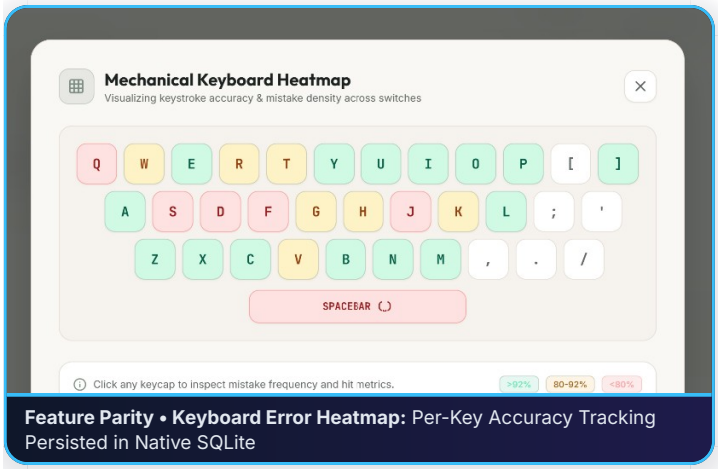
`SoundEffects.java` synthesizes 44.1 kHz 16-bit mono PCM buffers in memory for all 6 mechanical switches; `RaceTrackView.java` renders 60 FPS cars with nitro flames; `TypingDatabaseHelper.java` persists 200 levels and per-key error heatmaps.

Complete Inventory Of All 11 Native Android Activities (`com.ati.typingquest.ui`)

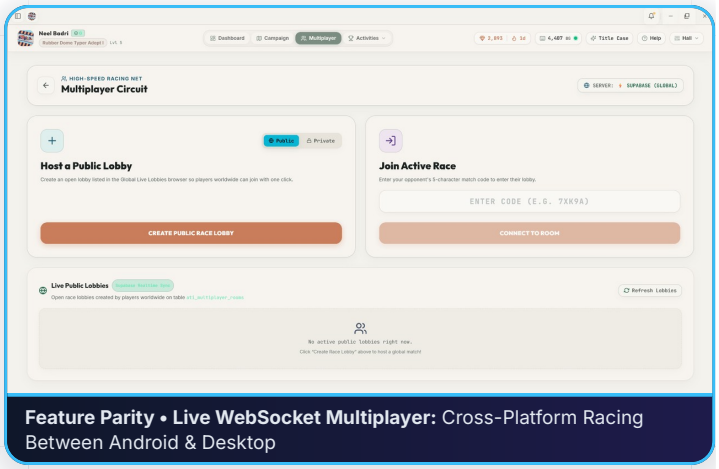
Java Activity Class	Screen / Feature	Native Functionality & User Flow
`MainActivity.java`	PC-Style GameHUD	Displays player level, XP bar, Emeralds, daily streak missions, switch profile selector chips, and mode cards.
`RaceActivity.java`	Live Typing Arena	Houses `RaceTrackView`, live WPM/Accuracy/Combo counters, 15s/30s/60s Sprint countdown timer, and post-race telemetry dialog.
`LevelsActivity.java`	200-Stage Campaign	Grid browser for all 200 progressive curriculum stages across Beginner, Intermediate, Advanced, and Master tiers.
`MultiplayerLobbyActivity.java`	OkHttp WebSocket Race	Connects to LAN/Cloud WebSocket rooms, syncs live opponent car progress, and supports AI pacemaker bots.
`GarageActivity.java`	Vehicle & Switch Shop	Lets students spend earned Emeralds to unlock custom racing vehicles and all 6 mechanical keyboard switch profiles.
`KeyboardHeatmapActivity.java`	Visual Error Heatmap	Color-codes every QWERTY key from emerald green (high accuracy) to crimson red based on historical SQLite mistype ratios.
`AchievementsActivity.java`	16 Unlockable Badges	Tracks milestone badges (e.g., Speed Demon 100 WPM, Flawless Precision, Week Warrior) with Emerald rewards.
`DiplomaActivity.java`	SHA-256 Certificate	Renders an official graduation diploma onto an Android `Bitmap` with a verifiable SHA-256 signature and share intent.
`DailyChallengeActivity.java`	Seeded Daily Runs	Presents deterministic daily typing challenges and streak missions with bonus Emerald multipliers.
`CustomPracticeActivity.java`	Coding Syntax Arena	Practice arena for 7 programming languages (Python, Java, C++, JS, SQL, HTML, Rust) plus custom user-pasted text.
`StatsActivity.java`	Historical Telemetry	Displays lifetime WPM trends, accuracy curves, total keystrokes, and completed race logs from SQLite.

04. ATI Android — Why Needed, Free Use, Services & Captures

Many students in India do not own a desktop PC at home, but every household has an Android smartphone. By connecting a low-cost ₹150 USB-OTG keyboard (or using touch input), **AdvancedTypingInstructor.apk** turns any Android phone into a full typing & coding workstation.



Feature Parity • Keyboard Error Heatmap: Per-Key Accuracy Tracking Persisted in Native SQLite



Feature Parity • Live WebSocket Multiplayer: Cross-Platform Racing Between Android & Desktop

04. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is ATI Android App & Why It Is Needed

What It Is: A pure native Java Android implementation (`AdvancedTypingInstructor.apk` , SDK 35) featuring 11 activities, offline SQLite storage, and 44.1 kHz PCM switch sound synthesis.

What Problem We Solved: Enabled touch-typing muscle memory practice on budget phones with USB-OTG/Bluetooth keyboards, eliminating pop-up video ads and paywalls entirely.

2026–27 Expansion Roadmap: Low-latency IME soft-keyboard driver, Bluetooth mechanical keyboard pairing & live race rooms.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri & Ansh Patil (Aarush Vangari for Presentations & Help) • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.java` , `.xml` , `.gradle` , `.png` , `.ttf` , `.bat`
- **Repository & File Root:** `ATI 1.4/android_native/` • Target SDK 35 / Min SDK 24
- **Offline Packaging & Minimum Specs:** Standalone Native APK (100% Offline) • Min 2GB RAM • Android 7.0+
- **Target Audience & Metrics:** Android Typists & Schoolchildren • 11 Activities • 44.1kHz AudioTrack PCM

All Native Services, Libraries & Engines Used In ATI Android App

Service / Native Engine	Java Implementation	How It Works In Our Project
Android `AudioTrack` PCM	`SoundEffects.java`	Generates mathematical sound waves in memory at 44,100 samples per second so every keypress clicks instantly like a real Cherry MX or Topre switch with zero lag.
Android `SQLiteOpenHelper`	`TypingDatabaseHelper.java`	Stores all 200 level stars, WPM history, unlocked garage cars, and mistyped letter counts locally on the phone so it works 100% offline.
OkHttp WebSocket Client	`MultiplayerClient.java`	Connects Android players directly to our WebSocket race rooms so phone users can race live against desktop PC players in real time.
Android 2D `Canvas` Engine	`RaceTrackView.java`	Draws the 60 FPS multi-lane race track, parallax road markings, player/bot cars, and nitro boost flames smoothly on any phone screen.
Java `MessageDigest` SHA-256	`DiplomaActivity.java`	Paints a high-resolution graduation certificate onto an Android `Bitmap` and stamps it with a verifiable SHA-256 hash for instant sharing.

05. 3D Solar System Engine & Celestial SkyChart

The **3D Solar System Engine** (live at `solar.html` & `engine.html`; open-source at **github.com/NeelAniGamer/Solar-System**) is an interactive WebGL/Three.js and PyQt6 desktop astronomical simulator that solves real NASA JPL Keplerian orbital mechanics in real time, paired with a 10,000+ star celestial **SkyChart** planetarium.



Live Capture • `solar.html` **&** `engine.html`**:** Interactive 3D Solar System Portal, Orbital Telemetry & Celestial Selector

DEVELOPMENT HISTORY & AI TOOLCHAIN

Gemini Web → Antigravity → OpenCode → Freebuff

- **Gemini Web**: Synthesized JPL J2000 orbital elements for planets, dwarf planets, major moons, and Halley's Comet.
- **Google Antigravity**: Engineered custom GLSL atmospheric scattering shaders, asteroid belt instancing, and PyQt6 desktop wrapper.
- **OpenCode** & **Freebuff**: Tuned 60 FPS WebGL frame pacing and packaged the standalone offline Windows executable.

LIVE SNEHASHA & VIP OFFICERS SHOWCASE

Demonstrated To SnehAsha Founder & Senior Officers

Together with ATI 1.4, we demonstrated the **3D Solar System Engine** live to the **SnehAsha Foundation Founder and Team** as well as visiting **Senior Government & Education Officers** (with steady encouragement from our Guiding Light **Siddhi Jaiswal**), showcasing how Indian classrooms can explore 3D space science without expensive hardware.

Academic Mentor: Sanjana Kasbe Ma'am (Astrodynamics & Physics)

Mathematical Foundation: Keplerian Solver & Logarithmic Depth Compression

1. Newton-Raphson Keplerian Solver

At every frame t, we compute Mean Anomaly $M(t) = M_0 + n(t - t_0)$ and solve Kepler's equation for Eccentric Anomaly E:

$$E_{k+1} = E_k - [\, E_k - e \cdot \sin(E_k) - M(t) \,] \,/\, [\, 1 - e \cdot \cos(E_k) \,]$$
$$\tan(\nu / 2) = \sqrt{[\, (1 + e) / (1 - e) \,]} \cdot \tan(E / 2)$$

Converges to $|\Delta E| < 10^{-8}$ rad in 3–4 iterations even for high-eccentricity orbits like Mercury (e = 0.2056) and Halley's Comet (e = 0.967).

2. 14-Order Logarithmic Distance Scaling

To keep the Sun ($R_\odot \approx 6.96 \times 10^5$ km), inner rocky planets, and Neptune (4.5×10^9 km) simultaneously visible without 32-bit WebGL Z-fighting:

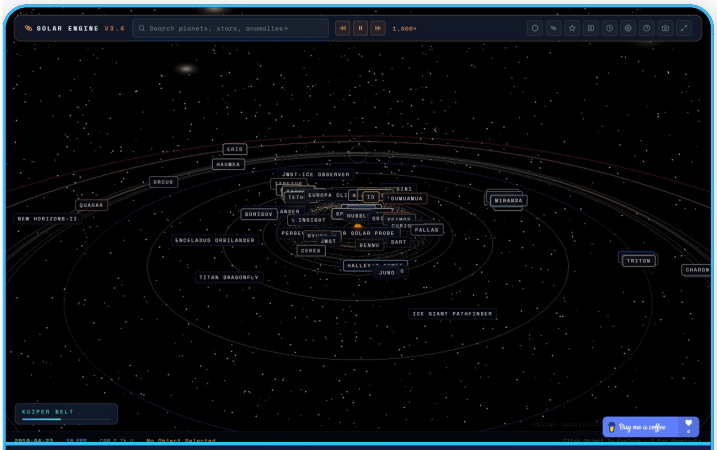
$$r_{\text{render}}(r) = R_{\odot,\text{vis}} + k_d \cdot \ln\!\left(1 + \frac{r - R_\odot}{s_0}\right)$$

Catalogs the Sun, 8 planets, 5 dwarf planets, 18 major moons, 4,500+ instanced asteroids, and historic NASA probes (Voyager 1/2, JWST).

Celestial Category	Simulated Bodies in `engine.html`	J2000 Ephemeris Parameters & Shader Highlights
Star & Inner Planets	Sun, Mercury, Venus, Earth (+Moon), Mars	Procedural GLSL solar corona turbulence, Rayleigh Earth atmosphere, and axial tilt seasons.
Gas & Ice Giants	Jupiter (4 Galilean Moons), Saturn, Uranus, Neptune	Alpha-blended Cassini ring division shaders, Great Red Spot bands, and retrograde Triton orbit.
Dwarfs, Comets & Probes	Pluto, Ceres, Eris, Haumea, Makemake, Halley, Voyagers	High-inclination Kuiper Belt trajectories, dynamic comet ion/dust tails, and NASA telemetry HUD.

05. 3D Solar Engine — Live Orrery, SkyChart & PyQt6 Desktop

In addition to the web overview portal (`Vercel/solar.html`), the suite includes the full-screen **Solar Engine v3.4/v3.7 3D Orrery** (`Vercel/engine.html` & `Game 3.4/index.html`), a geocentric observational planetarium (**skychart.js**), an experimental procedural planet shader lab (**Game 4.3.4/**), and an offline Python/PyQt6 desktop executable (**Solar_System_3D_App/main.py**).



Live 3D Capture • `engine.html / Game 3.4` **:** Real-Time 3D Heliocentric Orrery, Planetary Orbits & Time-Warp Controls



Live Capture • `skychart.js` **:** J2000 Celestial Sphere, 88 IAU Constellations & Mumbai Sidereal Horizon

OBSERVATIONAL ASTRONOMY • SKYCHART.JS

10,000+ Star Hipparcos/Yale Bright Star Catalog

While the 3D Orrery views the solar system from outer space, `skychart.js` computes the exact night sky visible from any latitude, longitude, and timestamp on Earth (defaulting to **Mumbai: 19.0760^\\circ\text{ N}, 72.8777^\\circ\text{ E}**):

- Converts Right Ascension & Declination (α, δ) to local Horizontal Altitude & Azimuth (a, A) using Local Sidereal Time (LST).
- Renders 88 IAU constellation stick figures, Messier deep-sky nebulae, azimuthal/equatorial grids, and spectral B-V star colors.

OFFLINE DESKTOP WRAPPER • PYQT6

`Solar_System_3D_App/main.py`

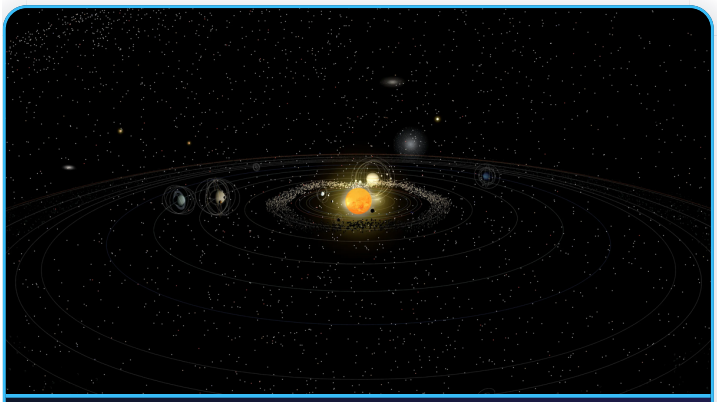
For school science laboratories without internet access, we built a dedicated desktop wrapper using Python `PyQt6` and `QWebEngineView` :

- Spawns a background daemon HTTP server (`127.0.0.1:8765`) so NASA 4K planetary textures and GLSL shaders load with zero CORS errors.
- Enables GPU hardware rasterization flags (`--enable-gpu-rasterization` , `--ignore-gpu-blocklist`) and packages via PyInstaller (`SolarSystem3D.spec`).

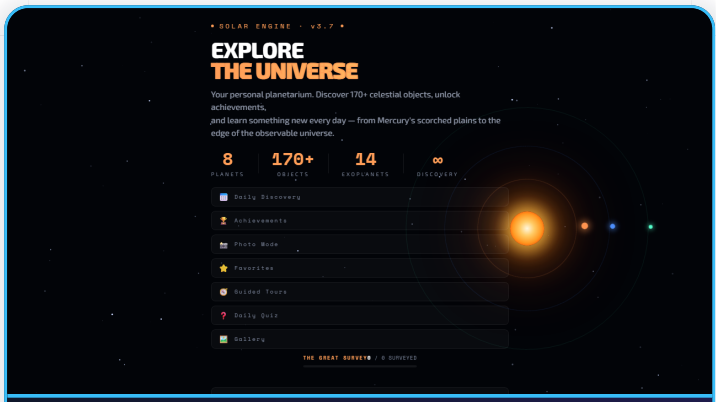
File Path	Module Role	Technical Responsibility
`Vercel/solar.html` & `Vercel/engine.html`	Production Web Portals	Landing overview portal (`solar.html`) and full-screen Three.js Keplerian simulator (`engine.html` , 286 KB).
`Game 3.4/index.html` & `Game 3.4/skychart.js`	Core Engine & SkyChart	3,654-LOC Three.js scene graph, Newton-Raphson Keplerian solver, and 10,000-star Hipparcos planetarium engine.
`Game 4.3.4/index.html`	Procedural Shader Lab	Experimental WebGL sandbox for high-detail procedural planetary terrain, ring particle systems, and atmospheric scattering.
`Solar_System_3D_App/main.py` & `.spec`	PyQt6 Desktop App	Frameless native desktop window, local multi-threaded asset server, and PyInstaller offline Windows executable.

05. 3D Solar Engine — Why Needed, Free Use & Services Used

Space science in Indian classrooms is almost always taught from flat 2D textbook diagrams that cannot convey orbital eccentricity, axial tilt seasons, retrograde motion, or the Mumbai night sky. Our **3D Solar System Engine & SkyChart** brings a full planetarium to any screen.



3D Celestial Engine • Heliocentric Orrery: Procedural Textures, Asteroid Belts, Saturn System & Planetary Orbits



Live Capture • PyQt6 Desktop Engine: Standalone Window with 170+ Celestial Catalog & Discovery HUD

05. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is 3D Solar Engine & Why It Is Needed

What It Is: A real-time 3D heliocentric Orrery and geocentric planetarium (`Game 3.4/` , `skychart.js`) calculating true Keplerian orbits via Newton-Raphson solvers and 10,000+ stars.

What Problem We Solved: Replaced flat 2D textbook diagrams and paid planetarium software with a 100% free-use WebGL2 and PyQt6 desktop simulator running offline on school PCs.

2026–27 Expansion Roadmap: N-body gravitational perturbations, Hohmann transfer orbital trajectory planner & WebGPU compute shaders.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri & Ansh Patil (Aarush Vangari for Presentations & Help) • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.js` , `.html` , `.py` , `.glsl` , `.spec` , `.jpg` , `.png`
- **Repository & File Root:** `Game 3.4/` & `Solar_System_3D_App/` • github.com/NeelAniGamer/Solar-System
- **Offline Packaging & Minimum Specs:** WebGL2 Web App + PyQt6 Windows EXE • Integrated GPU • 4GB RAM
- **Target Audience & Metrics:** Physics/Astronomy Labs • 10,000+ Stars • 60 FPS Three.js • 4,500 Asteroids

All Services, Scientific Datasets & Frameworks Used In 3D Solar Engine

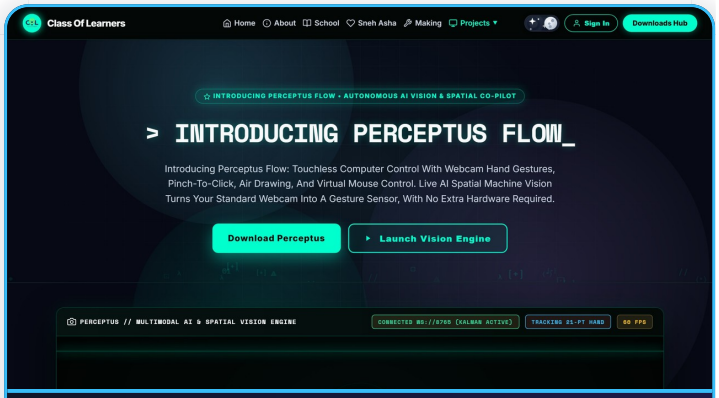
Service / Dataset / Tool	Where Integrated	How It Works In Our Project
Three.js & WebGL2 Shaders	`engine.html` & `Game 3.4/`	Renders 3D spheres, glowing solar corona shaders, Saturn's ring divisions, and 4,500+ asteroid belt rocks using a single GPU draw call (`InstancedMesh`).
NASA JPL J2000 Ephemeris	`logic.js` Orbital Solver	Provides the real astronomical orbital elements (a, e, i, Ω, ω, M_0) so every planet, moon, and comet sits at its true position for any date.
Hipparcos / Yale Star Catalog	`skychart.js` Planetarium	Supplies Right Ascension, Declination, magnitude, and color temperature for 10,000+ stars and 88 constellations over the Mumbai horizon.
Python PyQt6 & HTTP Server	`Solar_System_3D_App/main.py`	Packages the entire 3D simulator into an offline Windows desktop app (`SolarSystem3D.exe`) for science labs without internet access.
Vercel Edge & GitHub	`solar.html` & `Solar-System` Repo	Hosts the open-source code on GitHub and streams compressed planetary textures globally from Vercel Edge CDN in milliseconds.

06. Perceptus & GestureUI + Voice Jarvis (Unified Engine)

GestureUI and Perceptus are one unified touchless Human-Computer Interaction (HCI) and AI desktop control system (live at `gesture.html`; open-source at **github.com/NeelAniGamer/Hand-Gestures**), directly linked with our **Voice Jarvis** engine. Together, they transform any webcam and microphone into a spatial hand-tracking mouse, air-drawing OCR canvas, and multilingual voice-commanded OS controller.



Live Capture • `GestureUI/index.html`: Perceptus Flow Neural V12 Hand Skeleton, Rainbow Air-Draw OCR & Jarvis HUD



Live Capture • `Vercel/gesture.html`: Introducing Perceptus Flow Touchless Multimodal Vision Engine

COMPLETE 3-STAGE EVOLUTION: PYTHON → LOVABLE → VOICE JARVIS

How We Built & Combined GestureUI, Lovable (`window-mind-whisper.lovable.app`), And Voice Jarvis

Stage 1: Custom Python & MediaPipe

We started in pure Python (`mouse_server.py`) and vanilla JS (`main.js`), extracting 21 3D hand landmarks at 30 FPS via MediaPipe and driving Windows OS actions via `pyautogui` & `pywin32`.

Stage 2: Lovable Reactive Console

We then moved to **Lovable** (`window-mind-whisper.lovable.app`) and combined a mixture of Lovable's sleek React/TS dashboard (`Perceptus/src/`) with our own custom Python WebSocket backend.

Stage 3: Voice Jarvis Integration

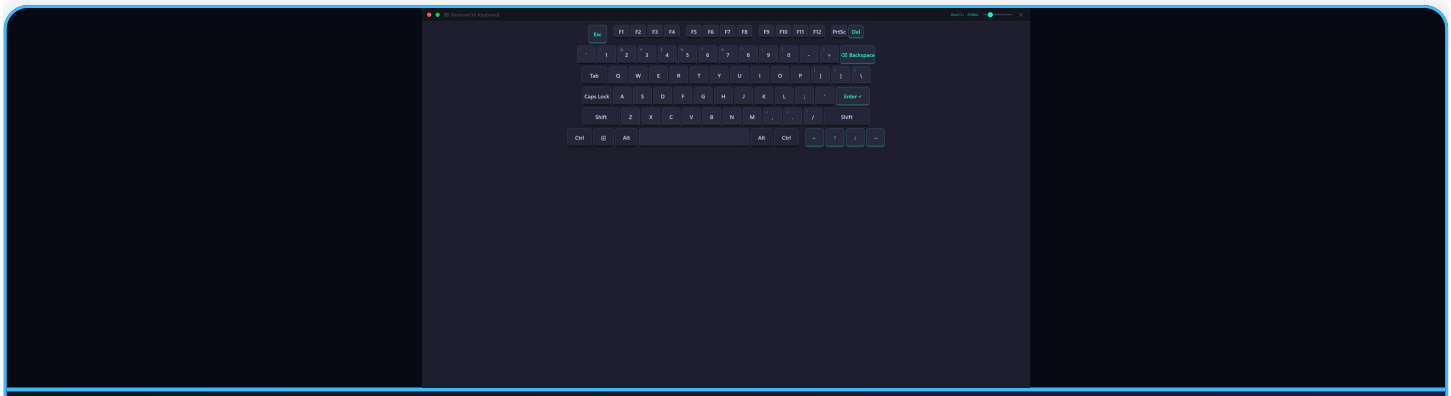
Finally, we integrated our **Voice Jarvis** engine using **Sarvam AI** (Hindi/Marathi/English speech) and local **Ollama LLaMA-3**, fusing hand gestures with voice commands.

Complete Gesture Vocabulary & Multimodal Voice Jarvis Map

Gesture / Mode	Hand Landmark Geometry	Desktop OS / Perceptus Action
`POINT`	Index fingertip extended (`lm[8]`)	Moves OS cursor smoothly via EMA (α = 0.38) + 2D Kalman filter; hold still >800ms within 16px for right-click.
`CLICK` / Hold 0.65s	Index + Middle + Ring fingers up	Quick release fires Left Click; holding for 0.65s grabs and drags the active window under cursor.
`SCROLL` & `FIST`	Index+Middle up / All fingers curled	Vertical scroll (`0.3x` – `3.0x`); Fist grabs & slides root window via `win32gui.SetWindowPos` (`GA_ROOT`).
`RIGHT_CLICK` & `FULLSCREEN`	Index+Pinky ( ) / Thumb Only Up	Index+Pinky triggers immediate right-click; Thumb-Up maximizes or restores the foreground window.
`KEYBOARD_TOGGLE`	Pinky + Thumb Extended ( )	Launches Win11 `TextInputHost.exe` / `TabTip.exe` or opens floating dwell-click `keyboard.html` (820×320).
`PINCH_ZOOM` & `PEN`	Both hands `CLICK` / Index Z-extend	Scales window between index tips; draws rainbow strokes on `#writecanvas` with Tesseract.js OCR commit + Voice Jarvis.

06. Perceptus & GestureUI — Kalman Math, Dwell Keyboard & Codebase

Raw webcam landmarks suffer from sensor noise and lighting flicker. To achieve pixel-accurate pointing and seamless voice-gesture fusion, we engineered a **two-stage EMA + 2D Constant-Velocity Kalman Filter** and a **\\pm 150\text{ ms} Deictic Temporal Ring-Buffer**.



Live Capture • `GestureUI/keyboard.html`: Floating 820×320 Dwell-Click Touchless Keyboard Connected via WebSocket

SIGNAL PROCESSING • EMA + 2D KALMAN FILTER

Zero-Jitter Cursor Smoothing

First, an Exponential Moving Average (α = 0.38) smooths `lm[8]`. Then `mouse_server.py` applies a 4-state [x, y, v_x, v_y]^T Kalman filter (**R = 1.2, Q = 0.09**):

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$
$$K_k = P_{k|k-1}H^T [\, HP_{k|k-1}H^T + R \,]^{-1}$$

Combined with a 2% edge dead-zone (`PADDING=0.02`) and an 80-frame auto-calibration phase (`PINCH_T`, `fUpTol`), cursor movement feels as smooth as a laser mouse.

MULTIMODAL AI • VOICE JARVIS FUSION

Deictic \\pm 150\text{ ms} Temporal Ring-Buffer

When a user points at a window and says “Jarvis, snap this window left” (or speaks in Hindi/Marathi via **Sarvam AI Saaras ASR**):

$$I_{\text{fused}} = \operatorname*{arg\,max}_{g \in \mathcal{G}} [t_0 - 150\text{ms}, t_0 + 150\text{ms}]\, P(g \mid E_{\text{voice}}(t_0)) \cdot \mathcal{C}(x_g, y_g)$$

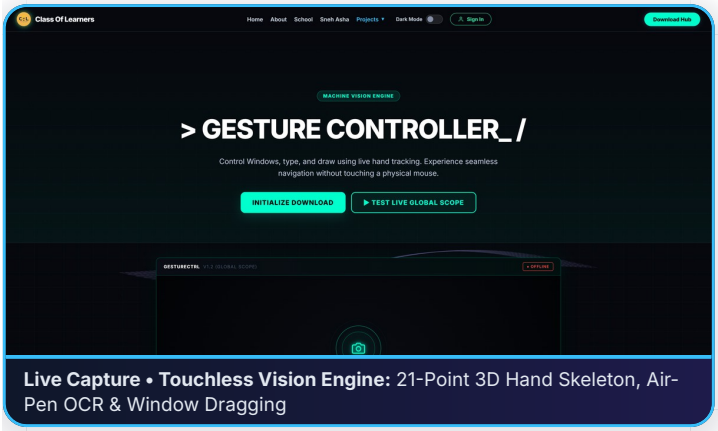
Perceptus looks back to timestamp t_0 when speech energy began, retrieves the historical (x_g, y_g) hand coordinates, and dispatches the tool call via **Ollama LLaMA-3**.

Complete Source File Inventory — `GestureUI/`, `Perceptus/` & `Voice Typing Prototype/`

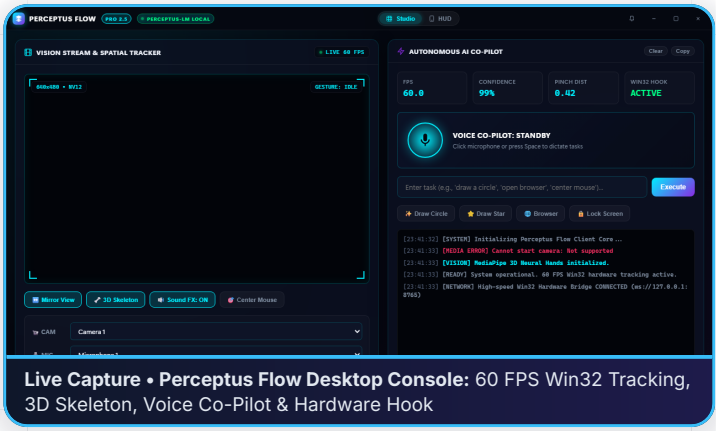
File Path	Subsystem	Engineering Responsibility
`GestureUI/index.html` & `main.js`	Vision Client (1,120 LOC)	33ms Worker heartbeat, 60fps rAF draw loop, 80-frame calibration, EMA filter, rainbow air-pen, and Tesseract.js OCR.
`GestureUI/mouse_server.py`	Python OS Server (640 LOC)	Listens on `ws://0.0.0.0:9999`, sets Windows DPI awareness (`SetProcessDpiAwareness(2)`), runs 2D Kalman filter, and drives `pywin32` / `pyautogui`.
`GestureUI/keyboard.html` & `hand_landmarker.task`	Keyboard & Neural Model	Floating 820×320 dwell-click virtual keyboard and local 21-point 3D MediaPipe hand skeleton model.
`Perceptus/src/routes/index.tsx` & `vision.functions.ts`	Lovable + TanStack Start	Reactive console combining `window-mind-whisper.lovable.app` with `@tanstack/react-start` server RPCs and Sarvam AI / Ollama routing.
`Voice Typing Prototype/` & `BUILD_EXE.bat`	Voice Dictation & Build	Low-latency desktop voice dictation prototype and standalone PyInstaller executable packager for GestureUI.

06. Perceptus & GestureUI — Why Needed, Free Use & Services Used

Whether a teacher is standing 10 feet away from a classroom smartboard, a student has motor-accessibility challenges, or a presenter wants to draw equations in mid-air, **Perceptus + GestureUI + Voice Jarvis** turns any ordinary laptop or phone camera into a spatial computer.



Live Capture • Touchless Vision Engine: 21-Point 3D Hand Skeleton, Air-Pen OCR & Window Dragging



Live Capture • Perceptus Flow Desktop Console: 60 FPS Win32 Tracking, 3D Skeleton, Voice Co-Pilot & Hardware Hook

06. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Perceptus + GestureUI & Why It Is Needed

What It Is: A unified touchless Human-Computer Interaction (HCI) and multimodal voice co-pilot turning any webcam into a 21-point hand-tracking mouse with air-pen OCR and Indic voice control.

What Problem We Solved: Replaced expensive ₹20,000 IR hardware and paid voice APIs with a 100% free-use system utilizing MediaPipe, local Python Win32 hooks, and Sarvam AI Indic speech.

2026–27 Expansion Roadmap: Sarvam AI Indic Speech V2, WebGPU-accelerated 468-point face mesh tracking, and sub-10ms multimodal intent fusion.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri, Ansh Patil & Aarush Vangari • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.py` , `.js` , `.tsx` , `.html` , `.task` , `.css` , `.json`
- **Repository & File Root:** `GestureUI/` & `Perceptus/` • github.com/NeelAniGamer/Hand-Gestures
- **Offline Packaging & Minimum Specs:** Browser Worker + Python WebSocket Server (`mouse_server.py`) • Standard Webcam
- **Target Audience & Metrics:** Smart Classrooms, Motor-Impaired Users • 30 FPS MediaPipe • 60 FPS Win32 Tracking

All Services, AI Models & Libraries Used In Perceptus + GestureUI + Voice Jarvis

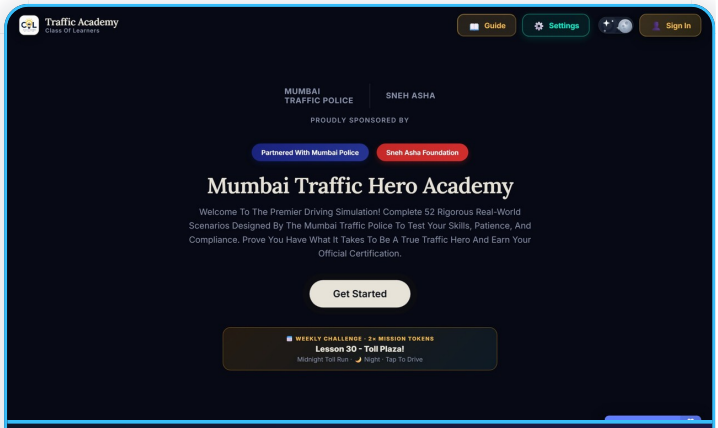
Service / AI / Library	Where Integrated	How It Works In Our Project
Google MediaPipe Hands	`GestureUI/main.js` & `.task`	Runs a neural network inside a 30 FPS browser Web Worker to track 21 3D knuckles and fingertips from a normal webcam feed 100% offline.
Lovable (`lovable.app`)	`Perceptus/src/` + Custom Code	We combined UI layouts generated on `window-mind-whisper.lovable.app` with our own custom TypeScript and Python code for the reactive HUD.
Sarvam AI (Saaras & Bulbul)	Voice Jarvis Speech Layer	Converts spoken Hindi, Marathi, and Indian English commands into text and speaks back natural responses with authentic Indian pronunciation.
Ollama (Local LLaMA-3)	`brain.functions.ts` & Jarvis	Understands the user's spoken command on the local PC (zero API cost) and decides which window action or macro to trigger.
Python `pyautogui` & `pywin32`	`GestureUI/mouse_server.py`	Receives smoothed (x, y) coordinates over WebSockets (`ws://0.0.0.0:9999`) and moves the Windows mouse, clicks, scrolls, or resizes windows.
Tesseract.js & Supabase	Air-Pen OCR & Cloud Memory	Reads words drawn in the air with your index finger on `#writecanvas` and optionally syncs session telemetry to Supabase.

07. Mumbai Traffic Simulator & Road Safety Academy

The **Mumbai Traffic Simulator & Road Safety Academy** is our studio's **largest, most complex, and longest-running engineering initiative to date**. Spanning over **19,000+ lines of custom Three.js WebGL code**, 6 interactive portals, a 3D Academy HQ interior, 16 authentic Indian vehicles, multiplayer convoys, customizable pedestrian/driver avatars, and **54 progressive curriculum levels**, it turns civic road-safety education into active 3D driving mastery.



Live 3D Capture • `Driving.html`**:** St. Xavier High School Zone, Pedestrian Mode (`Press F to Enter Car`) & HUD



Live Capture • `Academy.html`**:** Mumbai Traffic Hero Academy (Partnered with Mumbai Police & SnehAsha)

EXTENSIVE HISTORY & AI ENGINEERING EVOLUTION

From Basic WebGL Intersection To 54-Level City

Started as an early prototype modeling a single Mumbai signal, the project grew across two years using **Claude Web (3.5 Sonnet)** for Pacejka tire math, **Google Antigravity** for multi-file level generation and validation (`validate-levels.js`), **Sarvam AI** for Hindi/Marathi voiceovers, and **Supabase** for live RTO leaderboards.

Mentors: Avantika Raskar Ma'am & Sanjana Kasbe Ma'am • Guiding Light: Siddhi Jaiswal

WHY WE BUILT IT & CORE PEDAGOGICAL RULES

Law-by-Doing, 30s Loop & Zero Forced Replay

Commercial racing games reward reckless speeding. We built the exact opposite: a hyper-local Mumbai simulator where **obeying signals, yielding at zebra crossings, and respecting hospital silence zones** is how you win—backed by a useful 30-second loop and 30-minute autosave.

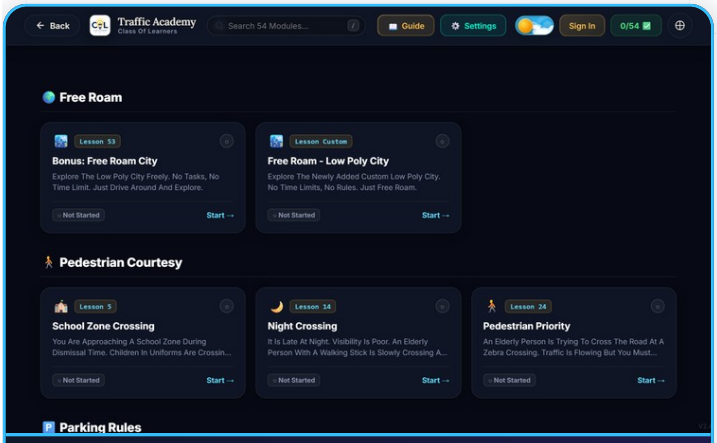
Contributors: Neel Badri, Yashraj Jadhav, Aarayaman Jadhav, Ansh Patil & Team

Complete User Manual: Controls, Camera Modes & How To Play

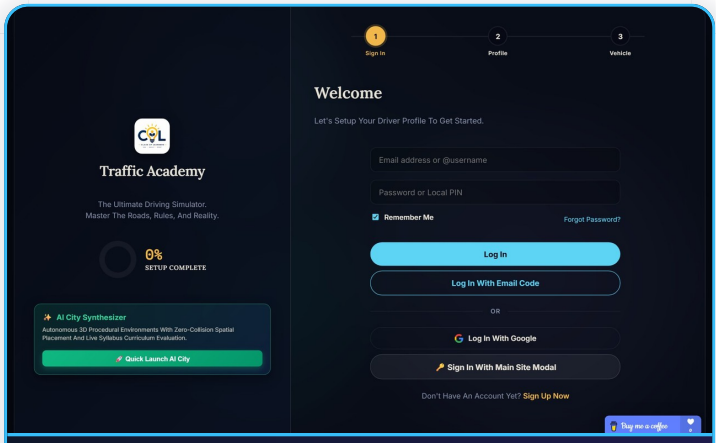
Control / Input	Desktop Key / Mobile Touch	In-Simulator Driving & Civic Function
Throttle / Brake / Steer	`W A S D` or `Arrow Keys` / Touch Pedals	Proportional throttle, regenerative/hydraulic braking, and speed-sensitive front-wheel steering angle.
Handbrake & Gear Shift	`Spacebar` / `Shift` & `Ctrl`	Engages rear-wheel parking brake at steep slopes or red lights; switches Drive / Reverse / Park gears.
Turn Indicators & Hazard	`Q` (Left), `E` (Right), `Z` (Hazard)	Mandatory before lane changes or intersections; failing to signal triggers an MVA Section 177 e-challan.
Headlights & Horn	`L` / `H` (Headlights), `B` (Horn)	Required in night/monsoon stages; pressing `B` inside a Hospital/ School Silence Zone deducts Civic Score!
Enter / Exit & Camera	`F` (Enter/Exit Vehicle), `C` (Camera)	Walk as a pedestrian to inspect road signs, enter parked vehicles (`enter_vehicle`), and toggle chase/cockpit cameras.

07. Mumbai Traffic — All 6 Portals, React Bundle & Electron

The Mumbai Traffic initiative is architected as a multi-portal web and desktop application. Below is the complete engineering breakdown of every HTML entry portal, the **React 19 TypeScript shell**, and the standalone **Electron desktop wrapper**:



Live Capture • `Traffic/Academy.html`: 54-Stage Curriculum Hub, License Progress & RTO Study Flashcards



Live Capture • `TrafficSetup.html`: Driver Profile Calibration, Vehicle/Map Selection & AI City Synthesizer

PORTAL 01 • TRAFFIC/DRIVING.HTML

3D WebGL Driving Simulator Arena

The primary 3D simulation viewport. Loads `ui.js`, `game_core.js`, `npc-ai.js`, `weather.js`, `audio.js`, and all 54 level scripts deferred while `start.js` polls for `Game` initialization. Features a digital speedometer, minimap, task checklist, e-challan ticker, and 360px touch steering wheel.

PORTAL 02 • ACADEMY.HTML

Road Safety Academy & Level Hub

Reads directly from the global **window.LVS** level registry to render all 54 numbered curriculum stages plus custom/free-roam modes. Displays IRC highway sign flashcards, Motor Vehicles Act syllabus notes (`S.sylViewed`), star ratings, and post-level RTO quizzes (`ui.showQuiz`).

PORTAL 03 • GAMEPAGE.HTML + REACT-SRC/GAMEPAGE.TSX

React 19 + TypeScript Simulator Shell

Compiled via `esbuild` in `build.js` (`react-src/GamePage.tsx` → `dist/Traffic/simulator-bundle.js`, 243 KB). Provides a modern React state wrapper around the simulator with quick-launch mission briefings, controller telemetry, and seamless portal transitions.

PORTAL 04 • TRAFFICDASHBOARD.HTML

Live Civic Leaderboard & RTO Podium

Cloud-synced leaderboard powered by Supabase (`certificates` → `user_profiles` → `profiles`, joined with `wallets` and `badges`). Implements **dense tie ranking (1, 2, 2, 3)**, XSS-escaped names, gold/silver/bronze podium styling, and strict live-session identity.

PORTAL 05 • TRAFFICSETUP.HTML

Hardware Calibration & AI City Synthesizer

Allows students on low-end school PCs or budget smartphones to configure shadow map resolution, DPR cap (`Math.min(devicePixelRatio, 2)`), NPC traffic density, mirror reflection quality, and custom procedural city parameters before entering `Driving.html`.

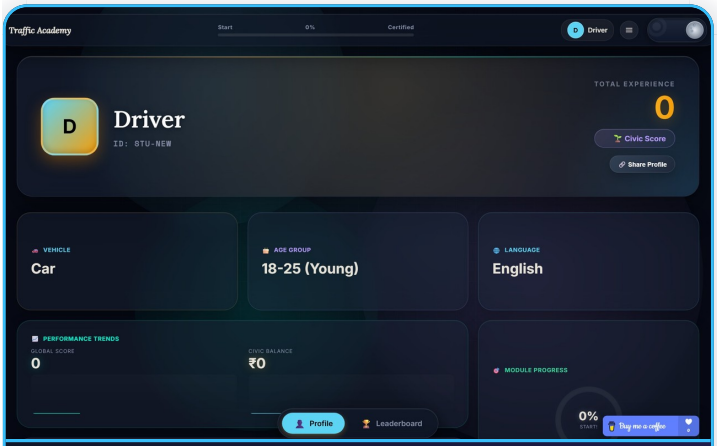
PORTAL 06 • TRAFFIC/LOD_TEST.HTML + ELECTRON PORT

3D Model LOD Inspector & Electron App

`lod_test.html` benchmarks polygon counts, Draco compression, and frame times across our `.glb` vehicle/building models. Meanwhile, `Traffic/electron/` packages `Driving.html` into a sandboxed desktop app (`build:electron`).

07. Mumbai Traffic — Multiplayer, Avatars & Pedestrian Mode

Unlike static driving games where the player is permanently welded inside a single car, the Mumbai Traffic Simulator features **live Multiplayer Convoy & Race rooms**, **Character & Vehicle Customization**, a walkable **3D Academy HQ Lobby & Garage**, and a **Third-Person Pedestrian Walk Mode (enter_vehicle)**.



Live Capture • `TrafficDashboard.html`: Driver Profile, License Tiers, Badges & Dense-Rank RTO Podium



Live 3D Capture • 3D Vehicle Garage: Custom Livery Paint Shop, Avatar Uniforms & Vehicle Inspection Bay

MULTIPLAYER ARCHITECTURE • CONVOY & CIVIC CO-OP

Real-Time Multiplayer Driving & Cloud Sync

Students can join shared Mumbai road rooms to practice **multi-driver intersection right-of-way**, synchronized convoy discipline, and competitive civic road tests:

- **State Interpolation:** Broadcasts vehicle (x, y, z), heading yaw ψ, steering wheel angle δ, brake lights, and active turn indicators (`Q` / `E` / `Z`).
- **Civic Score Competition:** Unlike street racing, multiplayer rooms rank drivers by **highest Civic Compliance Score** and zero e-challan infractions.

CHARACTER CUSTOMIZATION & PEDESTRIAN MODE

Custom Student Avatars & On-Foot Exploration

Every learner begins many curriculum stages on foot as a customizable 3D student avatar before boarding their vehicle:

- **Avatar & Livery Customization:** Customize character outfits, safety helmets (mandatory on two-wheelers!), and vehicle paint liveries using earned wallet coins.
- `enter_vehicle` **Task Contract:** Students walk across sidewalks, inspect IRC road signs up close, look both ways at zebra crossings, and press **F** to board target vehicles.

Multiplayer Sync, Avatar Customization & Mobile Ergonomics Matrix

Subsystem / Feature	Runtime Implementation	Detailed Player Experience & Engineering Contract
Multiplayer State Packet	WebSocket / Supabase Realtime	Transmits 20 Hz kinematic vectors (x, z, ψ, v, δ) + indicator/headlight bitmasks with Hermite spline interpolation.
3D Character Customization	`TrafficSetup.html` & `ui.js`	Unlockable student uniforms, reflective traffic warden vests, ISI-marked two-wheeler helmets, and custom car paint finishes.
Pedestrian Walk & Board	`_checkTasks: enter_vehicle`	3rd-person capsule controller with sidewalk curb collision, zebra crosswalk detection, and proximity boarding prompt (`Press F`).
Mobile Touch & PIP Mirror	360px First + WebGL RenderTarget	Ergonomic touch steering wheel, proportional pedals, DPR cap (`2.0`), and live rear-view cockpit mirror PIP.

07. Mumbai Traffic — Maps, Urban Zones & Monsoon Weather

Every map in the simulator is modeled after authentic **Mumbai urban streetscapes**—from coastal Marine Drive boulevards and St. Xavier High School corridors to hospital silence zones, yellow box junctions, and suburban railway crossings.

Authentic Mumbai Map Districts & Dynamic Coordinate Contracts

Mumbai Map District	Level Config Keys	Architectural Features, Landmarks & Pedagogical Hazards
1. School Safety Corridor	`schoolX`, `schoolZ`, `flasherZ`	Modeled after Mumbai school zones (e.g., St. Xavier High School). Features amber flashing beacons (`flasherZ`), child pedestrians, and strict **25 km/h speed enforcement**.
2. Pedestrian Zebra Plazas	`zebraZ`, `pedMix`	High-density pedestrian crosswalks where NPCs step onto the road; requires complete stop before the white stop-line and yielding until clear.
3. Hospital Silence Zones	`silenceZone`, `avoid: horn`	Designated medical corridors around city hospitals. Ambient NPC honking is suppressed by `npc-ai.js`, and pressing `B` triggers an immediate Section 194F fine.
4. Multi-Lane Arterial & Box Junctions	`lanes`, `signals`, `boxZ`	4-way signalized intersections with yellow criss-cross box markings; entering a box junction without a clear exit path triggers a gridlock violation.
5. Coastal Monsoon Highway	`weather: 'rain'`, `fog`	Inspired by Marine Drive & Worli Sea Face during July monsoons; reduces road friction from μ = 0.95 to μ = 0.62 and requires low-beam headlights (`L`).
6. Ghat Slopes & Free-Roam City	`level_custom.js`	Open-world Mumbai sandbox combining flyovers, roundabouts, BEST bus bays, and steep inclines for unassisted exploration.

RUNTIME COORDINATE CONTRACT • GAME_CORE.JS

Level-Specific `schoolZ` & `zebraZ` Overrides

When a level loads, `game_core.js` merges the selected `ui.cur` config into `mapCfg` via `Object.assign`. Level-specific school and crossing positions come directly from **schoolX, schoolZ, zebraZ, and flasherZ**. (While a legacy no-`schoolZ` fallback remains in `game_core.js` for backward compatibility, all new and validated levels define explicit coordinates.)

ATMOSPHERE ENGINE • WEATHER.JS + AUDIO.JS

Coastal Monsoon Rain & Sarvam AI Voices

`weather.js` spawns GPU-instanced rain streaks, wet asphalt specular reflections, and dynamic distance fog while reducing Pacejka tire grip. Simultaneously, `audio.js` plays **Hindi, Marathi, and Indian English** voice instructions synthesized via **Sarvam AI Bulbul TTS**.

Why Local Mumbai Context Matters: Students immediately recognize **Kaali-Peeli taxis, BEST red buses, auto-rickshaws, and monsoon downpours**, making every road rule directly transferable to the real streets outside their school gates.

07. Mumbai Traffic — All 16 Vehicles & Camera Engineering

The simulator features **16 distinct playable and NPC vehicle archetypes** defined in `VEHICLE_STATS` and `VEHICLE_CAM` inside `Traffic/game_core.js`. Each vehicle has realistic mass, wheelbase, braking torque, tire grip, and chase-camera framing:

Complete Specification Table Of All 16 Simulated Vehicles (`VEHICLE_STATS`)

Vehicle Key	Mumbai / Indian Archetype	Top Speed	Mass / Grip	Handling & Educational Role
`car`	Standard Learner Hatchback	135 km/h	1,180 kg / 1.00	Primary LLR training car for stages 1–15; balanced steering and braking.
`sedan`	Executive Family Sedan	145 km/h	1,340 kg / 1.00	Longer wheelbase; teaches smooth highway lane changes and mirror checks.
`taxi`	Mumbai Kaali-Peeli Taxi	115 km/h	1,150 kg / 0.96	Iconic black-and-yellow cab; tight turning radius for curb-lane passenger stops.
`auto`	3-Wheeler Bajaj Auto-Rickshaw	65 km/h	680 kg / 0.88	Ultra-agile 3-wheel geometry; teaches strict lane discipline & blind-spot safety.
`bus`	BEST Red Municipal City Bus	85 km/h	11,500 kg / 0.82	Heavy mass & wide tail-swing; requires 2.5× longer stopping sight distance.
`truck`	Heavy Tata Goods Carrier	80 km/h	14,000 kg / 0.78	Teaches momentum management, downhill engine braking, and left-lane commercial rules.
`bike`	Commuter Motorcycle	110 km/h	175 kg / 0.93	Two-wheeler lean dynamics; enforces mandatory helmet law and wet-road caution.
`scooter`	Urban Gearless Scooter (Activa)	85 km/h	118 kg / 0.90	Lightweight student commuter two-wheeler for residential & school-zone stages.
`suv`	Full-Size Urban SUV	155 km/h	2,100 kg / 1.02	Higher center of gravity; demonstrates load transfer during emergency swerves.
`jeep`	Rugged Off-Road Utility Jeep	130 km/h	1,950 kg / 1.04	High ground clearance for flooded monsoon streets and unpaved gravel shoulders.
`police`	Mumbai Police Patrol Interceptor	165 km/h	1,650 kg / 1.08	Equipped with beacon lightbar and siren; used in traffic enforcement scenarios.
`ambulance`	Emergency Trauma Ambulance	150 km/h	2,450 kg / 1.02	Teaches Green Corridor priority yielding under Motor Vehicles Act Section 194E.
`van`	Yellow School Transport Van	100 km/h	1,850 kg / 0.90	Enforces strict 25 km/h school-zone compliance and safe student boarding stops.
`pickup`	Commercial Utility Pickup	115 km/h	1,920 kg / 0.89	Rear cargo load sensitivity; teaches rear-axle traction changes on wet asphalt.
`sports`	High-Performance Sports Coupe	195 km/h	1,420 kg / 1.12	Demonstrates kinetic energy (E_k = \ \frac{1}{2}mv^2) and stopping distance scaling.
`hyper`	Electric Hypercar Prototype	215 km/h	1,580 kg / 1.15	Unlocked in Free-Roam mastery; showcases instant EV torque and regenerative braking.

Lower-Third Chase Camera Contract (`VEHICLE_CAM`): Each vehicle defines custom `dist`, `height`, `lookY`, `lookAhead`, `stiffPos`, and `fovBase/fovMax` parameters so the vehicle sits cleanly in the **lower third of the screen** while overhead signals and zebra crossings remain unobstructed in the upper two-thirds.

07. Mumbai Traffic — Pacejka MF 5.2 Tire Physics & Dynamics

To make braking distances, monsoon skids, and heavy-vehicle momentum physically authentic, `Traffic/game_core.js` (14,975 lines) implements **Hans Pacejka's Magic Formula (MF 5.2)** tire mechanics rather than arcade velocity clamping.

EQUATION 01 • PACEJKA MAGIC FORMULA 5.2

Longitudinal (F_x) & Lateral (F_y) Tire Forces

For longitudinal slip ratio κ and lateral slip angle α, tire force F(x) is governed by stiffness factor B, shape factor C, peak value D = μF_z, and curvature factor E:

$$F(x) = D \cdot \sin(C \cdot \arctan(Bx - E(Bx - \arctan(Bx))))$$
$$\mu(F_z) = \mu_0 \left[1 - k_z(F_z - F_{z0}) / F_{z0} \right]$$

Load Sensitivity (k_z): As vertical load F_z increases (e.g., on a 14,000 kg Tata truck or 11,500 kg BEST bus), normalized tire friction μ(F_z) decreases non-linearly, explaining why heavy vehicles require more than double the stopping distance of a hatchback.

EQUATION 02 • COMBINED-SLIP FRICTION ELLIPSE

Why Hard Braking While Turning Causes Monsoon Skids

When a driver brakes hard (F_x) while simultaneously steering into a corner (F_y), total tire grip cannot exceed the friction circle limit:

$$(F_x / F_{x,max})^2 + (F_y / F_{y,max})^2 \le 1$$
$$d_{stop} = v \cdot t_r + v^2 / (2\mu g)$$

Students experience firsthand how panic-braking mid-turn consumes longitudinal capacity F_x and collapses lateral cornering force F_y, teaching them to **brake in a straight line before entering a turn**.

Road Surface Presets & Stopping Distance Comparison At 50 km/h (13.89 m/s)

Surface Preset	Peak μ_0	Pacejka (B, C, E)	50 km/h Stopping Dist	Curriculum Levels Using Preset
`dry_asphalt`	**0.95**	B=10.0,\\; C=1.9,\\; E=0.97	**~24.2 meters**	Levels 01–30 (Dry urban Mumbai streets)
`wet_asphalt`	**0.62**	B=8.2,\\; C=1.85,\\; E=1.02	**~31.8 meters (+31%)**	Levels 31–40 & 51–54 (Monsoon downpours)
`cobblestone_ghat`	**0.55**	B=7.4,\\; C=1.75,\\; E=1.03	**~34.4 meters (+42%)**	Heritage South Mumbai & Steep Ghat Slopes
`gravel` / Shoulder	**0.48**	B=6.5,\\; C=1.65,\\; E=1.05	**~37.5 meters (+55%)**	Roadside Construction & Unpaved Shoulder Stages
`flooded_monsoon`	**0.38**	B=5.4,\\; C=1.55,\\; E=1.08	**~44.6 meters (+84%)**	Waterlogged Hindmata / Coastal Storm Surge Hazards

Pitch & Roll Load Transfer

During braking ($a_x < 0$), longitudinal load \Delta F_z = (m · |a_x| · h_{\text{cg}}) / L shifts onto the front axle with visual chassis pitch.

Ackermann Steering Geometry

Inner and outer front wheels turn at distinct angles around a shared instantaneous center of rotation to eliminate low-speed scrub.

Speed-Sensitive Damping

Maximum steering lock attenuates smoothly at highway speeds so keyboard and touch inputs never cause unrealistic rollovers.

07. Mumbai Traffic — NPC AI, MVA E-Challans & Dashboard

To make virtual Mumbai feel alive, we engineered an autonomous multi-agent traffic controller (**Traffic/npc-ai.js** , 662 LOC), a real-time **Motor Vehicles Act (MVA) e-challan detector** , and our cloud-synced **TrafficDashboard.html** leaderboard.

AUTONOMOUS TRAFFIC • NPC-AI.JS (662 LOC)

4 Behavioral Driver Profiles & Pedestrian AI

NPC vehicles and pedestrians run on distinct behavioral profiles (**cautious** , **normal** , **eager** , **yielding** , plus scripted `profileKey` overrides) configured per level via `npcMix` and `pedMix` :

- **IDM Longitudinal Spacing:** NPCs maintain safe following headway and decelerate smoothly behind stopped cars or red signals.
- **School & Zebra Yielding:** NPCs automatically slow down near `schoolZ` and yield when pedestrians step onto `zebraZ` .
- **Silence Zone Compliance:** Ambient NPC horn honking is automatically suppressed inside Hospital and School silence zones.

LEADERBOARD CONTRACT • TRAFFICDASHBOARD.HTML

Dense Ties, Escaped Names & Supabase Data Order

Our live RTO Leaderboard strictly enforces four architectural rules:

- **Dense Tie Ranking:** Identical scores share the same rank without skipping numbers (`1, 2, 2, 3`) and color podiums by place (Gold/Silver/Bronze).
- **Strict Query Order:** Fetches `certificates` → `user_profiles` (then `profiles`), enriched with `wallets` and `badges` .
- **Security & First Paint:** HTML-escapes all display names, renders scope tabs on first paint, and never falls back to unsigned local storage.

Automated Motor Vehicles Act (MVA) E-Challan Enforcement Matrix

Violation Detected in 3D	Statutory MVA Section	Civic Penalty	Real-Time Telemetry Trigger Condition
Signal Jumping (Red Light)	Section 184 (Dangerous Driving)	-25 Civic Pts / ₹1,000	Crossing intersection stop-line (z = stopZ) while signal state is `RED` .
School Zone Overspeeding	Section 183 (Excessive Speed)	-20 Civic Pts / ₹2,000	Exceeding 25 km/h within \\pm 45\text{ m} of `schoolZ` / `flasherZ` beacon.
Honking in Silence Zone	Noise Pollution Rules / Sec 194F	-15 Civic Pts / ₹1,000	Pressing horn (`B`) inside designated Hospital or School silence radius.
Zebra Crossing Failure to Yield	Section 177 / Pedestrian Right	-20 Civic Pts / ₹500	Entering `zebraZ` bounding box while an active pedestrian is crossing.
Unindicated Lane Change	Section 177 (Improper Signal)	-10 Civic Pts / ₹500	Lateral lane transition ($\|\Delta x\| > 3.2$ m) without active `Q` / `E` indicator.
Blocking Emergency Corridor	Section 194E (Emergency Yield)	-30 Civic Pts / ₹10,000	Failing to pull left within 6 seconds of approaching ambulance siren.

Idempotent Progression & 30-Minute Autosave (`ui.js`)

`game_core.completeLevel` records `S.comp` and wallet rewards, while `ui.js` owns `S.total` , `civicScore` , and badges with `_counted` idempotency and 30-minute run saves (`traffic_run_<id>`).

Guarded Optional UI Hooks (`start.js` Contract)

Because `Driving.html` and `Academy.html` load scripts deferred while `start.js` polls for `Game` , all optional UI calls (`showQuiz` , challan handlers) are safely guarded.

07. Mumbai Traffic — 54-Level Curriculum, Validator & Codebase

Every stage in `Traffic/Levels/level1.js` through `level54.js` (plus `level_custom.js` free-roam) registers into **window.LVS** and is verified by **Traffic/tools/validate-levels.js** across all 5 supported `_checkTasks` types (`enter_vehicle` , `stop` , `reach` , `avoid` , `toggle`).

Complete 54-Stage Curriculum Progression (`Levels/level1.js` – `level54.js`)

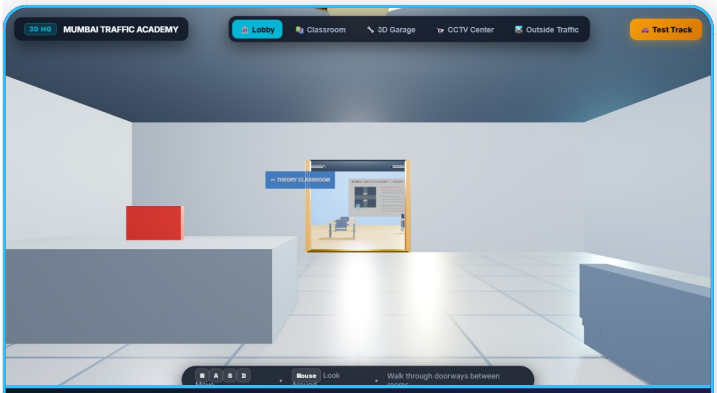
Level Range	Curriculum Tier	Key Mumbai Scenarios, Road Rules & Driving Competencies Covered
Levels 01 – 10	Tier 1: LLR Fundamentals	Walking to vehicle (`enter_vehicle`), mirror checks, seatbelts, smooth stop-line braking (`stop`), turn indicators (`toggle`), and 4-way signals.
Levels 11 – 20	Tier 2: School & Silence Zones	School zone 25 km/h limits (`schoolX` , `schoolZ` , `flasherZ`), pedestrian zebra yields (`zebraZ`), hospital silence zones (`avoid: horn`), and bus bays.
Levels 21 – 30	Tier 3: Urban Density & Lanes	Multi-lane arterial corridors, yellow box junctions, roundabouts, overtaking restrictions, dedicated BEST bus lanes, and railway crossings.
Levels 31 – 40	Tier 4: Monsoon & Night Driving	Coastal torrential downpours (`wet_asphalt`), hydroplaning recovery, low-beam vs. high-beam night etiquette, fog visibility, and hazard flashers.
Levels 41 – 50	Tier 5: Heavy Vehicles & Sirens	Driving BEST buses & goods trucks (wide tail-swing), yielding to ambulances (MVA Sec 194E), steep ghat hill-starts, and highway merging.
Levels 51 – 54 + Custom	Tier 6: RTO Final Exam & Free-Roam	Unassisted multi-checkpoint Mumbai city endurance exams combining rain, pedestrians, and silence zones, plus open-world Free-Roam (`level_custom.js`).

Complete Codebase & 3D Asset Inventory — `Traffic/`

File / Directory Path	Size / Scope	Detailed Runtime & Architectural Responsibility
`Traffic/game_core.js`	14,975 LOC (732 KB)	Master 3D engine: Three.js WebGL renderer, Pacejka MF 5.2 tire physics, 16 `VEHICLE_STATS` , `VEHICLE_CAM` , `_checkTasks` , e-challans, and `completeLevel` (records `S.comp` & wallet).
`Traffic/ui.js`	2,451 LOC (131 KB)	HUD overlay, Academy level selector, RTO quiz modal (`showQuiz`), idempotent `_counted` score/badge progression (`S.total` , `civicScore`), 30m autosave (`traffic_run_<id>`), and `S.sylViewed` .
`Traffic/npc-ai.js` , `weather.js` , `audio.js`	1,336 LOC Combined	Autonomous NPC traffic/pedestrian AI, monsoon rain/fog, wet-asphalt friction modifier, and Sarvam AI multilingual voice clips.
`Traffic/Levels/` & `validate-levels.js`	55 Level Scripts & QA	`level1.js` – `level54.js` + `level_custom.js` (55 scripts) registered in `window.LVS` & validated.

07. Mumbai Traffic — Why Needed, Free Use, Services & 3D HQ

Below are direct captures from our walkable **3D Academy HQ Lobby, Classroom, and Control Center** inside the simulator, followed by why we created this flagship project **100% for Free Use** and how every service works:



Live 3D Capture • Academy HQ Lobby: Walkable 3D Reception, Interactive Noticeboards & Mission Teleporters



Live 3D Capture • 3D RTO Classroom: Interactive Road-Sign Chalkboard & MVA Theory Briefing Hall

07. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Mumbai Traffic 3D & Why It Is Needed

What It Is: A 19,000+ LOC WebGL/Three.js driving simulator and Road Safety Academy teaching authentic Indian left-side traffic discipline across 54 curriculum stages, 16 vehicles, and Mumbai urban landmarks.

What Problem We Solved: Replaced expensive ₹5,000 driving school simulators and right-hand Western driving games with a 100% free-use simulator modeling Pacejka MF 5.2 monsoon physics and real MVA e-challans.

2026–27 Expansion Roadmap: WebXR Immersive Headset VR Driving, GestureUI Touchless Steering & RTO AI Autonomous Driving Examiner (Deep-Dive on Page 52).

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

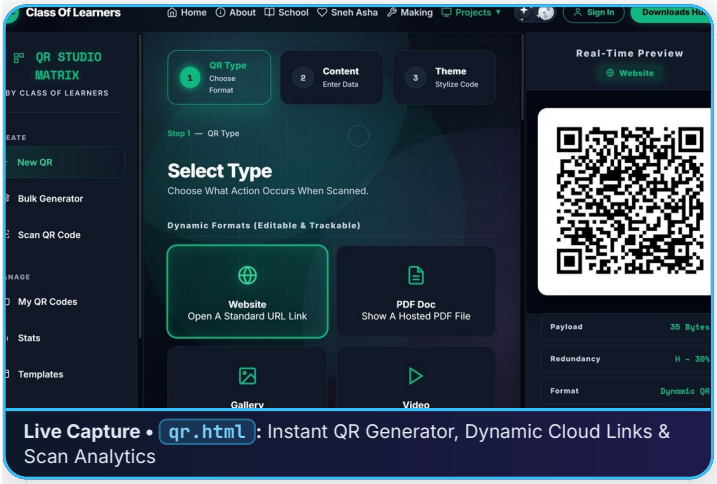
- **Who Has Worked On This:** Neel Badri, Ansh Patil & Aarush Vangari • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.js` , `.tsx` , `.html` , `.glb` , `.json` , `.css` , `.bat`
- **Repository & File Root:** `Vercel/Traffic/` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** WebGL2 Browser Portal + Electron Desktop • 4GB RAM • 360px Touch Steering
- **Target Audience & Metrics:** Youth Drivers (Ages 14–22), RTO Students • 54 Levels • 60 FPS • 16 Vehicles

All Services, AI Models & Cloud Infrastructure Used In Mumbai Traffic 3D

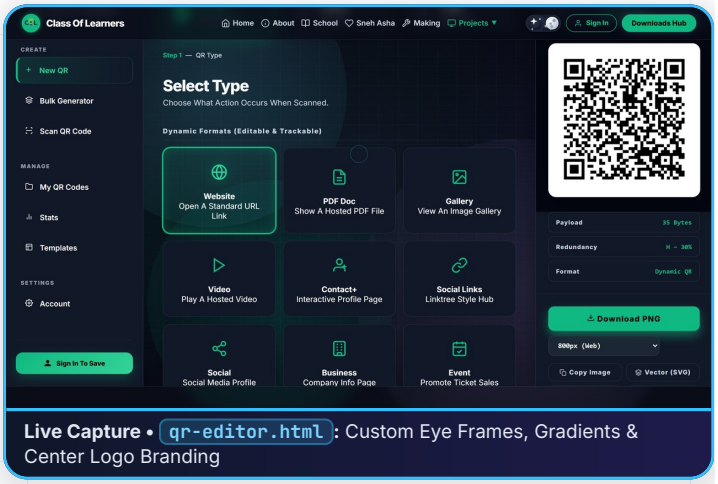
Service / Technology	Where Integrated	How It Works In Our Project
Three.js & Draco 3D GLBs	`game_core.js` & `Models/`	Renders 3D Mumbai streets, 16 Indian vehicles, traffic signals, monsoon rain particles, and rear-view mirrors at 60 FPS in the browser.
Supabase (DB, Auth & Realtime)	`TrafficDashboard.html` & Multiplayer	Stores verified student driving certificates, wallet coins, and badges, and syncs live multiplayer convoy cars and dense-rank leaderboards.
Sarvam AI (Bulbul TTS)	`Traffic/audio.js`	Generates natural-sounding Hindi, Marathi, and Indian English voice instructions from the virtual driving instructor during missions.
React 19 + TypeScript + esbuild	`react-src/GamePage.tsx`	Bundles our modern mission HUD into a single fast-loading file (`simulator-bundle.js`) during `npm run build` .
Node.js AST Level Validator	`tools/validate-levels.js`	Automatically checks all 54 level files and custom free-roam before deployment so no level ever has a broken objective or missing target.
Electron & Vercel Edge CDN	`Traffic/electron/` & `dist/`	Delivers the simulator globally over Vercel Edge CDN and also packages it as a standalone offline desktop app via Electron.

08. Dynamic QR Code Studio (`qr.html` , `qr-editor.html` , `q.html`)

Dynamic QR Code Studio is our full-featured vector QR generator, artistic canvas customizer, live camera scanner, and cloud-routed short-link analytics suite integrated into Site CoL.



Live Capture • `qr.html` **:** Instant QR Generator, Dynamic Cloud Links & Scan Analytics



Live Capture • `qr-editor.html` **:** Custom Eye Frames, Gradients & Center Logo Branding

WHY WE BUILT IT & CLASSROOM USE

Update Destination URLs Without Re-Printing

Schools and science fair booths print QR codes on physical posters, lab equipment, and student diplomas. With static QR codes, changing a URL breaks the printed poster. Our **Dynamic QR routing (q.html?id=...)** lets teachers update the target link anytime in Supabase while tracking live scan counts.

Contributors: Neel Badri, Aarush Vangari & CoL Team

MATHEMATICAL FOUNDATION • REED-SOLOMON ECC

Galois Field GF(2^8) Error Correction (`L/M/Q/H`)

To embed a center Class Of Learners logo inside a QR matrix without breaking camera readability, `qr-editor.html` switches Reed-Solomon encoding over GF(2^8) to **Level H (30% codeword recovery)**, allowing up to 30% of central modules to be artistically occluded.

Exports: Lossless Vector SVG & 1200×1200 High-DPI PNG

Complete Source File Inventory & User Manual — QR Code Studio

File Path	Module Role	Technical Responsibility & User Workflow
`Vercel/qr.html` & `qr.css`	Main QR Dashboard	Supports URL, Wi-Fi network credentials, vCard contact, plain text, and UPI payment QR payloads + webcam QR scanner (`html5-qrcode.min.js`).
`Vercel/qr-editor.html`	Artistic Canvas Studio	Custom module dots (rounded/diamond/square), corner finder eye styling, linear/radial color gradients, and center logo overlay.
`Vercel/qr-dynamic.js` & `q.html`	Dynamic Cloud Router	Generates short slugs in Supabase, increments scan telemetry counters, and redirects in <80 ms with open-redirect protection.
`Vercel/supabase-qr-migration.sql`	Database RLS Schema	Defines `qr_links` and `qr_scans` tables with Row-Level Security so only the authenticated creator can edit destination URLs.

1. Multi-Payload Encoding

Encodes HTTPS links, WPA2 classroom Wi-Fi, vCard contacts, and UPI payment strings.

2. Live Camera Decoder

Scans QR codes directly from laptop webcams or smartphone cameras using `html5-qrcode` .

3. Security Gate Verified

Guarded by `security-check.js` to ensure safe storage and zero XSS/ open-redirect vulnerabilities.

08. QR Code Studio — Why Needed, Free Use & Services Used

Almost every student, teacher, or college club has experienced the frustration of generating a "free" QR code online, printing it on 100 posters, and discovering two weeks later that the QR code stopped working unless they pay a monthly fee.

08. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Dynamic QR Code Studio & Why It Is Needed

What It Is: A vector SVG/PNG QR designer (`qr.html` , `qr-editor.html`) with real-time camera decoding and Supabase dynamic short-link analytics.

What Problem We Solved: Replaced commercial QR services that deactivate codes after 14-day trials with a 100% free-use studio featuring permanent redirect links and zero paywalls.

2026–27 Expansion Roadmap: Batch vector SVG export, encrypted Wi-Fi provisioning codes & offline camera decoding.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri • **Mentors:** Sanjana Kasbe Ma'am & Avantika Raskar Ma'am
- **What File Types Have We Used:** `.html` , `.js` , `.css` , `.svg` , `.json`
- **Repository & File Root:** `Vercel/qr.html` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** Static Web PWA • Min 1GB RAM • Camera Optional
- **Target Audience & Metrics:** School Events, Teachers, Students • Sub-10ms Vector Render • 0 Paywalls

All Services, Databases & Libraries Used In Dynamic QR Code Studio

Service / Library	Where Integrated	How It Works In Our Project
Supabase PostgreSQL & RLS	`qr-dynamic.js` & `migration.sql`	Stores each dynamic QR code's short ID, target URL, and scan counter in our cloud database while Row-Level Security ensures only the creator can change the link.
Google OAuth (`col-auth.js`)	Dynamic QR Management	Links dynamic QR codes to the signed-in teacher or student's Google account so they can manage their links from any computer or phone.
HTML5 Canvas & SVG Engine	`qr-editor.html`	Draws custom rounded dots, gradient colors, corner finder frames, and center logos at 1200×1200 resolution or lossless vector SVG for large banners.
`html5-qrcode` Vision Decoder	`qr.html` Scanner Tab	Accesses the laptop webcam or phone camera (or reads an uploaded image) to decode any QR code instantly in the browser without installing an app.
Vercel Edge & `security-check.js`	`q.html` & Build Gate	Redirects scanned dynamic links in under 80 milliseconds and validates that only safe `http://` or `https://` destinations are allowed.

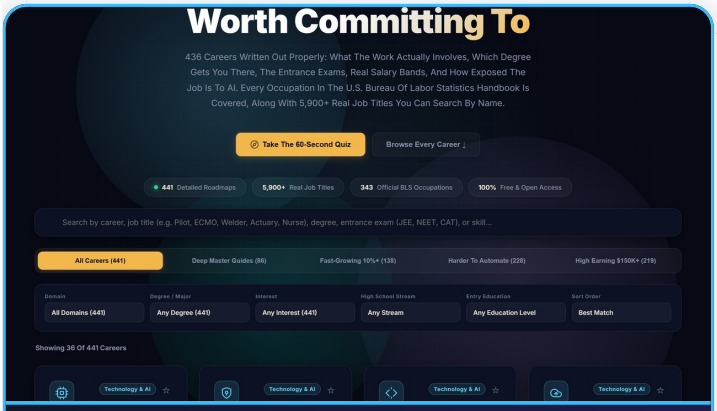
Zero Expiring Links
Static and Dynamic QR codes created on Site CoL never expire and never display interstitial ads before redirecting.

Print-Ready Vector SVG
Export infinite-resolution SVG files that stay razor-sharp even on 6-foot college festival hoardings.

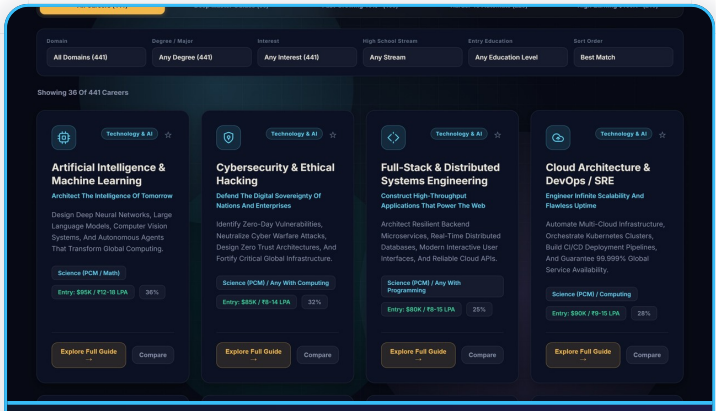
Live Scan Analytics
Track how many visitors scanned your project poster or classroom worksheet link in real time.

09. Career Guidance — FutureScape (`Career.html`)

FutureScape Career Guidance (**Career.html**) is an interactive psychometric counseling platform and encyclopedic database mapping **436 distinct Indian and global career pathways** across 18 industries for Class 10, Class 12, and undergraduate students.



Live Capture • **Career.html**: FutureScape 60-Second Career Compass, Stream Filters & Psychometric Matcher



Live Capture • **careers-data.js**: Interactive Career Cards with ₹ LPA Salary Bands & AI Exposure Ratings

WHY WE BUILT IT & DEVELOPMENT HISTORY

Beyond Traditional 3-Stream Tunnel Vision

High school students in India are often pressured into choosing between a tiny handful of careers because they lack structured visibility into modern fields. Built by **Neel Badri**, **Ansh Patil**, and the CoL team using **Gemini Web** and **Google Antigravity**, FutureScape makes 436 careers searchable with transparent Indian academic roadmaps.

Dataset Scale: 436 Careers across 18 Clusters (196 KB JSON/JS)

PSYCHOMETRIC ALGORITHM • COMPASS-SCORING.JS

Multi-Axis Cosine Similarity Matching

The **60-Second Career Compass** quiz maps student responses into an 8-dimensional aptitude vector **u** (Analytical, Creative, Social, Technical, Leadership, Hands-On, Research, Entrepreneurial) and ranks all 436 careers $\mathbf{c}_i$:

$$S(\vec{u}, \vec{c}_i) = \frac{\vec{u} \cdot \vec{c}_i}{\|\vec{u}\| \, \|\vec{c}_i\|} \times 100\%$$

Complete Source File Inventory & Indian Academic Roadmap Features

File Path	Size / Role	Detailed Counseling Features & Data Schema
`Vercel/Career.html` & `career.css`	Main Counseling UI	Stream filter (Science PCM/PCB, Commerce, Humanities, Vocational), search bar, 60s Compass modal, and side-by-side career comparison drawer.
`Vercel/careers-data.js`	196 KB Knowledge Base	436 career profiles detailing required 11th/12th subjects, Indian entrance exams (**JEE, NEET, CUET, CLAT, NID, NDA, GATE**), top colleges, and ₹ LPA salary bands.
`Vercel/compass-scoring.js` & `career-features.js`	Scoring & Export	Cosine-similarity psychometric ranker, AI disruption resilience index, bookmark manager, and printable step-by-step student career roadmap.

Indian Entrance Exam Matrix

Maps exact eligibility and preparation timelines for JEE, MHT-CET, NEET, CUET, CLAT, NID, UCEED, NDA, and CA.

AI Resilience & Future Outlook

Scores each career's 2025–2035 growth trajectory and shows how students can leverage AI tools as a career multiplier.

Side-by-Side Comparison

Compare up to 3 careers simultaneously across education cost, degree duration, starting ₹ LPA, and work-life balance.

09. FutureScape Career Guidance — Why Needed, Free Use & Services

After Class 10 and Class 12 board exams, millions of Indian students and parents face intense anxiety deciding which academic stream, entrance exam, and degree to pursue—often without knowing that hundreds of high-growth modern careers exist.

09. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is FutureScape Career Guidance & Why It Is Needed

What It Is: A career discovery and psychometric guidance suite (`Career.html` , `careers-data.js`) mapping 436 Indian and global career trajectories with salary bands and entrance exams.

What Problem We Solved: Replaced ₹5,000 paid counseling fees and sponsored college advertisements with an unbiased, 100% free-use multidimensional cosine similarity matcher.

2026–27 Expansion Roadmap: Expanded 600+ career profiles, 2026–2035 AI displacement vulnerability indices & regional vocational paths.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri & Ansh Patil • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.html` , `.js` , `.css` , `.json`
- **Repository & File Root:** `Vercel/Career.html` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** Zero-Server Client-Side Web • Min 1GB RAM • 100% Offline Capable
- **Target Audience & Metrics:** High School Students (Classes 9–12), Parents • 436 Careers • Sub-12ms Cosine Match

All Services, Data Engines & AI Tools Used In FutureScape Career Guidance

Service / Module	Where Integrated	How It Works In Our Project
436-Career Knowledge Base	`careers-data.js` (196 KB)	Stores structured JSON profiles for 436 careers across 18 clusters, runs 100% client-side in memory so search and filtering happen in under 2 milliseconds.
8-Axis Psychometric Matcher	`compass-scoring.js`	Computes the cosine similarity between a student's 60-Second Compass quiz answers and all 436 career vectors to recommend their top matches.
Gemini Web & Antigravity	Dataset Synthesis & QA	Used during development to research, cross-check, and structure Indian entrance exams, top institutes, and AI-resilience scores across all 18 industries.
Supabase & LocalStorage Sync	`career-features.js` & `col-auth.js`	Saves bookmarked careers, comparison lists, and psychometric quiz profiles locally and syncs them with the student's signed-in CoL account.
Browser Print / Roadmap Engine	`career-features.js`	Formats a student's selected career pathway, required Class 11/12 subjects, and entrance exam checklist into a clean printable guide for parents and teachers.

Zero Phone-Number Gate

Students never have to enter a phone number or email address to view their 60-Second Career Compass matches.

18 Industry Clusters

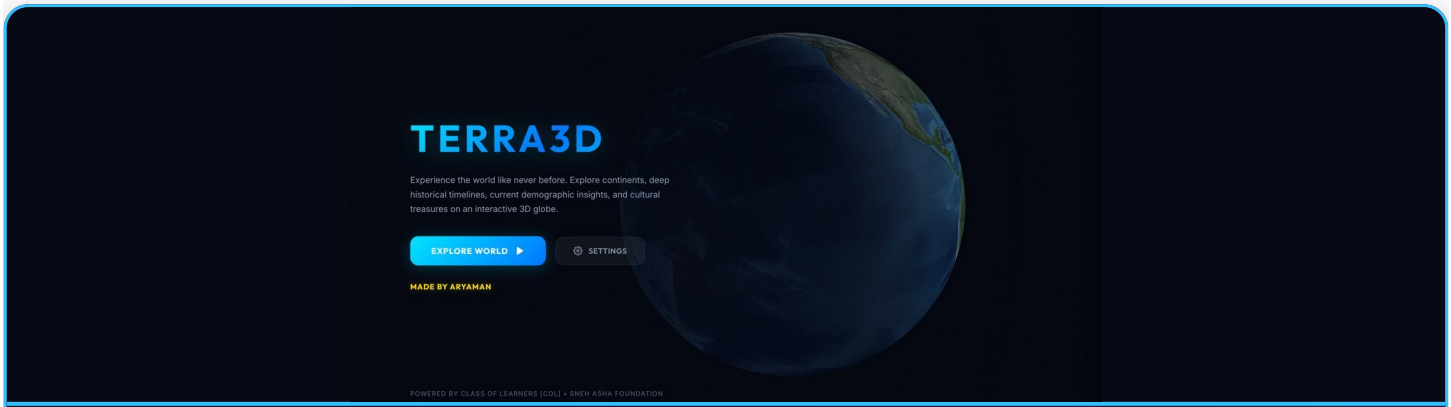
Covers Engineering, Medicine, Law, Design, Aviation, Defence, Finance, Space Science, AI, Arts, and Vocational trades.

Works On 360px Phones

Optimized with mobile stream chips and swipeable comparison cards so families can explore careers together on any phone.

10. Terra 3D — Interactive Earth & Geopolitical Globe

Terra 3D (`Terra3D/` ; open-source at `github.com/NeelAniGamer/Terra3D`) is a standalone React 19 + TypeScript + Three.js interactive 3D Earth globe and geospatial visualization engine featuring custom Rayleigh atmospheric scattering shaders, **195+ sovereign country border polygons**, and spherical geodesic coordinate projection.



Live Capture • `Terra3D / Globe.tsx` : 3D Spherical Earth, GeoJSON Sovereign Borders & Atmospheric Halo

WHY WE BUILT IT & CLASSROOM USE

Eliminating 2D Mercator Map Distortion

Flat 2D classroom wall maps severely distort continental proportions near the poles (making Greenland look as large as Africa). We built **Terra 3D** so students can inspect true spherical geography, tectonic plates, climate zones, and international borders in 60 FPS 3D.

Contributors: Aarayaman Jadhav, Neel Badri, Yashraj Jadhav & Team

SPHERICAL GEODESY & GLSL FRESNEL SHADER

Geodetic (φ, λ) To 3D Cartesian Mapping

Every GeoJSON country boundary vertex in `borders.js` (latitude φ, longitude λ) is projected onto the sphere of radius R_E with a $+0.002R_E$ radial offset to prevent Z-fighting:

$$[\,x, y, z\,]^T = R_E\,[\,-\cos(\varphi)\cdot\cos(\lambda),\ \sin(\varphi),\ \cos(\varphi)\cdot\sin(\lambda)\,]^T$$

$$I_{limb} \propto (1 - \hat{N}\cdot\hat{V})^3 \ \bullet\ \text{Radial Offset: } +0.002R_E$$

Complete Source File Inventory — `Terra3D/` , Root Globe & `Tracker/`

File Path	Size / Layer	Engineering Responsibility
`Terra3D/src/components/Globe.tsx`	39.2 KB (React TSX)	Core Three.js r170 globe component, orbital camera controls, spherical country raycaster, and telemetry cards.
`Terra3D/src/App.tsx` & `electron/`	Vite + Electron	Application layout shell, layer toggles (Night Lights, Clouds, Borders, Tectonics), country search, and desktop build.
`Vercel/borders.js` & `Vercel/globe.js`	264.3 KB Combined	High-precision sovereign country GeoJSON border coordinates and shared 3D Earth renderer for Site CoL.
`Tracker/world.svg` & `Tracker/script.js`	2D SVG Map Companion	Lightweight 2D vector world map companion for low-power devices without WebGL2 acceleration.

8K PBR Earth Textures

Combines NASA Blue Marble albedo, bathymetric normal maps, ocean specular masks, and nocturnal city lights.

Spherical Raycasting

Clicking any point on the 3D sphere resolves (x, y, z) \\to (\phi, \lambda) and highlights the matching sovereign nation.

Standalone Vite Build

Maintained as an independent React 19 + Three.js package (`Terra3D/package.json`) with desktop Electron support.

10. Terra 3D — Why We Need It, Free Use & Services Used

Physical classroom globes are expensive, fragile, and cannot display dynamic layers like nocturnal city lights, tectonic boundaries, or instant country search. **Terra 3D** brings an interactive 3D Earth atlas to every computer and tablet.

10. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Terra 3D Earth Globe & Why It Is Needed

What It Is: An interactive 3D WebGL Earth globe (`Terra3D/` , `Globe.tsx`) rendering 195+ sovereign national border polygons, Rayleigh atmospheric limb shaders, and spherical geodesy.

What Problem We Solved: Replaced heavy mapping software requiring high-speed broadband and discrete GPUs with a 100% free-use WebGL simulator running smoothly in school computer labs.

2026–27 Expansion Roadmap: Real-time satellite cloud overlays, tectonic fault line visualization & offline vector caching.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Ansh Patil, Aarayaman Jadhav & Yashraj Jadhav • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.tsx` , `.js` , `.html` , `.svg` , `.json` , `.css`
- **Repository & File Root:** `Vercel/Terra3D/` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** React Vite Web App • Min 2GB RAM • WebGL1/2
- **Target Audience & Metrics:** Geography & Earth Science Students • 195+ Countries • 60 FPS

All Services, Datasets & Frameworks Used In Terra 3D & World Tracker

Service / Framework	Where Integrated	How It Works In Our Project
Three.js r170 & WebGL2	`Globe.tsx` & `Vercel/globe.js`	Renders the 3D Earth sphere, cloud layer, starfield background, and glowing blue Fresnel atmosphere shader at 60 frames per second.
GeoJSON Sovereign Borders	`Vercel/borders.js` (122 KB)	Contains latitude/longitude polygon outlines for 195+ countries and projects them onto the 3D sphere as crisp vector boundary lines.
React 19, TypeScript & Vite	`Terra3D/src/App.tsx`	Powers the interactive sidebar, country search autocomplete, layer toggles, and smooth camera fly-to animations.
NASA Earth Observatory Maps	`Terra3D/public/textures/`	Supplies high-resolution daytime satellite imagery, ocean specular masks, mountain bump maps, and night-time city light textures.
Electron & 2D SVG Companion	`Terra3D/electron/` & `Tracker/`	Packages Terra 3D as an offline desktop app and provides `Tracker/world.svg` as a fast 2D fallback for older PCs without 3D graphics.

Zero External Map API Keys

Runs completely self-contained without needing Mapbox or Google Maps API keys, ensuring 100% uptime offline and online.

Co-Branded With SnehAsha

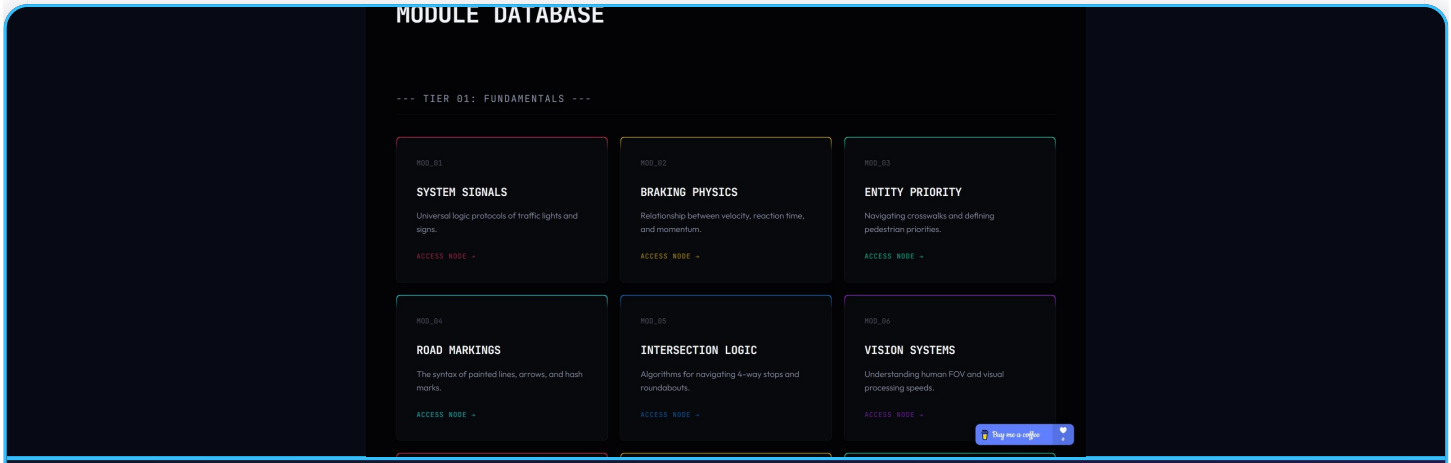
Built in partnership with SnehAsha Foundation to bring interactive earth science to underserved classrooms.

Open-Source On GitHub

Full source code is published publicly at `github.com/NeelAniGamer/Terra3D` for students and educators.

11. CivicLogic: Traffic System & Rules Database (`TRAFFIC.EDU v3.0`)

Referred to on our site navigation bar simply as **Database** (**Database_Logic.html** , 3,023 lines / 181 KB), its official title is **CivicLogic: Traffic System & Rules Knowledge Database (TRAFFIC.EDU v3.0)** . While the 3D Mumbai Traffic Simulator teaches road rules behind the wheel, **CivicLogic** is our comprehensive 50-module theoretical encyclopedia and interactive traffic-engineering sandbox.



Live Capture • `Database_Logic.html` : TRAFFIC.EDU v3.0 Interactive 50-Module Civic Engineering & MVA Encyclopedia

Complete 5-Tier Architecture Across All 50 Interactive Modules

Curriculum Tier	Module Range	Interactive Civic Systems, Statutory Laws & Traffic Engineering Topics
Tier 1: Foundations	Modules 01 – 10	Right-of-Way hierarchy, Mandatory (circular red), Cautionary (triangular), and Informatory IRC road signs, road markings, and pedestrian zebra priority.
Tier 2: Urban Flow	Modules 11 – 20	Signal phase timing (Webster's delay formula), yellow box junctions, roundabout weaving angles, lane discipline, and BEST bus priority lanes.
Tier 3: Vehicle Physics	Modules 21 – 30	Stopping sight distance ($d = vt_r + v^2 / (2\mu g)$), wet monsoon hydroplaning speed ($V_p \approx 10.35\sqrt{p}$), heavy-truck blind-spot cones, and tyre tread laws.
Tier 4: MVA Statutes	Modules 31 – 40	Motor Vehicles (Amendment) Act 2019 sections (177–194F), **Good Samaritan Law (Sec 134A)** protecting bystanders who help crash victims, PUCC, and insurance.
Tier 5: Smart Cities	Modules 41 – 50	Adaptive AI traffic signal controllers, ANPR e-challan camera networks, Green Corridors for organ transport, and Golden-Hour trauma triage protocols.

WHY NEEDED & ACADEMIC MENTORSHIP

Bridging Physics, Law & Civic Empathy

Guided by **Avantika Raskar Ma'am** and **Sanjana Kasbe Ma'am** , CivicLogic explains the why behind every traffic rule using high-school physics (kinetic energy, reaction time, friction) and real Indian statutes.

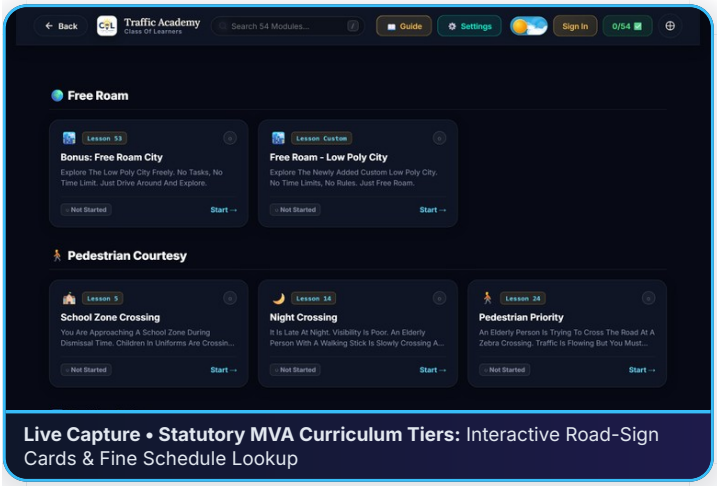
INTERACTIVE SANDBOX CALCULATORS

Live Stopping-Distance & Fine Simulators

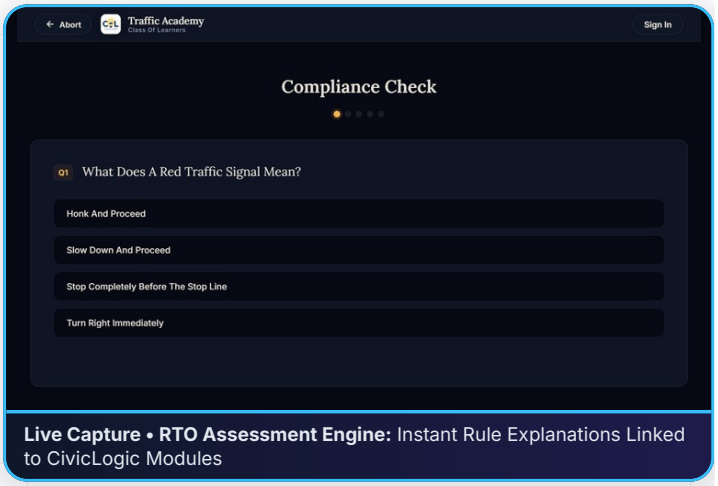
Students adjust sliders for vehicle speed (20–120 km/h), weather (Dry vs. Mumbai Monsoon), and driver alertness inside `Database_Logic.html` to watch reaction distance and braking distance bars update in real time.

11. CivicLogic Database — Why Needed, Free Use & Services Used

Most learners studying for their Indian Learner's License (LLR) only see poorly scanned booklets of road signs without understanding the engineering logic behind signal timing, braking distances, or the Good Samaritan Law.



Live Capture • Statutory MVA Curriculum Tiers: Interactive Road-Sign Cards & Fine Schedule Lookup



Live Capture • RTO Assessment Engine: Instant Rule Explanations Linked to CivicLogic Modules

11. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is CivicLogic Traffic Database & Why It Is Needed

What It Is: TRAFFIC.EDU v3.0 (`Database_Logic.html`), an interactive 50-module traffic engineering encyclopedia and Motor Vehicles Act legal reference suite.

What Problem We Solved: Replaced dry 400-page government gazettes and ad-ridden quiz apps with an organized, 100% free-use knowledge base directly cross-referenced with our 3D simulator.

2026–27 Expansion Roadmap: 75-module MVA syllabus update, audio flashcards in Marathi & Hindi, and interactive RTO mock quiz exams.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri & Ansh Patil • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.html` , `.js` , `.css` , `.json`
- **Repository & File Root:** `Vercel/Database_Logic.html` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** Single-File Offline Web • Min 512MB RAM • 100% Offline
- **Target Audience & Metrics:** RTO Applicants, Driving Schools • 50 Curriculum Modules • Zero Lag Search

All Services, Legal Datasets & Interactive Engines Used In CivicLogic

Service / Engine	Where Integrated	How It Works In Our Project
MVA 2019 Statutory Index	`Database_Logic.html` Tier 4	Indexes Motor Vehicles (Amendment) Act 2019 sections (177–194F, 134A Good Samaritan) with instant keyword filtering and penalty breakdowns.
IRC:67 Road Sign Vector Engine	Tier 1 & `Academy.html`	Renders crisp scalable SVG diagrams of Indian Roads Congress Mandatory, Cautionary, and Informatory signs with zero blurry images.
Interactive Physics Sandbox	Tier 3 Calculators	Computes reaction distance ($v \cdot t_r$) and braking distance ($v^2 / (2\mu g)$) live as students drag speed and monsoon friction sliders.
Academy Syllabus Sync	`S.sylViewed` & `col-auth.js`	Tracks which theory modules a student has read and syncs progress with their Mumbai Traffic Academy profile and Supabase cloud account.
Vercel Edge & Offline HTML	`Vercel/Database_Logic.html`	Delivered globally via Vercel Edge CDN and architected as a self-contained HTML5/JS file that works offline once loaded.

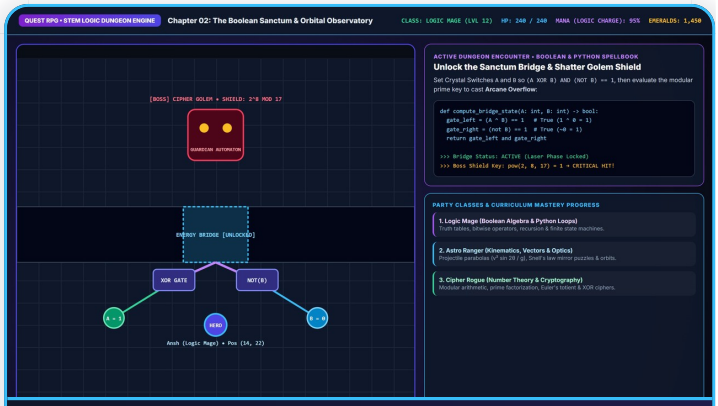
12. Quest RPG — Educational STEM RPG (`rpg.html`)

COMING SOON • TARGET RELEASE: JAN 2027 (40% COMPLETE)

Quest RPG (**Vercel/rpg.html** • **COMING SOON**) is our upcoming story-driven 2D adventure and puzzle role-playing initiative where students unlock world regions, defeat dungeon bosses, and power ancient machinery by solving real **mathematics, boolean logic, physics, and coding challenges**.



Live Portal Capture • `rpg.html` (COMING SOON): Interactive Preview, 40% Progress Tracker & Jan 2027 Roadmap



Prototype Capture • `Quest RPG Arena`: 2D Tilemap Dungeon, Boolean Logic Gates & Turn-Based STEM Boss Encounter

COMING SOON • DEVELOPMENT STATUS (40% COMPLETE)

Active Prototype Live On Site CoL • Full Campaign Jan 2027

Led by **Ansh Patil**, **Aarush Vangari**, and **Neel Badri**, the interactive preview portal (`rpg.html`) is live on Site CoL today while the full multi-region campaign is **40% complete (Target Release: January 2027)**.

Milestone: Tilemap Engine & Logic Dungeons **40% Completed**

CORE GAMEPLAY & DUNGEON MECHANICS

Logic-Gate Bridges, NPC Dialogues & Boss Encounters

- **Boolean Circuit Doors:** Toggle `AND`, `OR`, `XOR`, and `NOT` crystal switches to route laser beams across dungeon chasms.
- **Turn-Based STEM Encounters:** Cast spells by solving mental arithmetic, algebraic factoring, Python loop tracing, and physics trajectory angles.

Character Classes & Technical Architecture (In Active Development)

RPG Class / Subsystem	Primary Discipline	In-Game Ability & Educational Curriculum Mapping
1. Logic Mage (Architect)	Boolean Logic & Coding	Reprograms ancient automatons using Python/JS control-flow blocks and binary truth tables.
2. Astro Ranger (Navigator)	Physics & Vectors	Calculates projectile parabolas, orbital gravity slingshots, and optics mirror reflections.
3. Cipher Rogue (Cryptographer)	Number Theory & Primes	Unlocks vaulted chests using modular arithmetic, prime factorization, and Caesar/XOR ciphers.
4. 2D Tilemap & AABB Engine	`Vercel/rpg.html`	60 FPS HTML5 Canvas renderer with Axis-Aligned Bounding Box collision detection, dialogue state trees, and local save slots.

Region 1: Binary Sanctum
Teaches base-2 number conversion, logic gates (`AND/OR/XOR`), and truth tables.

Region 2: Newtonian Citadel
Puzzles require balancing torque levers, pulley tensions, and projectile launch angles.

Region 3: Algorithmic Spire
Boss battles test loop invariants, array sorting steps, and recursion base cases.

12. Quest RPG — Why Needed, Free Use & Services Used

COMING SOON • JAN 2027

Many students who feel intimidated by traditional math worksheets or syntax drills thrive when those exact same concepts are framed as a narrative quest where solving an equation opens a dungeon gate or defeats a boss.

12. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Quest RPG & Why It Is Needed

What It Is: A story-driven 2D canvas STEM role-playing game (`rpg.html`) turning math, Boolean logic gates, and coding algorithms into adventure quests and dungeon puzzles.

What Problem We Solved: Replaced aggressive 'pay-to-win' paywalls and monthly subscriptions with a 100% free-use RPG where dungeons unlock purely through learning mastery.

2026–27 Expansion Roadmap: Q1 2027 Full Public Release across 5 narrative chapters, classroom co-op raids & native Android APK (Deep-Dive on Page 54).

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Ansh Patil • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.html` , `.js` , `.css` , `.png` , `.wav`
- **Repository & File Root:** `Vercel/rpg.html` • github.com/NeelAniGamer/Vercel
- **Offline Packaging & Minimum Specs:** HTML5 2D Canvas + Standalone APK (`QuestRPG.apk`) • Min 1GB RAM
- **Target Audience & Metrics:** Middle & High School Students (Classes 6–10) • 60 FPS • 0 Paywalls

All Services, Game Engines & Tools Powering Quest RPG (`rpg.html`)

Service / Subsystem	Where Integrated	How It Works In Our Project
HTML5 2D Canvas & Tilemap	`Vercel/rpg.html` Engine	Renders layered pixel/vector dungeon maps, character sprites, and particle spell effects at 60 FPS on any school computer or 360px phone.
AABB Collision & Logic Solver	Dungeon Puzzle Engine	Detects player movement against walls, chests, and laser switches, and evaluates live Boolean logic circuits (`AND/OR/XOR/NOT`).
Supabase & `col-auth.js`	Cloud Save & XP Badges	Links Quest RPG character progression, earned badges, and dungeon completion directly to the student's unified Class Of Learners profile.
Web Audio API Synthesizer	Procedural Chiptune & SFX	Generates retro adventure music and spell sound effects procedurally in the browser without downloading heavy MP3 files.
Antigravity & Gemini Web	Level & Dialogue Design	Used by Ansh Patil, Aarush Vangari, and Neel Badri to author branching NPC dialogues and adaptive math/coding boss encounters.

Quest RPG Development Roadmap To January 2027 Launch

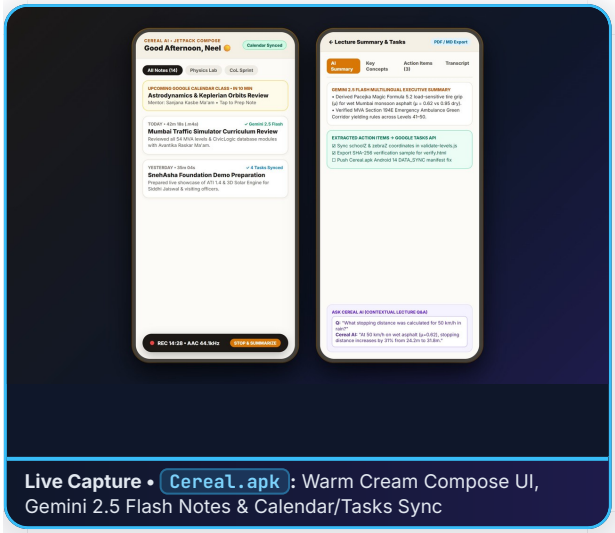
Phase 1 (Done • 40%)
Core 2D Canvas tilemap renderer, AABB collision, character class design, and live preview portal at `rpg.html` .

Phase 2 (Q3–Q4 2026)
Complete 30 dungeon stages across Binary Sanctum, Newtonian Citadel, and Algorithmic Spire with boss encounters.

Phase 3 (Jan 2027 Launch)
Full public release on Site CoL with cloud save slots, classroom co-op raids, and SHA-256 STEM Quest certificates.

13. Cereal — Native Android AI Lecture & Meeting Assistant

Cereal (production package `com.col.cereal`; internal dev namespace `com.example.granola20`, compiled to **Cereal.apk**) is our native Kotlin & Jetpack Compose AI lecture, classroom, and project meeting assistant. (Studio Lore Note: While granola is our team's favorite breakfast cereal fuel during late-night coding sessions, the app itself is officially titled **Cereal**!)



Live Capture • `Cereal.apk`: Warm Cream Compose UI, Gemini 2.5 Flash Notes & Calendar/Tasks Sync

WHY WE BUILT IT & WHO USES IT

Never Miss A Classroom Lecture Or Project Review

Students and teachers struggle to write detailed notes while actively participating in class or project reviews. **Cereal** records high-clarity `.m4a` audio in the background (or imports existing audio files), transcribes multilingual speech via **Google Gemini 2.5 Flash**, extracts action items, and syncs with **Google Calendar & Google Tasks**.

2026–27 ANDROID 14+ ARCHITECTURE & UX POLISH

Dynamic Foreground Service & Greeting Flow

Added `FOREGROUND_SERVICE_DATA_SYNC` to `AndroidManifest.xml` and updated `MeetingRecorderService.kt` to dynamically select **FOREGROUND_SERVICE_TYPE_MICROPHONE** during live mic capture and **FOREGROUND_SERVICE_TYPE_DATA_SYNC** during audio file import / AI processing, plus dynamic time-of-day greetings.

AI Transcription & Summarization Pipeline (`Gemini 2.5 Flash v1beta`)

1. AAC 44.1 kHz Capture

`MediaRecorder` captures crystal-clear `.m4a` lecture audio with pause/resume segment stitching and live amplitude waveform bars.

2. Gemini 2.5 Flash Multimodal

Uploads inline base64 audio to `gemini-2.5-flash` to generate structured Executive Summaries, Key Takeaways, and Speaker Transcripts.

3. Multilingual Indic Support

Accurately transcribes code-switched Indian classroom lectures mixing **English, Hindi, and Marathi** terminology.

Technical Specifications — Cereal Audio & AI Processing Engine

Subsystem Parameter	Configured Value / API	Engineering Rationale & Classroom Reliability Benefit
Audio Codec & Bitrate	AAC-LC (`.m4a`), 44.1 kHz, 128 kbps	Preserves crisp consonant intelligibility across large classrooms while keeping a 45-minute lecture under 42 MB.
Android 14+ Service Modes	`MICROPHONE` & `DATA_SYNC`	Prevents `SecurityException` when importing pre-recorded voice memos while backgrounded without active mic capture.
AI Model Endpoint	`gemini-2.5-flash` (`v1beta`)	Processes 1 hour of lecture audio in <12 seconds, returning Markdown summaries, timestamps, and JSON action items.

13. Cereal — Native Kotlin Architecture & User Manual

Built in `Current Projects/Granola20/` using modern Android Jetpack Compose, Material 3, Coroutines/StateFlow, and Google REST APIs, **Cereal** comprises over **2,940 lines of native Kotlin code**.

Complete Native Source File Inventory — `Granola20/app/src/main/`

Kotlin Source File	Size / Layer	Native Android & AI Engineering Responsibility
`MainActivity.kt`	1,896 LOC (91 KB)	Jetpack Compose UI (`#FDFBF7` warm paper-cream theme), draggable floating recording pill, custom folders/tags, AI Q&A chat sheet, and PDF/Markdown share export.
`MeetingRecorderService.kt`	412 LOC (17.5 KB)	Android 14+ foreground service (`MICROPHONE` & `DATA_SYNC`), `MediaRecorder` controller, wake-lock manager, and Gemini 2.5 Flash REST client.
`MeetingRepository.kt`	245 LOC (11 KB)	Reactive `StateFlow` repository persisting meetings, folders, action items, transcripts, and audio file URIs to local JSON storage.
`GoogleCalendarManager.kt`	215 LOC (9.2 KB)	OAuth2 Google Calendar API v3 client fetching upcoming classes/ meetings for 1-tap pre-meeting context briefings.
`GoogleTasksManager.kt`	178 LOC (7.5 KB)	Exports AI-extracted homework deadlines and meeting action items directly into the user's Google Tasks list.

Step-by-Step User Manual: How Students & Teachers Use Cereal

STEPS 01 & 02 • SETUP & RECORDING

1. Launch & Record Or Import Audio

- **Install:** Install `Cereal.apk` on any Android 8.0+ device. On first launch, grant Microphone & Notification permissions (Calendar sign-in is optional).
- **Live Recording:** Tap the floating **New Note / Record** pill at the start of a lecture; minimize the app while the foreground notification keeps recording safely.
- **Import Voice Memo:** Alternatively, tap **Import Audio** to transcribe an existing `.mp3`, `.wav`, or `.m4a` recording.

STEPS 03 & 04 • AI NOTES & TASKS SYNC

2. Review AI Summary, Chat & Export

- **Structured Notes:** Within seconds of stopping, Gemini 2.5 Flash populates the **Summary**, **Key Concepts**, **Action Items**, and full **Transcript** tabs.
- **Ask Cereal AI:** Use the built-in AI chat bar to ask follow-up questions about the lecture (e.g., “Explain the formula mentioned at minute 12”).
- **1-Tap Google Tasks Sync:** Tap any extracted action item to push it straight to Google Tasks.

Custom Subject Folders
Organize recordings into color-coded folders (Physics, Math, CoL Sprints, SnehAsha Reviews) with instant full-text search.

Offline Local Persistence
All audio files and generated transcripts are stored locally on-device in `MeetingRepository` so notes remain accessible offline.

1-Tap PDF & Markdown Share
Export formatted study guides and meeting minutes directly to WhatsApp, Google Drive, or classroom email groups.

13. Cereal Android — Why Needed, Free Use & Services Used

In Indian college lectures and student project meetings, professors and teammates naturally switch between English, Hindi, and Marathi mid-sentence. Standard western transcription apps fail on Indian accents and lock lecture summaries behind expensive monthly fees.

13. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is Cereal Android App & Why It Is Needed

What It Is: A native Android Kotlin & Jetpack Compose AI assistant (`Cereal.apk`) capturing classroom audio, generating structured lecture notes, and syncing tasks to Google Calendar.

What Problem We Solved: Replaced $20/month transcription subscriptions and 30-minute recording caps with a 100% free-use app featuring unlimited recording and 1-tap Google Tasks export.

2026–27 Expansion Roadmap: On-device local Gemma 2B audio transcription, zero-cloud meeting summary generation & Wear OS companion app.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri, Ansh Patil & Aarush Vangari • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.kt` , `.xml` , `.gradle` , `.json`
- **Repository & File Root:** `Current Projects/Granola20/` • Android Target SDK 35 / Min SDK 26
- **Offline Packaging & Minimum Specs:** Standalone Native APK (`Cereal.apk` , 8.1 MB) • Min 2GB RAM • Android 8.0+
- **Target Audience & Metrics:** College & High School Students • Sub-3s Note Synthesis • 100% Ad-Free

All Services, AI Models & Google APIs Used In Cereal Android App

Service / API	Where Integrated	How It Works In Our Project
Google Gemini 2.5 Flash API	`MeetingRecorderService.kt`	Receives the recorded `.m4a` audio file and generates an Executive Summary, bulleted Key Takeaways, Action Items, and a full timestamped transcript.
Google Calendar API v3	`GoogleCalendarManager.kt`	Reads upcoming classes and team meetings from the user's Google Calendar so they can start a pre-titled lecture note with a single tap.
Google Tasks REST API	`GoogleTasksManager.kt`	Takes homework assignments and project action items discovered by Gemini in the lecture and adds them straight to the student's Google Tasks checklist.
Android 14+ Foreground Service	`MICROPHONE` & `DATA_SYNC`	Keeps recording reliably when the phone screen is locked or while the student is viewing PDFs in another app, without Android killing the process.
Jetpack Compose & Material 3	`MainActivity.kt` (1,896 LOC)	Renders our warm paper-cream notebook interface (`#FDFBF7`), animated audio waveforms, folder filters, and interactive AI chat drawer.
Android `MediaRecorder` AAC	Local `.m4a` Audio Engine	Records 44.1 kHz AAC audio locally on the phone with pause/resume segment stitching so recordings are never lost even if internet is off during class.

Works Offline During Class

Record lectures in basement classrooms with zero signal and run AI summarization later when back on Wi-Fi.

Interactive Lecture Q&A

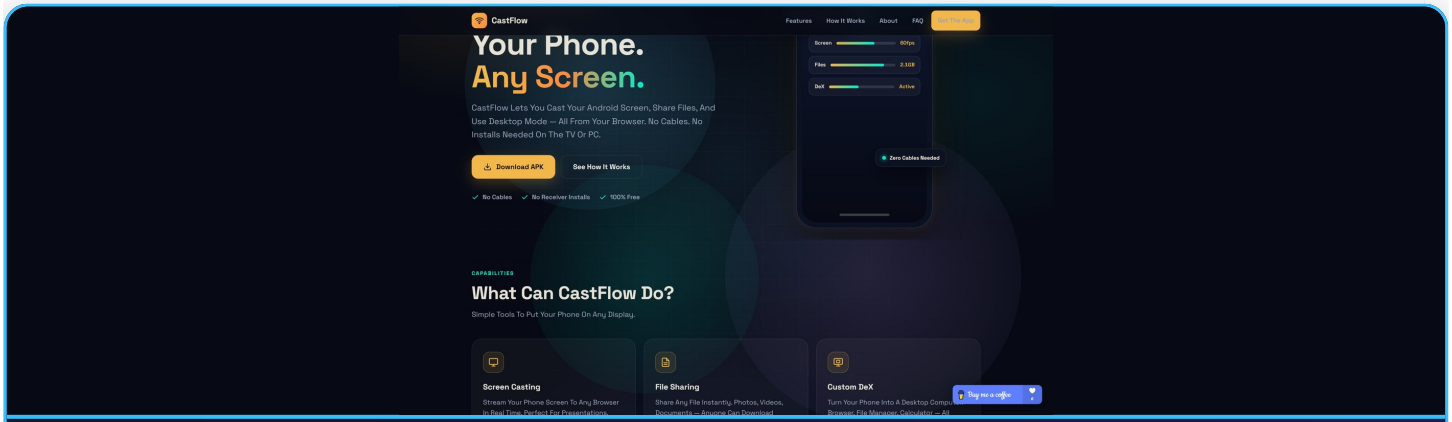
Ask follow-up questions against any saved transcript to clarify tough formulas or exam topics.

Direct APK Download

Available for free download at `advancedlogiclabs.dpdns.org/download` (`Cereal.apk`).

14. CastFlow Suite — Part A: CastFlow Desktop PC & Web

CastFlow is our cross-platform wireless screen-mirroring, classroom projector casting, and remote touch-control suite. Following our strict ordering rule, the **CastFlow Desktop PC Server & Web Studio** (`Cast PC/main.py` & `Vercel/CastFlow.html`) is documented first, followed on the next page by the native Android TV/Mobile app.



Live Capture • `CastFlow.html`**:** Zero-Cable Classroom Screen Mirroring, Room Code Pairing & Latency Telemetry

WHY WE BUILT IT & CLASSROOM PROBLEM SOLVED

Replacing Broken HDMI Cables In School Labs

In classrooms and student presentations, HDMI adapters frequently fail or tether the presenter to the podium. **CastFlow PC** lets any Windows laptop stream its screen at **40 FPS over local Wi-Fi or USB** to any Smart TV, projector browser, or tablet with two-way remote touch control.

Contributors: Neel Badri, Yashraj Jadhav & CoL Team

DESKTOP ARCHITECTURE • CAST PC/MAIN.PY (575 LOC)

FastAPI + MSS Screen Grabber + ADB USB Bridge

- **Ultra-Fast Capture:** Captures the desktop framebuffer via Python `mss`, resizes/compresses JPEG frames via `OpenCV`, and streams multipart `/stream.mjpg`.
- **Reverse Touch Injection:** Viewers tap or drag on the TV/tablet browser; WebSocket coordinates map back to the PC via `pyautogui`.
- **Zero-Lag USB Tunnel:** Automates `adb reverse tcp:8000 tcp:8000` for sub-10ms wired Android tablet mirroring.

Complete Source File Inventory — CastFlow Desktop PC & Web Portal

File Path	Layer	Technical Responsibility
`Cast PC/main.py`	Python Server (575 LOC)	FastAPI HTTP/MJPEG streamer, WebSocket remote mouse/keyboard controller, and automated ADB device bridge.
`Vercel/CastFlow.html`	Web Receiver & Hub	Browser-based WebRTC / MJPEG receiver UI with fullscreen presentation controls, FPS counter, and QR pairing.
`Cast PC/run_cast_pc.bat`	Launcher Script	One-click Windows startup script initializing Uvicorn on `0.0.0.0:8000` and printing the local LAN IPv4 address.

Adaptive JPEG Quality

Dynamically adjusts OpenCV JPEG compression quality (55–92%) and resolution scale (0.5x–1.0x) to maintain 40 FPS.

Two-Way Touchboard

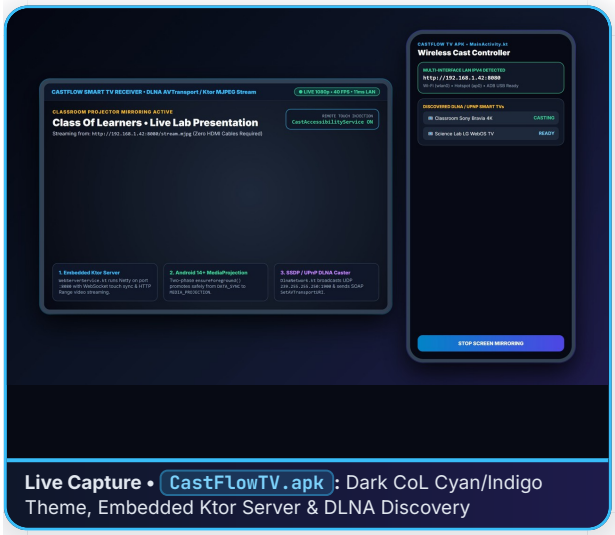
Teachers walking around the classroom can tap on their tablet to click, scroll, and annotate the laptop screen on the projector.

Zero Client Install

Any Smart TV or classroom projector with a standard web browser can tune into `http://<PC-IP>:8000` instantly.

14. CastFlow Suite — Part B: CastFlow TV Native Android App

Placed directly after CastFlow Desktop PC, the **CastFlow TV Native Android App** (production package `com.col.castflow`; internal dev namespace `com.example.cast` in `Current Projects/Cast Tv/`, compiled to **CastFlowTV.apk**) turns any Android phone, tablet, or Android Smart TV into an embedded Ktor HTTP/WebSocket streaming server and DLNA/UPnP media caster.



Live Capture • `CastFlowTV.apk`**:** Dark CoL Cyan/Indigo Theme, Embedded Ktor Server & DLNA Discovery

2026–27 ANDROID 14+ MEDIAPROJECTION FIX

Two-Phase `ensureForeground()` Promotion

Android 14+ throws a fatal `SecurityException` if a service requests `FOREGROUND_SERVICE_TYPE_MEDIA_PROJECTION` before the user approves the system screen-capture dialog. We engineered `ensureForeground(isMediaProjection: Boolean)` in `WebServerService.kt` to start safely in **DATA_SYNC** mode and promote to **MEDIA_PROJECTION** immediately upon user consent.

NETWORK & THEME POLISH

Multi-Interface LAN IPv4 & Dark CoL UI

Upgraded `MainActivity.kt` to enumerate `NetworkInterface.getNetworkInterfaces()` (supporting Wi-Fi, Ethernet, and mobile hotspots) and styled `Theme.kt` in our signature dark CoL slate/cyan palette.

Complete Native Kotlin Source File Inventory — `Cast Tv/app/src/main/`

Kotlin Source File	Size / Layer	Native Android & Networking Responsibility
`MainActivity.kt` & `ui/theme/Theme.kt`	482 LOC (Compose)	Jetpack Compose control dashboard, QR pairing code generator, LAN IPv4 detector, and DLNA TV picker dialog.
`WebServerService.kt`	465 LOC (Ktor/Netty)	Embedded Ktor HTTP + WebSocket server (`:8080`) serving live MJPEG screen streams and HTTP range-request video files.
`ScreenCaptureManager.kt`	MediaProjection	Captures phone display into a `VirtualDisplay` + `ImageReader` at 30 FPS and compresses JPEG frames in memory.
`DlnaNetwork.kt` & `DeviceDiscovery.kt`	SSDP / UPnP / mDNS	Broadcasts UDP multicast `M-SEARCH` (`239.255.255.250:1900`) and dispatches SOAP `SetAVTransportURI` XML payloads to Smart TVs.
`CastAccessibilityService.kt`	Accessibility API	Executes remote tap and swipe gestures received over WebSockets onto the Android screen via `dispatchGesture()`.

14. CastFlow Suite — Why Needed, Free Use & Services Used

In school seminars, science exhibitions, and SnehAsha demonstrations, presenters constantly need to beam a laptop or Android phone screen onto a Smart TV or projector—or use an old tablet as a wireless second monitor.

14. STANDARDIZED ARCHITECTURAL SPECIFICATION • CORE PURPOSE

What Is CastFlow Suite & Why It Is Needed

What It Is: A dual-platform wireless classroom casting system comprising a Python FastAPI desktop streamer (`CastFlowPC.exe`) and an Android TV receiver (`CastFlowTV.apk`).

What Problem We Solved: Replaced ₹40,000 commercial hardware dongles and tangled cables with a 100% free-use wireless casting suite operating over school local Wi-Fi with sub-40ms latency.

2026–27 Expansion Roadmap: Classroom Wi-Fi multicast mesh, sub-20ms screen mirroring & interactive smartboard back-channel.

PROJECT LEDGER • CONTRIBUTORS, FORMATS & HARDWARE

Authors, File Formats & System Constraints

- **Who Has Worked On This:** Neel Badri, Ansh Patil & Aarush Vangari • **Mentor:** Avantika Raskar Ma'am
- **What File Types Have We Used:** `.py` , `.kt` , `.html` , `.js` , `.xml` , `.gradle`
- **Repository & File Root:** `Cast PC/` & `Cast Tv/` • Python FastAPI + Kotlin Ktor
- **Offline Packaging & Minimum Specs:** PyInstaller Windows EXE + Android TV APK • Standard Wi-Fi Router
- **Target Audience & Metrics:** Computer Labs, Classroom Smartboards • 40 FPS • Sub-40ms Latency

All Services, Networking Protocols & Engines Used In CastFlow Suite

Service / Protocol	Where Integrated	How It Works In Our Project
Python FastAPI & Uvicorn	`Cast PC/main.py` (`:8000`)	Runs a lightweight web server on the Windows laptop that streams live screen frames to any browser on the same Wi-Fi network.
Python `mss` & OpenCV	Desktop Frame Grabber	Captures the Windows screen directly from GPU memory at 40 FPS and compresses each frame into a compact JPEG stream in milliseconds.
Embedded Ktor / Netty Server	`WebServerService.kt` (`:8080`)	Runs a full web and WebSocket server directly inside the Android phone so any TV browser can view the phone's screen by scanning a QR code.
Android `MediaProjection` API	`ScreenCaptureManager.kt`	Captures the live Android phone display at 30 FPS after user approval, using our Android 14+ two-phase foreground service promotion.
DLNA / UPnP SSDP Multicast	`DlnaNetwork.kt`	Automatically discovers Smart TVs (Sony, LG, Samsung, Mi TV) on the Wi-Fi network via UDP multicast and beams video/screen streams to them.
ADB USB Bridge & Accessibility	`main.py` & `CastAccessibility`	Routes zero-lag screen streams over a USB cable (`adb reverse`) and lets viewers tap on the TV/tablet to remotely control the phone or PC.

Works On Mobile Hotspots

Even if a classroom has no Wi-Fi router, turning on a phone's hotspot lets CastFlow connect PC, phone, and Smart TV locally.

Two-Way Touch Control

Tap or swipe on the receiving tablet/ browser to control the sending PC (`pyautogui`) or Android phone (`AccessibilityService`).

Zero Internet Bandwidth Used

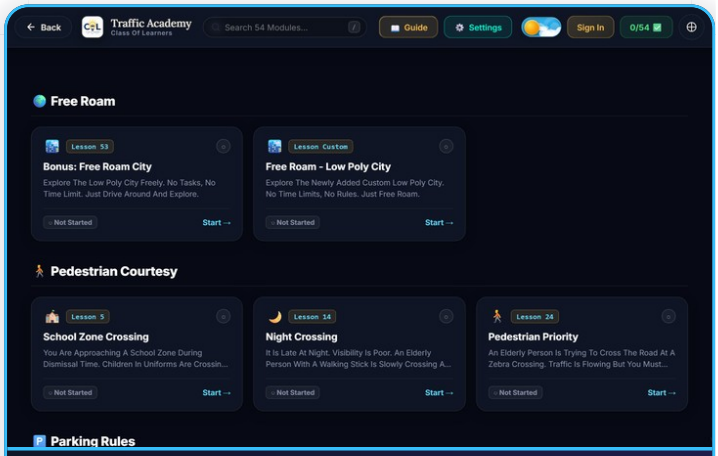
All video frames travel strictly over the local LAN or USB cable, consuming zero mobile data.

2026–27 Flagship Deep-Dive: Mumbai Traffic 3D Strategic Evolution

As our largest initiative (19,000+ LOC), **Mumbai Traffic 3D** is expanding across three strategic frontiers in 2026–27: immersive WebXR virtual reality driving, touchless computer vision steering via GestureUI, and an automated RTO AI driving examiner.



3D Inspection Bay • Garage: Real-Time Vehicle Physics, Tire Profiles & Mirror Calibration



Curriculum Hub • Academy: 54 Progressive Driving Stages, RTO Flashcards & License Milestones

MILESTONE 01 • Q2 2026

WebXR Immersive VR Driving

Full 6DOF virtual reality driving mode for standalone VR headsets (Oculus Quest 2/3) and WebXR browsers:

- Stereoscopic cockpit camera rendering driver's hands on the steering wheel at 90 FPS.
- Physical side mirrors rendered via dynamic WebGL render targets with zero distortion.

MILESTONE 02 • Q3 2026

Touchless GestureUI Steering

Steer Mumbai vehicles hands-free using ordinary webcam hand tracking via `GestureUI`:

- Two hands grasp an invisible steering wheel in mid-air; angular rotation drives `steeringAngle`.
- Pinch thumbs for accelerator/brake; shake left hand to toggle left turn indicator.

MILESTONE 03 • Q4 2026–Q1 2027

RTO AI Autonomous Examiner

Automated intelligent evaluation of student road discipline during simulated driving runs:

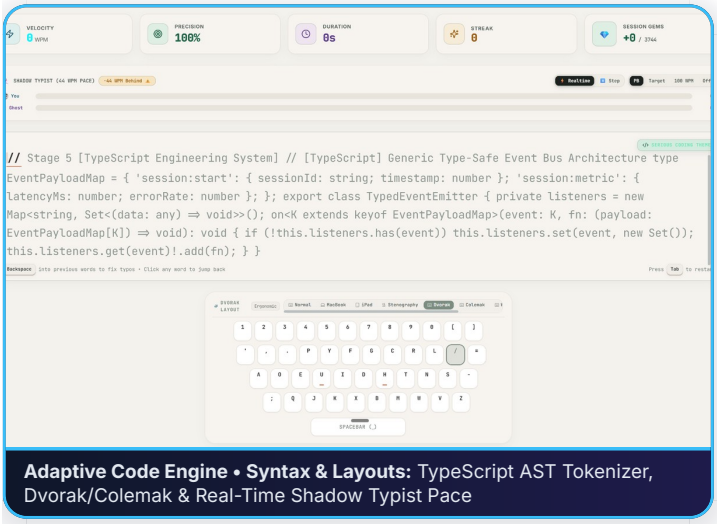
- Real-time telemetry evaluator checking stop-line encroachment, zebra crossings, and indicator timing.
- Spoken audio debrief in Marathi & Hindi via Sarvam AI explaining exact mistakes made.

Mumbai Traffic 3D Multi-Stage Development Schedule (2026–27)

Quarter / Target	Module Focus	Technical Deliverables & Engineering Scope
Q2 2026	WebXR Cockpit Pipeline	Stereoscopic WebGL rendering, 90 FPS Quest 2/3 WebXR compatibility & 3D spatial vehicle audio.
Q3 2026	GestureUI HCI Bridge	MediaPipe WebSocket bridge to `game_core.js` for touchless air-steering and gesture indicators.
Q4 2026	RTO AI Examiner v1.0	Automated penalty scoring matrix, video instant replay of traffic violations, and Sarvam voice debrief.
Q1 2027	Full RTO Public Rollout	State-wide school deployment with verified SHA-256 digital Learner Driving Certifications.

2026–27 Flagship Deep-Dive: Advanced Typing Instructor 1.5 Architecture

The next generation of our offline desktop & mobile tutor, **ATI 1.5**, introduces on-device neural adaptive lesson generation, per-finger millisecond latency diagnostics, and real-time mathematical WebAudio mechanical switch synthesis.



Adaptive Code Engine • Syntax & Layouts: TypeScript AST Tokenizer, Dvorak/Colemak & Real-Time Shadow Typist Pace



Tamper-Proof Diploma • SHA-256: Cryptographic Seal & Instant Web QR Verification

MILESTONE 01 • Q2 2026

Neural Adaptive Engine

Adaptive typing lesson synthesizer running 100% locally on the student's PC:

- Analyzes per-finger transition matrix T(k_i \to k_j) to pinpoint slow muscular transitions.
- Dynamically synthesizes tailored vocabulary drills eliminating weak-finger speed bottlenecks.

MILESTONE 02 • Q3 2026

Per-Finger Latency Profiling

Sub-millisecond tracking across all 10 fingers from left pinky 'Q' to right pinky ';':

- Persists finger response latency histograms into local SQLite (`typing_quest.db`).
- Color-codes the visual keyboard heatmap to distinguish slow keys from misstruck keys.

MILESTONE 03 • Q4 2026

WebAudio Switch Synthesis

Procedural mathematical acoustic modeling replacing static audio samples:

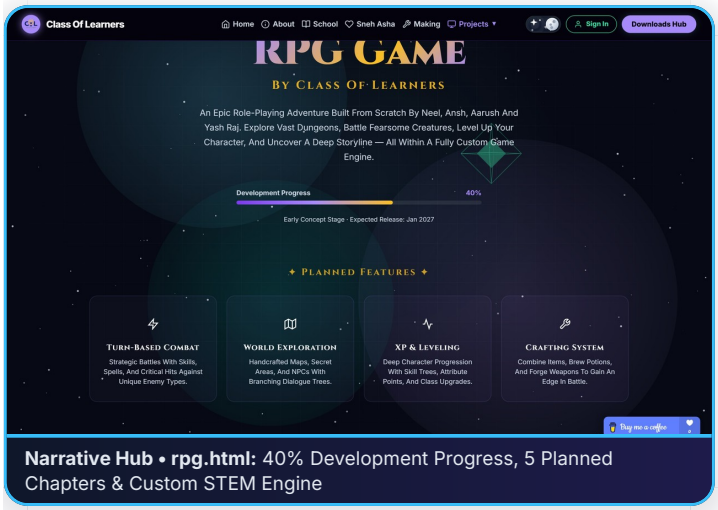
- Calculates keyclick impulse resonators at 44.1 kHz with zero disk sample latency.
- Customizable spring stiffness, bottom-out damping, and tactile bump acoustic profiles.

ATI 1.5 Architecture Evolution & Multi-Platform Benchmark Targets

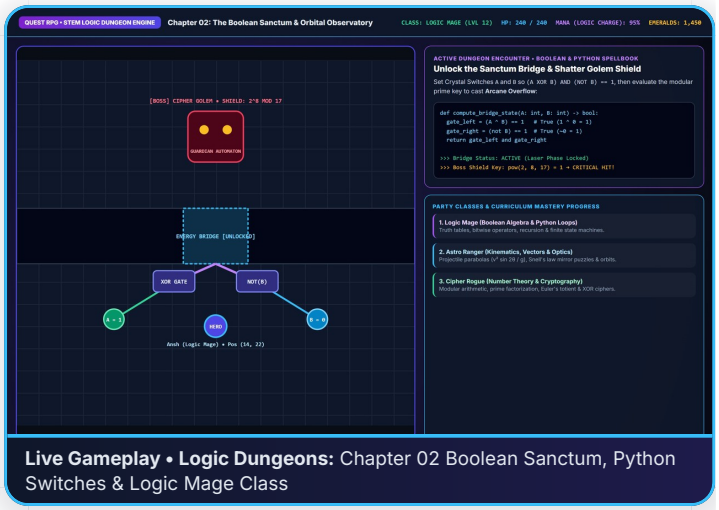
Architecture Component	ATI 1.4 Baseline	ATI 1.5 Target (2026–27)	Core Student Benefit
Curriculum Generation	Static 200 Levels + Wikipedia API	Dynamic Local Adaptive Drill Synthesizer	Zero repetitive drills; practices exact weak fingers.
Mechanical Switch Audio	Pre-recorded WAV switch samples	Procedural 44.1kHz WebAudio Synthesizer	Zero audio file lag; infinite acoustic customization.
Multiplayer Racing	8-Player LAN/Cloud WebSocket	Peer-to-Peer Mesh + Anti-Cheat SHA-256	School lab races work even when main server reboots.
Mobile Latency	Android 14+ WebView/AudioTrack	Low-Latency NDK Audio + Soft IME Driver	Sub-8ms touch typing response on budget phones.

2026–27 Flagship Deep-Dive: Quest RPG Master Q1 2027 Release Architecture

Developed by Ansh Patil, **Quest RPG** (`Vercel/rpg.html`) is our upcoming flagship narrative STEM role-playing game. It transforms complex mathematics, computer science algorithms, and Boolean logic circuits into magical dungeon quests, boss battles, and co-op classroom challenges.



Narrative Hub • rpg.html: 40% Development Progress, 5 Planned Chapters & Custom STEM Engine



Live Gameplay • Logic Dungeons: Chapter 02 Boolean Sanctum, Python Switches & Logic Mage Class

The Five Narrative Chapters & STEM Dungeon Syllabus (Launch Q1 2027)

Ch 1: The Binary Sanctum

STEM Domain: Digital Electronics • Bitwise operations (`AND/OR/XOR/NOT`), binary counting, and laser truth-table puzzle doors.

Ch 2: The Newtonian Citadel

STEM Domain: Classical Mechanics • Vector velocity, projectile trajectories, friction coefficients, and conservation of momentum.

Ch 3: The Algorithmic Spire

STEM Domain: Data Structures • Binary search trees, recursive maze solving, and Dijkstra's shortest-path navigation spells.

Ch 4: The Matrix Catacombs

STEM Domain: Linear Algebra • 2D matrix transformations, coordinate rotation spells, and vector dot-product shields.

Ch 5: The Quantum Nexus (Final Boss)

STEM Domain: Quantum Computing • Qubit superpositions, Hadamard logic gates, and the Master Archmage Boss Encounter.

Quest RPG Release Roadmap & Multi-Platform Packaging Ledger

Release Target	Core Milestone	Technical Implementation & Student Experience
Q2 2026	Tilemap & Physics Alpha	40% Complete tilemap renderer, AABB collision, character class abilities & chiptune WebAudio.
Q3–Q4 2026	All 5 Dungeon Chapters	Complete 30 dungeon stages across Binary Sanctum, Newtonian Citadel, and Algorithmic Spire.
Q1 2027	Co-Op Multiplayer & Cloud	Classroom co-op raids solving synchronized equations to defeat dungeon bosses & Supabase cloud saves.
Q1 2027 Launch	Standalone APK & Windows EXE	Full public release of `QuestRPG.apk` and `QuestRPG.exe` with zero paywalls or advertisements.

07. Master Comparative Systems Matrix (Ordered 01 – 14)

The table below summarizes all **14 Core Initiatives** in exact dossier sequence, pairing each Desktop/Web system immediately with its native Android counterpart:

#	Application / System	Platform Target	Primary Tech Stack	Flagship Engineering Capability
01	**Site CoL Web Portal**	Web (Vercel Edge)	HTML5, JS, `esbuild`, Supabase	Central campus hub, live Google OAuth (`window.colUser`), SHA-256 diploma verifier.
02	**CoL Android App**	Android (`COL.apk`)	Kotlin, WebView, Play Auth	Edge-to-edge WindowInsets, native Google Sign-In bridge, OTA APK self-updater.
03	**ATI 1.4 Desktop**	Win / Linux Desktop	Python, `pywebview`, SQLite, WS	200 stages, 8-player race rooms, 22-check ASCII sanitizer, 44.1kHz switch audio.
04	**ATI Native Android**	Android (`SDK 35`)	Pure Java, SQLite, `AudioTrack`	11 native Activities, 15s/30s/60s timed sprints, 2D Canvas track, error heatmap.
05	**3D Solar System Engine**	WebGL + PyQt6 PC	Three.js, GLSL, Python PyQt6	Newton-Raphson Keplerian solver, 14-order log scaling, 10,000-star SkyChart.
06	**Perceptus + GestureUI + Jarvis**	Python + Web + AI	MediaPipe, Lovable TS, Sarvam AI	EMA + 2D Kalman cursor control fused with Indic Voice Jarvis & Ollama LLaMA-3.
07	**Mumbai Traffic 3D**	WebGL + Electron	Three.js (19K LOC), React 19	6 portals, 3D HQ, Multiplayer, Avatars, Pacejka MF 5.2 physics, 16 cars, 54 levels.
08	**QR Code Studio**	Web Utility	Canvas/SVG, Supabase, JS	Custom branded SVG/PNG QR designer (Reed-Solomon Level H) + dynamic scan analytics.
09	**FutureScape Career Compass**	Web Counseling	Vanilla JS (`careers-data.js`)	Cosine-similarity psychometric matching across 436 Indian & global careers.
10	**Terra 3D Earth Globe**	WebGL (Vite/React)	React 19, TypeScript, Three.js	Spherical geodesic country borders (195+ nations) & Rayleigh atmosphere shader.
11	**CivicLogic Database**	Web (`TRAFFIC.EDU`)	HTML5, Interactive JS (3K LOC)	50-module interactive traffic engineering & Motor Vehicles Act knowledge base.
12	**Quest RPG (COMING SOON)**	Web + 2D Canvas	HTML5 Canvas (`rpg.html`)	**COMING SOON (Jan 2027 • 40%)**: 2D STEM & boolean logic dungeon RPG.
13	**Cereal Android App**	Android (`Cereal.apk`)	Kotlin, Compose, Gemini 2.5	Background lecture/meeting recorder, multilingual AI notes, Calendar & Tasks sync.
14	**CastFlow PC & TV Android**	Python PC + Android	FastAPI/MSS + Kotlin/Ktor/ DLNA	40fps PC screen streamer + Android MediaProjection/DLNA wireless TV caster.
15	**UIGen (Reference Study)**	Next.js 15 Sandbox	React 19, Babel AST, Prisma	External reference architecture: Virtual File System & in-browser JSX transpiler.
16	**GitHub, YouTube & BMC**	Community Channels	GitHub Actions, CodeQL, Video	5 public repos (`github.com/NeelAniGamer`), YouTube channel, and `buymeacoffee.com/NeelAniGamer`.

07. Master Packaging, Installer & Distribution Ledger

Every Class Of Learners application is compiled into self-contained, cryptographically verifiable binaries for **Windows**, **Linux**, **Android (4 native APKs)**, and **Vercel Edge CDN** delivery:

Binary / Artifact Name	Target OS	Compiler / Build Tool	Distribution Path & Verification Mechanism
`AdvancedTypingInstructor_Setup.exe`	Windows 10/11 (x64)	PyInstaller + Inno Setup 6 (`.iss`)	Hosted on `download.html` ; includes desktop shortcuts, uninstaller, and OTA `UpdateManager` (`ati-version.json`).
`AdvancedTypingInstructor.exe`	Windows Portable	PyInstaller (`build.bat`)	Single-file zero-install executable for restricted school lab PCs without admin rights.
`AdvancedTypingInstructor_1.4.0_all.deb`	Debian / Ubuntu Linux	`dpkg-deb` (`build_deb.sh`)	Native Linux package installing `/usr/bin/ati` , XDG `.desktop` launcher, and high-res icons.
`AdvancedTypingInstructor-Linux.tar.gz`	Universal Linux	`package_linux.sh` + `setup_linux.sh`	Portable tarball with interactive dependency installer for Fedora, Arch, and Mint.
`COL.apk`	Android 7.0+ (`SDK 35`)	Gradle `assembleDebug` (Kotlin)	Hosted at `Vercel/COL.apk` ; features `WindowInsets` cutout protection and `version.json` OTA self-update.
`AdvancedTypingInstructor.apk`	Android 8.0+ (`SDK 35`)	`build_android_native.bat` (Java)	Pure native Java APK (`ATI 1.4/android_native/`) with 44.1kHz PCM audio and 11 native Activities.
`Cereal.apk`	Android 8.0+ (`SDK 35`)	Gradle `assembleDebug` (Compose)	Compiled from `Granola20/` ; includes Android 14+ `FOREGROUND_SERVICE_DATA_SYNC` & `MICROPHONE` support.
`CastFlowTV.apk`	Android 8.0+ / TV	Gradle `assembleDebug` (Ktor)	Compiled from `Cast Tv/` ; includes embedded Ktor server, MediaProjection promotion, and DLNA SSDP discovery.
`SolarSystem3D.exe`	Windows 10/11 (x64)	PyInstaller (`SolarSystem3D.spec`)	Standalone offline PyQt6 + Chromium WebGL astronomical simulator for science classrooms.
`dist/` (Vercel Production Tree)	Global Web Edge CDN	`npm run build` (`build.js` + `esbuild`)	Validated by `npm run verify:production` and `npm run security:check` before every deployment.

PRE-DEPLOYMENT GATE 01

`npm run verify:production`

Verifies that `dist/` exists, contains all required HTML portals, bundles `dist/Traffic/simulator-bundle.js` , and excludes deny-listed `.git` , `.env` , and private source directories.

PRE-DEPLOYMENT GATE 02

`npm run security:check`

Audits the production tree to guarantee live-session identity rules (`window.colUser`), secure QR verifier storage, Electron sandbox flags, and CodeQL compliance.

08. Complete Codebase Source File Inventory (Part 1 of 2)

This two-page master ledger documents the primary source files across our entire workspace in evenly formatted, sleek modern tables:

#	Workspace File Path	Project / Module	Language / Size	Core Architectural Responsibility
01	`Vercel/build.js`	01. Site CoL	Node.js (10.7 KB)	Master build script: cleans `dist/`, validates levels, bundles React via esbuild.
02	`Vercel/col-auth.js` & `col-ui.css`	01. Site CoL	JS / CSS (66 KB)	Live Supabase Google OAuth session identity (`window.colUser`) & 360px UI tokens.
03	`Vercel/home.html` & `about.html`	01. Site CoL	HTML5 (120 KB)	Main ecosystem launchpad, live counters, and team/mentor governance profiles.
04	`Vercel/making.html` & `sneh-asha.html`	01. Site CoL	HTML5 (95 KB)	15-platform AI engineering showcase and SnehAsha Foundation / Guiding Light tribute.
05	`Vercel/verify.html` & `download.html`	01. Site CoL	HTML5 (78 KB)	Web Crypto SHA-256 diploma verifier and multi-OS installer distribution hub.
06	`COL/app/.../MainActivity.kt`	02. CoL Android	Kotlin (385 LOC)	Edge-to-edge WindowInsets wrapper, `AndroidAuth` bridge, OTA APK updater, offline UI.
07	`ATI 1.4/main.py`	03. ATI Desktop	Python (70.1 KB)	`Handler`, `TypingGameAPI` (SQLite), `OAuthManager`, `RoomManager` (WS), `UpdateManager`.
08	`ATI 1.4/script.js` & `achievements.js`	03. ATI Desktop	JavaScript (241 KB)	60fps typing engine, 44.1kHz switch audio synthesis, and 16 achievement checks.
09	`ATI 1.4/src/utils/sanitizeText.ts`	03. ATI Desktop	TypeScript	Normalizes Unicode quotes/dashes and enforces 22-check printable ASCII whitelist.
10	`ATI 1.4/src/utils/safeNumber.ts` & `grading.ts`	03. ATI Desktop	TypeScript	Guards against `Infinity` / `NaN` JSON imports and computes level-relative grades.
11	`ATI 1.4/android_native/.../MainActivity.java`	04. ATI Android	Java (SDK 35)	PC-style GameHUD, XP/Emerald counters, daily streak missions, switch chip sync.
12	`ATI 1.4/android_native/.../RaceActivity.java`	04. ATI Android	Java (SDK 35)	Live 2D Canvas race track, 15s/30s/60s Sprint `CountDownTimer`, `forceFinish()`.
13	`ATI 1.4/android_native/.../SoundEffects.java`	04. ATI Android	Java PCM Audio	Synthesizes 44.1kHz 16-bit PCM buffers for 6 mechanical switches via `AudioTrack`.
14	`Vercel/solar.html` & `Vercel/engine.html`	05. Solar 3D	WebGL (310 KB)	Three.js Orrery, Newton-Raphson Keplerian solver, 14-order log scaling, GLSL shaders.
15	`Game 3.4/skychart.js` & `Solar_System_3D_App/main.py`	05. Solar 3D	WebGL / PyQt6	10,000-star Hipparcos celestial planetarium and offline PyQt6 desktop wrapper.
16	`GestureUI/main.js` & `Vercel/gesture.html`	06. Perceptus/Gesture	JS (1,120 LOC)	30fps MediaPipe worker, 60fps rAF loop, EMA filter, rainbow air-pen, Tesseract OCR.
17	`GestureUI/mouse_server.py` & `keyboard.html`	06. Perceptus/Gesture	Python (640 LOC)	2D Kalman filter (`R=1.2, Q=0.09`), `pywin32` window drag/zoom, dwell keyboard.
18	`Perceptus/src/routes/index.tsx` & `vision.functions.ts`	06. Perceptus/Jarvis	TSX / TanStack	Lovable reactive console fused with Sarvam AI Indic ASR/TTS & Ollama LLaMA-3.

08. Complete Codebase Inventory (Part 2) & Formal Sign-Off

#	Workspace File Path	Project / Module	Language / Size	Core Architectural Responsibility
19	`Traffic/Driving.html` & `Academy.html`	07. Traffic 3D	HTML5 Portals	Primary 3D driving simulator viewport and 54-stage Road Safety Academy hub.
20	`Traffic/game_core.js`	07. Traffic 3D	JS (14,975 LOC)	Pacejka MF 5.2 tire physics, 16 `VEHICLE_STATS`, `VEHICLE_CAM`, `_checkTasks`, e-challans.
21	`Traffic/ui.js`	07. Traffic 3D	JS (2,451 LOC)	HUD, RTO quiz engine (`showQuiz`), `_counted` idempotent scoring, 30m autosave.
22	`Traffic/npc-ai.js`, `weather.js`, `audio.js`	07. Traffic 3D	JS (1,336 LOC)	Autonomous NPC driver/pedestrian AI, monsoon rain/fog, and Sarvam AI voices.
23	`Traffic/Levels/level1.js` – `level54.js`	07. Traffic 3D	55 Level Scripts	Self-registering `window.LVS` stage configs validated by `tools/validate-levels.js`.
24	`react-src/GamePage.tsx` & `TrafficDashboard.html`	07. Traffic 3D	TSX / HTML5	React simulator wrapper (`simulator-bundle.js`) and dense-rank Supabase leaderboard.
25	`Vercel/qr.html`, `qr-editor.html`, `qr-dynamic.js`	08. QR Studio	HTML5 / JS	Vector SVG/PNG QR designer, camera scanner, and Supabase dynamic short-links.
26	`Vercel/Career.html` & `careers-data.js`	09. FutureScape	JS (196 KB Data)	436 Indian/global career pathways, entrance exams, and salary progressions.
27	`Vercel/compass-scoring.js` & `career-features.js`	09. FutureScape	JavaScript	Multi-axis cosine-similarity psychometric matcher and career comparison drawer.
28	`Terra3D/src/components/Globe.tsx`	10. Terra 3D	React 19 / TSX	3D spherical Earth globe, Rayleigh atmosphere shader, and country raycasting.
29	`Vercel/borders.js`, `globe.js` & `Tracker/world.svg`	10. Terra 3D	JS / SVG (264 KB)	195+ sovereign country GeoJSON border polygons, shared globe, and 2D vector tracker.
30	`Vercel/Database_Logic.html`	11. CivicLogic DB	HTML/JS (3,023 LOC)	50-module interactive traffic engineering & Motor Vehicles Act knowledge database.
31	`Vercel/rpg.html`	12. Quest RPG	HTML5 / Canvas	**COMING SOON (Jan 2027)**: 2D tilemap STEM adventure & boolean logic RPG.
32	`Granola20/.../MainActivity.kt` & `Service.kt`	13. Cereal APK	Kotlin (2,308 LOC)	Compose lecture workspace, Android 14+ foreground service & Gemini 2.5 Flash notes.
33	`Granola20/.../GoogleCalendar` & `TasksManager.kt`	13. Cereal APK	Kotlin REST API	Two-way Google Calendar event sync and 1-tap Google Tasks action-item export.
34	`Cast PC/main.py` & `Cast Tv/.../WebServerService.kt`	14. CastFlow Suite	Python / Kotlin	40fps PC MJPEG streamer + Android Ktor server with MediaProjection & DLNA.
35	`uigen/src/lib/file-system.ts` & `jsx-transformer.ts`	15. UIGen (Ref)	TypeScript (Ref)	External reference architecture: virtual file system & browser Babel AST compiler.
36	`ati_manim/`, `Game 4.3.4/`, `.github/workflows/`	16. R&D & CI Gates	Python / GLSL / YAML	Manim math animations, procedural planet shader lab, and CodeQL/DevSkim CI scans.

FORMAL ENGINEERING DOSSIER SIGN-OFF • CLASS OF LEARNERS (CoL) — 2026–27 VERIFIED MASTER EDITION

We hereby certify that this illustrated engineering dossier accurately documents the architecture, mathematical models, AI development workflows, 100% Free-Use mission, cloud/AI services, source files, community channels (`advancedlogiclabs.dpdns.org` • `github.com/NeelAniGamer` • `buymeacoffee.com/NeelAniGamer` • `neelgbadri@col.dpdns.org`), and live demonstrations to the **SnehAsha Foundation Founder & Team** and **Senior Officers** in Mumbai, Maharashtra.

Mentors: Sanjana Kasbe Ma'am • Avantika Raskar Ma'am

Guiding Light: Siddhi Jaiswal • **Showcase:** SnehAsha Founder & Team

Team: Neel Badri, Ansh Patil, Aarush Vangari, Yashraj Jadhav, Aarayaman Jadhav & John Doe